

Glaucoma — Case Cards

M.S. Ophthalmology practical examination · Tamil Nadu Dr M.G.R. Medical University · 16 October 2026

Every card follows the same headings. The generic long-case proforma is in 00 · Index and master case format.

★ kept at JEH last year · Long case · Short case · Task · Chart

Contents

No.	Card	Type	Last year
GX	Glaucoma treatment toolkit — every drug, laser and operation the glaucoma cards use	Toolkit	—
G1	Primary open-angle glaucoma, including the glaucomatous optic disc	Long + short case	★ Kept 6 times (one with pseudoexfoliation glaucoma)
G4	Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy and the acute attack	Long case	★ Kept 4 times (once as the long case)
G5	Neovascular glaucoma, including rubeosis iridis	Long + short case	★ Kept once
G6	Pseudoexfoliation syndrome and pseudoexfoliation glaucoma	Long + short case	★ Kept once (with POAG)
G7	CRVO with POAG, including hemi-CRVO with POAG	Short case	★ Kept twice (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case)
G8	Applanation tonometry — perform and record	Task	★ Done in 8 of the 12 glaucoma cases
G9	Gonioscopy — perform, grade and draw	Task	★ Done in about 8 of the 12 glaucoma cases
G10	Humphrey visual field in glaucoma — reading the printout	Chart	★ Done in 3 of the 12 glaucoma cases
G2	POAG with trabeculectomy — bleb assessment and the failed bleb	Long case	—
G3	POAG with a glaucoma drainage device	Long case	—
G11	Other glaucomas the examiner may ask — childhood and secondary glaucomas	Viva sheet	—

GX Glaucoma treatment toolkit — every drug, laser and operation the glaucoma cards use

Read before the glaucoma case cards · Toolkit

How to say a management plan in this subject

Keywords the examiner listens for: target pressure · prostaglandin analogue · uveoscleral outflow · aqueous suppressant · fixed combination · laser peripheral iridotomy · selective laser trabeculoplasty · trabeculectomy with mitomycin C · glaucoma drainage device · cyclophotocoagulation

SAY IT

I will first set a target pressure from the baseline pressure and the stage of damage. I will start one drug, a prostaglandin analogue at bedtime, and review after two months. If the target is not met, I will switch, or add an aqueous suppressant, preferably as a fixed combination. In angle closure, laser peripheral iridotomy comes first. If drops cannot be used reliably, I will offer selective laser trabeculoplasty. If pressure stays above target on maximal tolerated therapy, or the fields progress, I will advise trabeculectomy with mitomycin C. A glaucoma drainage device is kept for failed filtering surgery or scarred conjunctiva. Cyclophotocoagulation is for eyes with poor visual potential.

Drugs

Prostaglandin analogues — first line; intraocular pressure (IOP) falls 25–32%

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Latanoprost 0.005% once, at bedtime Travoprost 0.004% once Bimatoprost 0.01%, 0.03% once (a prostamide) Tafluprost 0.0015% once	Increases uveoscleral outflow : relaxes the ciliary muscle and remodels its extracellular matrix. The effect peaks 10–14 hours after the drop, so it is given at bedtime	Ocular: conjunctival hyperaemia; longer, darker lashes; darker lid skin; iris pigmentation (more tyrosinase in melanocytes); uveitis; cystoid macular oedema; reactivated herpetic keratitis. Systemic: almost none	Uveitis and neovascular glaucoma: more inflammation. Sturge–Weber syndrome: no benefit against a high episcleral venous pressure; uveal effusion. Drug of choice in angle recession

Beta-blockers — IOP falls 20–30%

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Timolol 0.25%, 0.5% twice a day, the first dose in the morning Betaxolol 0.5% twice a day (cardioselective) Levobunolol 0.5%, carteolol 1% once or twice a day	Aqueous suppressant : blocks cyclic adenosine monophosphate production in the ciliary epithelium, cutting aqueous production by 20–50%	Ocular: irritation, allergy, punctate epithelial erosions, dry eye. Systemic: bradycardia , hypotension, heart failure, bronchospasm , depression, sleep disturbance	Asthma , chronic obstructive pulmonary disease, heart failure, second- or third-degree heart block, bradycardia. Works poorly in a patient already on systemic beta-blockers

Alpha-2 agonists — IOP falls 15–20%

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Brimonidine 0.1%, 0.15%, 0.2% three times a day, sometimes twice Apraclonidine 0.5–1% twice a day; before and after laser	Brimonidine lowers aqueous production, raises uveoscleral outflow and is neuroprotective. Apraclonidine also raises trabecular outflow and lowers episcleral venous pressure	Ocular: allergic blepharoconjunctivitis , lid dermatitis; apraclonidine dilates the pupil. Systemic: dry mouth , somnolence , bradycardia, postural hypotension	Children under 2 years : crosses the blood–brain barrier (apnoea, hypotension, coma). Caution in cerebral or coronary insufficiency. Not after a drainage device: it vascularises the bleb capsule

Carbonic anhydrase inhibitors — IOP falls 15–20%

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Dorzolamide 2% three times a day Brinzolamide 1% two or three times a day	Aqueous suppressant: inhibits carbonic anhydrase II in the ciliary epithelium	Ocular: burning, stinging, superficial punctate keratitis, periorbital dermatitis. Systemic: thrombocytopenia	Sulpha allergy , renal failure, chronic liver disease
Acetazolamide 250 mg tablet two to four times a day; 500 mg slow-release capsule twice a day	The same action, given by mouth	Ocular: transient myopia. Systemic: paraesthesia , metallic taste, nausea, metabolic acidosis , potassium loss, renal stones ; aplastic anaemia, Stevens–Johnson syndrome	Sulpha allergy, renal failure or transplant, chronic liver disease

Miotics — IOP falls 15–20%

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Pilocarpine 0.5–4% two to four times a day	Open angle: the longitudinal ciliary muscle pulls the scleral spur, tightening the meshwork and raising outflow. Angle closure: miosis pulls the iris off the meshwork and relieves pupillary block	Ocular: browache, induced myopia , dim vision in poor light, congestion, more inflammation, cataract, retinal holes and detachment. Systemic: salivation, urination	Neovascular, uveitic and phacolytic glaucoma . Young patients tolerate it poorly. Ineffective in angle recession. Never with a prostaglandin analogue

Hyperosmotic agents — for an acute rise

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Mannitol 20% intravenous, 1–2 g/kg over 20–30 minutes Glycerol 50% oral, 1–1.5 g/kg Isosorbide 45% oral, 1–2 g/kg	Raise plasma osmolarity, so water leaves the eye, mainly from the vitreous; vitreous volume and IOP fall	Ocular: rebound IOP rise, intraocular haemorrhage. Systemic: nausea, vomiting, thirst, confusion, diuresis, dehydration, hyperglycaemia, pulmonary oedema, heart failure	Anuria, severe dehydration, heart failure, pulmonary oedema, low blood pressure. Glycerol: not in diabetes (isosorbide is safe)

Drugs in the acute attack

1. **Mannitol 20% intravenously, 1–2 g/kg over 20–30 minutes**, first, because miotics fail at very high pressure.
2. If no nausea or vomiting: **acetazolamide 250–500 mg stat, then 250 mg four times a day**. Or **glycerol 50% 1–1.5 g/kg**, after ruling out diabetes. Add **apraclonidine 0.5%**.
3. Once IOP is **below 40 mm Hg**: **pilocarpine 2%** four times in 30 minutes, then 6-hourly.
4. **Timolol 0.5% twice a day** with **brimonidine 0.2% three times a day**, 10 minutes apart.
5. **Prednisolone acetate 1% four times a day**; ibuprofen 500 mg twice a day.
6. Review hourly. When the cornea clears, laser peripheral iridotomy in both eyes (card G4).

Supporting drugs — steroid, cycloplegic, bevacizumab

Drug (strength · frequency)	Mechanism of action	Adverse effects (ocular · systemic)	Contraindications and cautions
Prednisolone acetate 1% four times a day after laser or an angle-closure attack; 6–8 times a day in neovascular glaucoma; steroid hourly for a failing bleb	Blocks phospholipase A2, so fewer prostaglandins are made; suppresses fibroplasia under the bleb	Ocular: IOP rise in steroid responders , posterior subcapsular cataract, reactivated herpes simplex, secondary infection, corneal melting	Bleb infection: only 12–24 hours after antibiotics. Watch IOP: glaucoma patients respond more often
Atropine 1% three times a day	Cycloplegic : paralyses the ciliary muscle, so the lens–iris diaphragm tightens and the chamber deepens. Also relieves ciliary spasm and prevents posterior synechiae	Ocular: the strongest cycloplegic; the pupil stays dilated 7–10 days. Systemic: flushing, tachycardia, fever, delirium	Can precipitate angle closure in an occludable angle. Used in neovascular and uveitic glaucoma, a shallow chamber after trabeculectomy and malignant glaucoma
Bevacizumab 1.25 mg in 0.05 ml intravitreal, through the pars plana	Blocks vascular endothelial growth factor (VEGF) , so iris new vessels regress within 24–48 hours	The effect lasts some weeks	An adjunct in neovascular glaucoma, given 2–3 days before panretinal photocoagulation (PRP) or surgery, for less bleeding. PRP remains the mainstay

Combinations

A fixed combination needs complementary mechanisms and similar dosing.

Combination	Why use it	Disadvantages
Timolol 0.5% + dorzolamide 2%	Lowers IOP more than either alone; two suppressants that still add; no electrolyte upset	Contraindications of both apply: asthma, heart block, sulpha allergy
Prostaglandin analogue + timolol	Complementary: more uveoscleral outflow and less inflow; IOP falls 13–37%	Timing clash: timolol works best in the morning, the prostaglandin at bedtime
Brimonidine + timolol	Two suppressants; a choice after iridotomy and in uveitic glaucoma	Brimonidine allergy; beta-blocker cautions
Prostaglandin + brimonidine, or + topical carbonic anhydrase inhibitor	Outflow drug plus inflow suppressant, with no beta-blocker	Brimonidine or sulpha allergy
Never prostaglandin + pilocarpine	—	Pilocarpine contracts the ciliary muscle and cuts the uveoscleral outflow the prostaglandin needs

Overall, for: simple dosing, adherence, less preservative and systemic toxicity, lower cost. **Against:** fixed timing, altered potency, and exposure to a component that may not help.

Lasers

Procedure (indications)	Key steps and settings	Advantages	Disadvantages and complications
Neodymium:yttrium-aluminium-garnet (Nd:YAG) laser peripheral iridotomy: angle closure at every stage, the fellow eye, pupillary block, pigment dispersion, combined mechanism	Pilocarpine 1%; apraclonidine or brimonidine before and after. Abraham lens (+55 dioptre). Outer third, 11–1 o'clock, in a crypt. 2–5 mJ, 1–3 pulses; 150–200 µm. End point: gush of pigment. IOP at 1 hour; steroid for 1 week (card G4)	Photodisruption works on any iris colour, with fewer late closures than argon	IOP spike, iritis, hyphaema, corneal burn, lens damage, glare; rarely malignant glaucoma. Not with corneal oedema, a flat chamber, synechial closure, uveitis
Argon laser iridoplasty (gonioplasty): plateau iris syndrome still occludable after iridotomy; nanophthalmos; narrow angle with primary open-angle glaucoma (POAG)	Argon burns on the peripheral iris: 200–400 mW, 0.1 s, 20–24 spots over 360°. The burns shrink and flatten the peripheral iris, so the angle deepens	Opens an angle that iridotomy alone leaves closed	Not a substitute for iridotomy: most such eyes also have pupillary block, so iridotomy comes first
Selective laser trabeculoplasty (SLT): POAG, pseudoexfoliation, pigmentary and steroid-induced glaucoma; poor compliance; failed argon trabeculoplasty	Frequency-doubled Q-switched Nd:YAG, 532 nm; SLT lens or Goldmann three-mirror. 0.5–1.5 mJ, 400 µm, 3 ns; 50–100 adjacent spots over 180°. Target: melanin in pigmented meshwork cells. End point: tiny bubbles	No heat spread to nearby tissue. Good control for 3–5 years; best results in pseudoexfoliation	IOP spike, mild uveitis, pigment dispersion, hyphaema, peripheral anterior synechiae (PAS). Half need drops again. Not in uveitic, neovascular, iridocorneal endothelial (ICE) syndrome, synechial or angle-recession glaucoma
Argon laser trabeculoplasty: as for SLT	200–1200 mW, 50 µm, 0.1 s; 40–50 spots over 180°, at the junction of pigmented and non-pigmented meshwork; blanching or a tiny bubble	IOP falls about 30%; half controlled at 5 years	IOP spike, uveitis, PAS; later blebs encapsulate up to 3 times more often
Panretinal photocoagulation (PRP): neovascular glaucoma with ischaemic retina; works while synechial closure is 270° or less	Green 532 nm; 500 µm, 0.1 s; mild white burns one burn-width apart; 1200–1600 burns in 3–4 sittings (card G5)	Treats the cause: less VEGF, so new vessels regress in 4–6 weeks	Poorer night, colour and peripheral vision; glare; corneal burns, iritis, shallow chamber
Transscleral diode cyclophotocoagulation: refractory glaucoma after failed or high-risk surgery; painful blind eye; patient too ill for surgery	Local anaesthesia. 1500–2000 mW for 1.5–2 s; 12–24 burns behind the limbus over 360°, sparing 3 and 9 o'clock; fewer in a seeing eye. Steroid for 2 weeks. Alternative: non-contact Nd:YAG, 30–40 spots over 360°	Quick; repeatable after 1 month; two-thirds need no repeat	Pain, inflammation, cystoid macular oedema, hypotony, phthisis bulbi ; rarely malignant glaucoma, suprachoroidal haemorrhage, sympathetic ophthalmia
Laser suture lysis: IOP above target up to 6 months after trabeculectomy; the tie on a non-valved tube (card G3)	Argon, Hoskins lens; 50 µm, 0.02–0.1 s, 250–1000 mW on a flap suture	Flap sutured tight, then opened: less hypotony	Conjunctival burn, leak, flat chamber, hypotony maculopathy

Surgery

Procedure (indications)	Key steps and settings	Advantages	Disadvantages and complications
Trabeculectomy: not at target on maximal tolerated drugs, or progressing; side effects or poor compliance; baseline IOP over 40 mm Hg; one eye lost; difficult follow-up	Stop timolol 2 weeks and carbonic anhydrase inhibitors 1–2 days before, so a bleb forms without hypotony. Superotemporal or superonasal. Scleral flap 3 × 4 mm , two-thirds thick; antimetabolite; Kelly punch, ostium 1.5 × 2 mm ; peripheral iridectomy, so the iris cannot block the ostium; 10-0 nylon flap; watertight 8-0 vicryl conjunctiva (card G2)	Plain operation lowers IOP about 41% . The flap gives steadier IOP and less hypotony than full-thickness surgery	Shallow chamber (commonest), leak, overfiltration, malignant glaucoma, fibrosis and bleb failure, cataract, bleb infection and endophthalmitis
Surgical peripheral iridectomy: acute attack with a hazy cornea and closure under two-thirds	Iridectomy through a limbal incision, mostly at 12 o'clock; Nd:YAG iridotomy for the fellow eye	Possible when the cornea is too hazy for laser. Clean, triangular margins; it cannot close	An open intraocular operation. Closure over two-thirds needs trabeculectomy instead

Procedure (indications)	Key steps and settings	Advantages	Disadvantages and complications
Combined cataract surgery and trabeculectomy: visually significant cataract with poor IOP control, drug intolerance or poor compliance, advanced glaucoma, pseudoexfoliation, one-eyed patient	Both operations in one sitting; an antimetabolite can be used	One operation; earlier visual recovery; fewer early IOP spikes	More hyphaema, uveitis, shallow chamber, hypotony; slower recovery than cataract alone; weaker long-term control than trabeculectomy alone
Cataract surgery alone: IOP controlled on 1–2 drugs; no significant field or disc damage; older; good compliance	Cataract surgery through a temporal clear-corneal incision	Simpler, shorter, fewer complications; the conjunctiva is spared for later surgery	Early IOP spike; long-term control uncertain; later filtering surgery fails more often
Glaucoma drainage device (Ahmed, Molteno, Baerveldt): refractory glaucoma (neovascular, uveitic, congenital, post-keratoplasty); failed trabeculectomy; scarred conjunctiva	Superotemporal; plate at the equator, 8–10 mm behind the limbus; non-valved tube tied for 2–3 weeks; tube 1.5–2 mm inside, parallel to the iris; patch graft (card G3)	An option when filtering surgery has failed or is likely to fail	Smaller IOP fall; hypotony, tube block or migration, tube erosion and endophthalmitis, corneal decompensation, diplopia. Needs useful visual potential
Goniotomy and trabeculotomy: primary congenital glaucoma (card G11)	Goniotomy (clear cornea): under a direct goniolens, a Swan knife cuts the anterior meshwork just behind Schwalbe's line, over about 120°. Trabeculotomy (hazy cornea): Schlemm's canal is deroofed ab externo; a probe is rotated into the chamber	Aqueous reaches Schlemm's canal directly; trabeculotomy needs no view of the angle	Hyphaema, iridodialysis, shallow chamber. Failure: combined trabeculotomy–trabeculectomy
Non-penetrating surgery: deep sclerectomy, viscocanalostomy, canaloplasty; for POAG	Deep sclerectomy: 5 × 5 mm superficial flap; deeper 4 × 4 mm flap removed with the roof of Schlemm's canal; space maintainer. Viscocanalostomy: canal dilated with viscoelastic; tight flap	Chamber not entered; inner meshwork kept, so less overfiltration and hypotony	Not for neovascular glaucoma or angle closure: there the iris root lies against the meshwork, so filtration fails
Cyclocryotherapy: painful blind eye; repeated surgical failure; aphakic or post-keratoplasty glaucoma	2.5 mm probe, anterior edge 1 mm from the limbus; –60 to –80 °C for 60 seconds ; 3–4 freezes per quadrant, under 180° per session; repeat after at least 1 month	Also destroys corneal nerves, so it relieves pain	Pain, uveitis, hyphaema, transient IOP rise, hypotony and phthisis , anterior segment ischaemia, sympathetic ophthalmia
Microinvasive glaucoma surgery (MIGS) (extra): mild to moderate glaucoma	Schlemm's canal micro-stent; goniotomy or trabeculotomy; canal dilatation with viscoelastic; suprachoroidal micro-stent	Minimal tissue handling	Mild to moderate disease only

Antimetabolites with trabeculectomy

Feature	Mitomycin C (MMC)	5-fluorouracil (5-FU)
Action	Alkylating agent; acts in all phases of the cell cycle; stops fibroblasts and vessel ingrowth; much more potent	Pyrimidine analogue; acts in the S phase ; stops fibroblast proliferation only
Dose and method	0.2–0.5 mg/ml for 1–5 minutes on a 4.5 × 4.5 mm sponge under conjunctiva and Tenon's capsule; rinse with balanced salt solution; tighter flap	50 mg/ml for 5 minutes at surgery; or subconjunctival 5 mg injections 90°–180° away, twice daily in week 1, daily in week 2 (21 injections, 105 mg)
Toxicity	More hypotony, bleb leak, bleb infection; endothelial ; cystic overfiltering bleb (commonest); corneal and scleral necrosis	Epithelial ; punctal occlusion
Used in	Almost all eyes; essential after failed trabeculectomy, conjunctival or cataract surgery, secondary glaucoma, young or dark-skinned patients, drops for over 3 years	After MMC complications in a previous operation, or risk of hypotony maculopathy

Both go in the sub-Tenon's space, because subconjunctival fibrosis is the commonest cause of failure. Prevent toxicity by thorough irrigation, careful dissection and watertight closure.

Lens extraction in angle closure: phacoemulsification with an intraocular lens deepens the chamber and opens the angle. Details are on card G4. (*Kanski — not in your practical books*)

Choosing between them

Trabeculectomy versus drainage device

Point	Trabeculectomy	Drainage device
Usual place	First operation after failed medical or laser treatment	Failed filtering surgery, scarred conjunctiva, refractory secondary glaucoma
IOP fall	Larger, about 41%	Smaller; comparable in neovascular glaucoma
Bleb	At the limbus, with microcysts	Far back over the plate; organised capsule; no microcysts
Main risks	Leak, overfiltration, bleb infection	Tube erosion, tube–cornea touch, diplopia

Valved versus non-valved implants

Point	Valved (Ahmed)	Non-valved (Molteno, Baerveldt)
Flow	One-way valve opens only above 8–10 mm Hg (Venturi effect)	No valve; the capsule around the plate gives the resistance
Surgery	One stage; works at once	Two stages: tube tied for 2–3 weeks while the capsule forms
Outcome	Safety and efficacy appear equal; Ahmed better than Molteno in Sturge–Weber syndrome	Larger plates for younger, healthier eyes; a hypertensive phase is seen with the Baerveldt

Viva questions

Drugs

Q What is the aim of medical treatment, and what are its advantages and disadvantages?

Target pressure: I aim for an IOP that will arrest or prevent optic nerve damage and field progression. For: serious side effects are rare, and one or two drugs control most patients. Against: drugs lower IOP less than surgery, are costly and lose effect with time. Compliance is hard, and long use may harm later surgery.

Q How would you classify antiglaucoma drugs by mechanism?

Aqueous suppressants: beta-blockers, alpha-2 agonists, carbonic anhydrase inhibitors. **Outflow drugs:** prostaglandin analogues (uveoscleral) and miotics (trabecular). **Hyperosmotics** shrink the vitreous. Brimonidine both suppresses and raises outflow.

Q Why is a prostaglandin analogue the first choice, and why is it given at bedtime?

Prostaglandin analogues give the largest fall, 25–32%, once daily, with almost no systemic effects. They raise **uveoscleral outflow** by relaxing the ciliary muscle and remodelling its matrix. The effect peaks 10–14 hours after the drop, and bedtime use means fewer vasodilatation symptoms.

Q Why is timolol best given in the morning, and why may its effect fade?

Timolol suppresses aqueous more in the morning; production falls naturally at night. A morning dose blunts the early-morning rise and avoids hypotension in sleep. Its effect can fade by **short-term escape** and **long-term drift**, as beta receptors respond to constant blockade.

Q Which beta-blocker is cardioselective, and where is it useful?

Betaxolol, a beta-1 blocker with minimal respiratory side effects. It is preferred in normal-tension glaucoma for optic-nerve blood flow.

Q How do you reduce systemic absorption of eye drops?

Lacrimal occlusion: press the inner corner of the eye and keep the eyes closed for 3 minutes after each drop.

Q Why are alpha-2 agonists avoided in infants?

Brimonidine crosses the blood–brain barrier in children. Under 2 years it can cause somnolence, hypotension, bradycardia, apnoea, seizures and coma.

Q In an acute attack, why wait until IOP is below 40 mm Hg before pilocarpine?

Sphincter ischaemia: at very high pressure the iris sphincter is paralysed, so miotics fail. I first lower the pressure with mannitol and acetazolamide.

Q Which drugs are contraindicated in uveitic and neovascular glaucoma?

Miotics and prostaglandin analogues: they break down the blood–aqueous barrier and increase inflammation. I use an **aqueous suppressant** with steroids and atropine.

Q How do hyperosmotic agents act, and what do you tell the patient after mannitol?

Plasma osmolarity rises and draws water out of the vitreous. After intravenous mannitol he must not get up at once: hypotension, rarely brain coning.

Lasers

Q Why is Nd:YAG preferred to argon for iridotomy, and when is argon used?

Photodisruption: Nd:YAG (1064 nm) cuts any iris colour, with fewer late closures; argon is thermal and needs pigment. Argon suits thick brown irides, anticoagulated patients and bleeding disorders.

Q How do you know an iridotomy is patent?

Lens capsule seen through the opening confirms it. At the time: a gush of aqueous or pigment, and the chamber deepens. Retro-illumination alone is not proof.

Q How does selective laser trabeculoplasty work, and why is it called selective?

Selective: the 532 nm, 3-nanosecond pulse hits only melanin in pigmented meshwork cells, without heat spread. Cytokines then raise matrix metalloproteinases, making the meshwork more porous.

Q How does cyclophotocoagulation lower IOP, and why spare 3 and 9 o'clock?

Ciliary epithelium destruction cuts aqueous production; outflow through the pars plana may also rise. The long posterior ciliary vessels and nerves lie at 3 and 9 o'clock.

Q How do you treat malignant glaucoma, and where does the laser come in?

Aqueous misdirection: aqueous is forced backwards into the vitreous. I start strong topical mydriatics, then intravenous mannitol. If these fail, **Nd:YAG hyaloidotomy** opens the anterior hyaloid face through a patent iridectomy. If the laser fails, pars plana vitrectomy removes the trapped fluid.

Surgery

Q What is the principle of trabeculectomy, and why is the name a misnomer?

Partial-thickness filtration: aqueous passes through a limbal fistula, around the scleral flap, into the subconjunctival space. The block removed need not include meshwork; it is a peripheral posterior keratectomy.

Q Which eyes are at high risk of trabeculectomy failure?

Poor-prognosis factors: age under 40 years, a previous failed filter, aphakia or pseudophakia, and neovascular glaucoma. Active uveitis, congenital glaucoma, Stevens-Johnson syndrome and ocular pemphigoid also count. In these eyes I use mitomycin C, or consider a drainage device.

Q Why do long-term drops lower the success of trabeculectomy?

Conjunctival changes: more fibroblasts and inflammatory cells, and fewer goblet cells, so the bleb scars. Drops for over 3 years are an indication for mitomycin C.

Q Compare mitomycin C with 5-fluorouracil.

Mitomycin C is an alkylating agent that acts in all phases of the cell cycle. It is far more potent, and toxic to the endothelium. In contrast, **5-fluorouracil** acts in the S phase only. It is toxic to the epithelium.

Q When will you advise a glaucoma drainage device rather than trabeculectomy?

Refractory glaucoma: neovascular, uveitic, congenital, post-keratoplasty, aniridia; failed trabeculectomy with antimetabolite; heavily scarred conjunctiva. Useful visual potential is a must. In primary glaucoma trabeculectomy comes first: its IOP fall is larger.

Q How do you manage a painful blind eye, and how does cyclocryotherapy help?

Pain relief is the aim, because vision cannot return. I start antiglaucoma drops, topical steroid and atropine. Then cyclocryotherapy: ice crystals and ischaemic necrosis destroy the ciliary processes and the corneal nerves, so pain eases. If pain persists: retrobulbar alcohol injection, then enucleation or enucleation.

Read more: Baidya p.148–152, 156–158, 161, 164–166, 172–180, 374, 434–442, 578 · Namrata p.172–176, 179, 182–184, 186, 189, 194–195, 198–200, 205 · Aravind 1.4, 4.6, 4.7, 4.8, 4.9, 4.10, 4.12, 4.15–4.21, 6.4, glaucoma model case sheet p.665–667 · Kanski p.377

G1 Primary open-angle glaucoma, including the glaucomatous optic disc

★ Kept 6 times at JEH last year (one with pseudoexfoliation glaucoma) · Long case and short case

At a glance

- **Primary open-angle glaucoma (POAG):** a chronic, progressive optic neuropathy with an open, normal-looking angle and reduced outflow. Usually silent; found at a routine check.
- Kept often for its stable signs: cupped disc, measurable pressure, open angle. The **glaucomatous optic disc** is also a short case.
- Examiners test the reasons, the examination order, the disc description and the ladder.

Keywords the examiner listens for: Goldmann applanation tonometry · central corneal thickness · Van Herick · relative afferent pupillary defect · gonioscopy · vertical cup–disc ratio · ISNT rule · neuroretinal rim · bayoneting · disc haemorrhage · retinal nerve fibre layer · Humphrey · target pressure · prostaglandin analogue · trabeculectomy

History — positive and negative, with reasons

Chief complaint: “Mr X, a 70-year-old man, has come for regular follow-up of glaucoma.”

History of present illness (house wording): “Patient was apparently normal ___ years before. Then he had defective vision in both eyes, gradual in onset, progressive in nature, not associated with pain or headache. He visited an ophthalmologist at his place and was found to have increased pressure in both eyes. He was started on eye drops (1 drop at bedtime). He came to our institution for further follow-up.”

Positive history

Question	Why I ask — what the answer tells me
When did the blurring start? Was it gradual?	Gradual, painless loss fits POAG. A sudden loss points elsewhere.
Is it getting worse? How fast?	POAG is slow. Rapid field loss is an indication for neuroimaging.
One eye or both?	POAG may be one-sided or both-sided. One eye only: I look for pseudoexfoliation (PXF) or angle recession.
Worse at one time of day, with heaviness?	Early-morning or afternoon blur matches the intraocular pressure (IOP) peak.
What was the pressure when glaucoma was found?	The untreated pressure is the baseline for the target.
Any old field charts or scans (optical coherence tomography, OCT)?	The baseline for judging progression.
How many drops now? Is the pressure controlled?	Not at target on 3 drugs is an indication for trabeculectomy.
Do you use them daily? Show me how.	Poor compliance is an indication for laser or surgery.

Negative history

Question	A “yes” would point to	How to tell them apart
Any headache with vomiting and eye pain?	An acute angle-closure attack .	Shallow peripheral anterior chamber (AC), closed angle, mid-dilated vertically oval pupil. Glaukomflecken mark a past attack.
Any coloured haloes?	Corneal epithelial oedema from a pressure rise: angle closure. Other causes: mucus on the cornea in conjunctivitis, incipient cataract, vitreous opacities, snow blindness, a tilted intraocular lens.	Fincham (stenopaeic slit) test: a slit is moved across the pupil. The glaucoma halo stays intact; the halo of incipient cataract breaks into its component colours.
Any redness?	Angle closure; a glaucomatocyclitic crisis (Posner–Schlossman syndrome); drop toxicity or allergy.	Corneal oedema, shallow AC (closure); keratic precipitates, open angle (crisis); inferior congestion (drops).
Any watering?	Drop side effects: pilocarpine lacrimation, brimonidine allergy, surface toxicity.	Fluorescein staining shows toxicity; lid allergy points to the drug.
Any eye pain?	Severe: angle closure or neovascular glaucoma. Mild: a crisis. POAG may ache.	Gonioscopy: closed angle, new vessels, or open angle with keratic precipitates.
Any injury to the eye?	Angle-recession glaucoma , usually in one eye.	Angle recession on gonioscopy; iridodialysis, phacodonesis, traumatic cataract.
Frequent change of glasses?	Raised pressure weakens accommodation; field loss is mistaken for poor vision. Also cataract, keratoconus.	Nuclear sclerosis or a cone at the slit lamp; a cupped disc in glaucoma.

Question	A “yes” would point to	How to tell them apart
Any night blindness?	Poor dark adaptation in open-angle glaucoma; miotics make it worse.	Pilocarpine in the drug history; a small pupil.
Any sudden loss of vision?	An acute pressure rise; central retinal vein occlusion (CRVO); snuff-out (sudden loss after trabeculectomy in advanced glaucoma).	Corneal oedema with high IOP; haemorrhages in all quadrants (CRVO); recent surgery.

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Diabetes?	Raises POAG risk; also matters when planning surgery.
Hypertension? Which tablets?	Raises POAG risk. On a systemic beta-blocker, a topical one works poorly, so it is not my first choice.
Thyroid disease?	Raises POAG risk; thyroid eye disease raises episcleral venous pressure.
Asthma or chronic obstructive pulmonary disease (COPD)?	No beta-blocker (bronchospasm).
Heart failure, heart block or a slow pulse?	No beta-blocker . Brimonidine: caution in coronary insufficiency.
Kidney or liver disease?	No carbonic anhydrase inhibitor (CAI) in renal failure, transplant or chronic liver disease.
Sulpha allergy?	No CAI .
Any other drug allergy?	Avoid that drug.
Head injury?	A past optic insult can mimic normal-tension glaucoma (NTG).
Stroke (cerebrovascular accident, CVA)?	A haemodynamic crisis worsens nerve damage. Brimonidine: caution.
Migraine?	Vasospasm: a risk factor, above all for NTG.
Raynaud's phenomenon?	Vasospasm: a risk factor, above all for NTG.
Peripheral vascular disease (PVD)?	Cardiovascular disease raises POAG risk.
Major blood loss or shock?	Severe hypotension damages the optic nerve.
Steroids: drops, nasal spray, tablets?	Steroid-induced glaucoma ; ask about transplant, keratoplasty, Cushing's.
Cataract surgery: when, which eye?	Pseudophakia. Past conjunctival or cataract surgery makes mitomycin C (MMC) essential at trabeculectomy.
Other eye laser or surgery?	Records an iridotomy, bleb, tube or graft.
Refractive surgery?	Thinner cornea: applanation reads falsely low.
Short sight (myopia)?	Raises POAG risk: stretching makes the myopic disc more oval. Decreased scleral rigidity can give a faulty pressure reading.
Treatment: drug, strength, frequency, duration	“E/D (eye drops) bimatoprost 1 drop HS (at bedtime), both eyes, 2 years”. Over 3 years of drops: MMC essential, lower surgical success.
Smoking? Alcohol?	Tobacco raises IOP; alcohol lowers it.
Glaucoma in the family?	Risk 4-fold in siblings, 2-fold in offspring ; screen relatives.

Examination — must-do steps and their viva

General and systemic (this case only)

- **Pulse, blood pressure, chest:** bradycardia, heart block or wheeze bars beta-blockers. Diastolic perfusion pressure below 55 mm Hg is a risk factor.
- **Face and skin:** haemangioma (Sturge-Weber), neurofibromatosis, carotid-cavernous fistula: raised episcleral venous pressure.
- **Nervous system:** a neurological lesion can mimic glaucoma.

Ocular examination — RE | LE template

Right eye (RE) and left eye (LE), in the department's recording order. The steps below follow the order I examine.

Structure	Look for in this case	How to record (example)
Visual acuity	Central vision goes last	6/18, PH (pinhole) 6/12; N6
Extraocular movements (EOM)	Seniors' addition	Full

Structure	Look for in this case	How to record (example)
Tension	Goldmann applanation tonometer (GAT); asymmetry ≥ 4 mm Hg is significant	Tn (tension) by GAT 18 mm Hg at 10 am
Lids	Prostaglandin (PG) analogue: pigmentation, long lashes	Normal
Conjunctiva	Drug congestion, worse inferiorly; bleb	Clear
Cornea	Krukenberg spindle; PXF; keratic precipitates	Clear
Anterior chamber	Van Herick (VH) grade; cells, flare	VH grade IV (ND, normal depth)
Iris	PXF; transillumination defects; peripheral iridotomy (PI)	CPN (colour, pattern normal)
Pupil	Relative afferent pupillary defect (RAPD)	RTL (round, reacting to light) 3 mm
Lens	PXF; glaukomflecken; cataract	PCIOL (posterior chamber intraocular lens)
Gonioscopy	No peripheral anterior synechiae (PAS), recession or new vessels	Scleral spur (SS) in all quadrants
Fundus	Disc in the slide order; exclude pallor, tilted disc, drusen, pit	Media clear; disc vertically oval, well-defined margins, cup-disc ratio (CDR) 0.8; neuroretinal rim (NRR) thinning + (superior, inferior); nasalisation +; bayoneting +; peripapillary atrophy (PPA) +; tessellated fundus
Fields	Confrontation (seniors' addition)	Full in all quadrants

Step 1 — Visual acuity, pinhole and near vision

Do: Each eye unaided, with pinhole and with glasses; near vision at 33 cm; colour vision.

Record: "RE 6/18, PH 6/12; LE 6/9; N6 at 33 cm both eyes."

Q Why is central vision kept until late in glaucoma?

Central vision fibres arise closest to the disc, lie most superficially and leave by its centre. Arcuate fibres at the poles suffer first.

Q Why test colour vision?

Asymmetric dyschromatopsia suggests an intracranial lesion, not glaucoma: an indication for neuroimaging.

Step 2 — Pupils, for a relative afferent pupillary defect (before any drop)

Do: The direct and consensual reflexes of each eye; then the swinging torch light test.

Record: "RTL 3 mm, no RAPD", or "RE: RAPD present".

Q Why check the pupils before any drop?

A **relative afferent pupillary defect** cannot be judged in a pupil made small or large by drugs. Hence the mnemonic PART B.

Q What does an RAPD mean in glaucoma?

Advanced, asymmetric optic nerve damage. With only mild cupping it suggests another cause, so I order neuroimaging.

Step 3 — Slit-lamp examination of the anterior segment, with Van Herick grading

Do: Lids to lens. Van Herick: a thin slit 1 mm inside the temporal limbus at 60°; peripheral AC depth against corneal thickness.

Record: "AC: VH grade IV (ND). Iris CPN. Lens PCIOL. No PXF."

Q How do you grade the anterior chamber by Van Herick?

Van Herick compares peripheral AC depth with corneal thickness. Grade 4 ≥ 1 ; grade 3 $\frac{1}{4}$ – $\frac{1}{2}$; grade 2 = $\frac{1}{4}$; grade 1 $< \frac{1}{4}$; grade 0 contact. Below $\frac{1}{4}$, the angle may be occludable.

Q Which secondary causes do you look for at the slit lamp?

Pseudoexfoliation flakes on the pupil and lens. A **Krukenberg spindle** (pigment dispersion), keratic precipitates (uveitis), iris new vessels, signs of trauma.

Step 4 — Applanation tonometry

Do: I explain the test and instil topical anaesthetic and fluorescein. Under cobalt blue light, I turn the dial until the inner edges of the two green semicircles touch. Dial $\times 10$ = mm Hg. I note the time.

Record: "Tn by GAT: RE 18 mm Hg, LE 16 mm Hg at 10 am". Central corneal thickness (CCT) is an investigation.

Q What is the normal IOP, and why is 21 mm Hg the upper limit?

Normal IOP is 10–21 mm Hg. The mean is 15.5 ± 2.57 mm Hg. About 95% lie within 2 standard deviations of the mean; mean plus 2 standard deviations is about 20.5 mm Hg. An asymmetry of 4 mm Hg or more is significant.

Q Why measure the pressure before gonioscopy and dilatation?

Gonioscopy presses on the cornea and lowers the reading. Dilatation raises the pressure by 4–5 mm Hg.

Q What is the principle of applanation tonometry?

Imbert-Fick law: pressure = flattening force $\div$ flattened area ($P = F/A$). At 3.06 mm, corneal rigidity and tear capillary attraction cancel out.

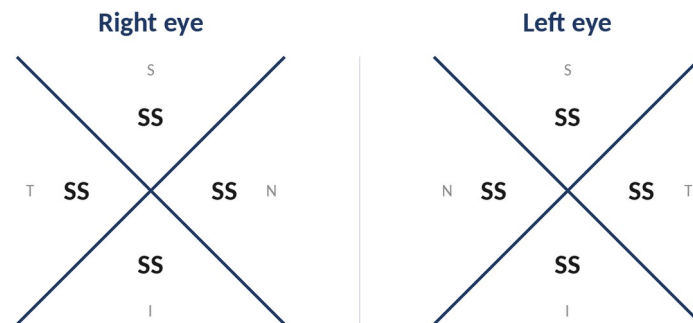
Q How does central corneal thickness affect the reading?

Goldmann applanation tonometry is calibrated for a mean central corneal thickness of 520 μm . A thinner cornea reads falsely low, a thicker one falsely high, by about 0.7 mm Hg per 10 μm .

Step 5 — Gonioscopy

Do: After tonometry, before dilatation: a Goldmann lens with a fluid bridge on the anaesthetised cornea, through all four quadrants. Zeiss lens indentation if narrow.

Record: a cross for each eye, with the deepest structure seen in each quadrant.



Q Why is gonioscopy essential in POAG?

Gonioscopy proves the angle is open. It excludes angle closure and shows recession, pseudoexfoliation, pigment, PAS or new vessels.

Q What is the principle of gonioscopy?

Light from the angle undergoes **total internal reflection** at the tear-air interface. The gonioscopy lens replaces that interface.

Q Name the angle structures, and the Shaffer grades of an open angle.

From behind forward: **ciliary body band, scleral spur, trabecular meshwork, Schwalbe's line**. Shaffer grades 3 (25–35°) and 4 (35–40°) are open; closure is impossible.

Step 6 — Dilated disc examination: direct ophthalmoscope, +90 dioptre lens, red-free light

Do: Direct ophthalmoscope: red glow at arm's length, then media, disc and macula. Slit lamp: a +90 dioptre (D) lens, then red-free (green) light for nerve fibre layer defects. Indirect ophthalmoscope for the periphery.

Record: each eye separately, in this order. Red glow, media. Disc: size, shape, cup shape, margins, colour, CDR, NRR. Vessels, arteriovenous (AV) ratio, PPA, nerve fibre layer defects. Background, macula and foveal reflex, periphery (indirect). "Confirmed with +90 D."

Q What are the optic nerve head changes in POAG?

Papillary: enlarged, vertically elongated or asymmetric cup; focal rim narrowing (notch); saucerisation (shallow bowing, not deep excavation). **Peripapillary:** disc haemorrhage, nerve fibre layer defects, peripapillary atrophy. **Vascular:** nasalisation, baring of circumlinear vessels, bayoneting.

Q What is the best way to examine and to document the disc?

Slit-lamp biomicroscopy with a Hruby, 60, 78 or 90 D lens. Disc photography is the best documentation.

Q What do you assess on the disc, qualitatively and quantitatively?

Qualitative: rim contour, disc haemorrhage, parapapillary atrophy, bared vessels, nerve fibre layer. **Quantitative:** disc size, cup-disc ratio, rim-disc ratio.

Step 7 — Confrontation fields

Do: Face to face, 60 cm apart; his right eye fixes my left, the other eyes closed. A finger comes in from the periphery, midway, in each quadrant.

Record: "Fields (confrontation test): full in all quadrants."

Q Where do glaucomatous field defects begin?

In **Bjerrum's area**, 10–25° from fixation, arching from the blind spot. Confrontation is rough, so I confirm with automated perimetry.

Q How does a glaucomatous field defect differ from a neurological one?

Glaucomatous defects respect the horizontal meridian: they follow the arcuate nerve fibres, which end at the horizontal raphe. They are relative and incongruous. **Neurological** defects respect the vertical meridian, and are absolute and congruous.

Q What is the extent of the normal monocular field?

The normal monocular field reaches about 60° superiorly, 75° inferiorly, 60° nasally and 100° temporally. My own field must be normal for confrontation.

Summary and complete diagnosis

Mr ___, ___ years, on ___ drops for ___ years: gradual, painless, progressive defective vision; no haloes, trauma or steroids. Visual acuity ___ / ___; Tn by GAT ___ / ___ mm Hg at ___; VH grade ___; RAPD ___; lens ___; deepest angle structure ___; vertical CDR ___ / ___ with ___; fields ___.

eye → disease → type / stage / grade → aetiology → complications or status → other eye

- “Both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management.”
- “Right eye primary open-angle glaucoma, moderate stage, not at target on two drugs; left eye primary open-angle glaucoma, mild stage.”

Differential diagnosis

Condition	Points for	Points against	How to tell apart
Ocular hypertension (OHT)	IOP > 21 mm Hg twice; open angle	Normal disc, rim and field	Normal retinal nerve fibre layer (RNFL) on OCT; normal Humphrey field
Normal-tension glaucoma	Glaucomatous disc and field; open angle	Pressure never above normal	Diurnal phasing: IOP always < 22 mm Hg. More disc haemorrhages, notches, beta-zone PPA
Pseudoexfoliation glaucoma	Open angle; raised IOP	Higher IOP, wider swings; often one eye	White flaky material on the pupil margin and anterior capsule after dilatation
Pigmentary glaucoma	Wide-open angle; raised IOP	Younger, 20–30 years	Krukenberg spindle; mid-peripheral iris transillumination defects; dark, heavy trabecular pigment
Steroid-induced glaucoma	Open angle; raised IOP	Steroid use or Cushing's	Posterior subcapsular cataract; IOP settles weeks to months after stopping
Angle-recession glaucoma	Open angle; raised IOP	Past blunt trauma; one eye	Recession on gonioscopy; phacodonesis, traumatic cataract

Exclude first: primary angle-closure glaucoma — Van Herick grade and gonioscopy (open angle in all quadrants).

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Pachymetry: CCT of each eye	True pressure	Thin < 500 µm, thick > 570 µm	A thin cornea hides a higher true IOP and carries more risk
Diurnal phasing: IOP every 3 hours for 24 hours	Baseline, peak, swing	Normal swing 3–6 mm Hg; ≥ 8 mm Hg significant	Times drops to the peak; explains progression; separates NTG
Humphrey 24-2 (10-2 if advanced)	Stage	Paracentral, arcuate or nasal-step defects; hemifield test abnormal	Mean deviation (MD) gives the stage, Z value and target
OCT of the RNFL and ganglion cell complex (GCC)	Structure; pre-perimetric diagnosis	Superior and inferior RNFL thinning; eyes differ by > 9 µm	Confirms early disease when fields are normal; tracks progression
Gonioscopy (step 5), yearly	Open angle	No PAS, recession or new vessels	Excludes angle closure and secondary glaucoma
Disc photography	Baseline	Cupped disc	Shows change over time
Neuroimaging , only if atypical	Compressive lesion	Normal	For pallor more than cupping or a vertical-meridian field
Blood pressure, blood sugar	Fitness for surgery	Controlled	Low diastolic perfusion adds risk

Reading the Humphrey field

- **Reliability first:** fixation losses < 20%. False positives > 15% (white scotomas) or false negatives > 33% (clover leaf) make it unreliable.
- **Pattern deviation plot:** shows focal defects hidden by general depression from cataract or a small pupil. Anderson-Patella criteria: card G10.
- **Indices:** MD is normally within 2 decibels (dB) of 0 and grows more negative as the field worsens. Pattern standard deviation (PSD) shows focal loss.

Reading the OCT

- **What it measures:** RNFL thickness on a circle around the disc. At the macula, the GCC: RNFL, ganglion cell and inner plexiform layers.

- **Normal:** an hourglass map; a double hump on the TSNIT (temporal–superior–nasal–inferior–temporal) graph. Red means below the 1st percentile.
- **Glaucoma:** inferior and superior thinning first (7 o'clock right eye, 5 o'clock left); the double hump flattens.
- **Advanced disease:** the RNFL reaches a floor; I follow the GCC and 10-2 fields.

Say: "I will proceed with pachymetry and diurnal phasing to confirm the true pressure. Then I will stage the damage with Humphrey fields and OCT."

How I will manage

Aim

To reach and keep a **target pressure**, at or below which further damage is unlikely.

The ladder

1. **I set the target.** $TP = IP (1 - IP/100) - Z \pm 2$ (TP target pressure, IP initial pressure, Z from the stage). Or I lower IOP by 20% (OHT), 30% (mild damage, NTG) or 40% (advanced).
2. **I start one drug:** a PG analogue at bedtime. If unsuitable, a beta-blocker or an alpha-2 agonist.
3. **I review at 2 months.** Not at target: I switch to a better drug before adding one.
4. **I add a second drug,** as a fixed combination where possible.
5. **I offer laser trabeculoplasty** when drops cannot be used reliably: cost, memory, instillation, intolerance.
6. **I advise trabeculectomy with MMC** when any of these applies:
 - not at target on 3 drugs, or fields progressing on maximal tolerated therapy;
 - side effects or poor compliance;
 - baseline IOP over 40 mm Hg, one eye already lost, or difficult follow-up.
7. **I choose a glaucoma drainage device** after failed filtering surgery with antimetabolites, or for a badly scarred conjunctiva.

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Latanoprost 0.005% at bedtime (travoprost 0.004%; bimatoprost 0.01%, 0.03%). IOP fall 25–32%	PG analogue: more uveoscleral outflow	Hyperaemia, long lashes, iris and lid pigmentation, uveitis, cystoid macular oedema. No systemic effects	Uveitis
Timolol 0.25%, 0.5% twice daily, first dose in the morning (betaxolol 0.5%, cardioselective). Fall 20–30%	Beta-blocker: less cyclic adenosine monophosphate in the ciliary epithelium, so less aqueous	Dry eye, allergy. Bradycardia, hypotension, heart failure, bronchospasm, depression	Asthma, COPD, heart failure, second- or third-degree heart block, bradycardia
Brimonidine 0.1%, 0.15%, 0.2% three times, sometimes twice daily. Fall 15–20%	Alpha-2 agonist: less aqueous, more uveoscleral outflow	Allergic blepharoconjunctivitis, dry mouth, somnolence	Children under 2 years
Dorzolamide 2% three times daily (brinzolamide 1% twice; acetazolamide 250 mg tablet 2–4 times daily). Fall 15–20%	CAI: blocks carbonic anhydrase II in the ciliary epithelium	Drops: burning, stinging. Tablets: paraesthesia, acidosis, renal stones, aplastic anaemia	Sulpha allergy, renal failure, chronic liver disease
Pilocarpine 0.5–4% 2–4 times daily. Fall 15–20%	Miotic: ciliary muscle pulls the scleral spur, opening the meshwork	Browache, dim vision, induced myopia, retinal detachment	Uveitic and neovascular glaucoma; not with a PG analogue

Fixed combinations (timolol with dorzolamide; PG analogue with timolol) give simpler dosing and less preservative. Full profiles: card GX.

Laser and surgery in this case

Procedure (indication in this case)	Key steps	Advantages	Disadvantages and complications
Selective laser trabeculoplasty (SLT) , or argon laser: drops not reliable	Q-switched frequency-doubled Nd:YAG (neodymium-doped yttrium aluminium garnet) laser, 532 nm; 400 µm, 3 ns, 0.5–1.5 mJ; 50–100 spots over 180°	Saves drops; SLT hits only pigmented cells, with no heat spread	IOP spike, mild uveitis, hyphaema; effect wanes (argon: half controlled at 5 years). Not in uveitic, neovascular or angle-recession glaucoma
Trabeculectomy with MMC: not at target, progression, poor compliance	Conjunctival flap; scleral flap 3 × 4 mm, 2/3 depth; MMC 0.2–0.5 mg/ml for 1–5 minutes; ostium 1.5 × 2 mm under the flap; peripheral iridectomy; 10-0 nylon	Lowers IOP more than drugs; few or no drops	Early: shallow AC, hypotony, hyphaema, snuff-out. Late: bleb leak, bleb infection, endophthalmitis, cataract, failure

Procedure (indication in this case)	Key steps	Advantages	Disadvantages and complications
Combined cataract surgery and trabeculectomy: cataract with IOP on several drugs, field loss, PXF or one eye	Cataract extraction and filtering surgery at one sitting	One operation; early and long-term IOP control; fewer drops	More complications; slower recovery; less IOP control than trabeculectomy alone
Glaucoma drainage device (Ahmed valve or non-valved): failed trabeculectomy; scarred conjunctiva	Plate at the equator superotemporally; tube 1.5–2 mm into the AC; scleral patch graft	Works where a trabeculectomy would scar and fail	Tube erosion, corneal decompensation, diplopia, hypotony, endophthalmitis

Cataract with glaucoma. Expect a miotic pupil, synechiae, weak zonules in PXF, bleeding and a pressure rise after surgery. I do **cataract surgery alone** when IOP is controlled on 1–2 drugs without significant damage. A temporal clear-corneal incision spares the conjunctiva. Risks: an early pressure spike, and a later trabeculectomy more prone to fail.

Follow-up and counselling

- IOP and compliance 3-monthly (mild to moderate) or monthly (advanced). Fields 6-monthly or 3-monthly.
- Gonioscopy yearly: a thickening lens can add angle closure, needing an iridotomy.
- The damage is permanent, but treatment stops further loss. Drops are for life.
- PG analogue at bedtime, timolol in the morning. After each drop, close the eyes and press the inner corner for 3 minutes.

What you must know

Definition

Glaucoma is “a chronic progressive optic neuropathy caused by a group of disorders leading to irreversible damage to optic nerves and corresponding visual field defects with or without high intraocular pressure.”

POAG: “In open-angle glaucoma the iridocorneal angle is open (unobstructed) and normal in appearance but aqueous outflow is diminished.” Primary means no other eye disease; adult onset is after 40 years.

In plain words: the nerve is slowly lost behind an open drain that works poorly. Diagnosis rests on the **disc**, **field** and **open angle**.

- **Ocular hypertension:** IOP above normal (often > 21 mm Hg), open angles, no glaucomatous damage. About 1 in 10 develop glaucoma over 10 years.
- **Normal-tension glaucoma:** optic nerve damage without IOP exceeding the normal range (below 22 mm Hg on a diurnal curve).
- **Glaucoma suspect:** a normal white-on-white field, but other risk factors such as disc changes.
- **Target pressure:** the IOP at which further glaucomatous damage is unlikely.
- **Juvenile open-angle glaucoma:** POAG starting at 3–40 years; linked to the myocilin gene.

Classification

By the angle: open-angle or angle-closure, each primary or secondary (secondary open-angle: pseudoexfoliation, pigment dispersion).

Grading of glaucomatous optic nerve head damage, with the Z value for the target formula. Stage the patient from his Humphrey MD.

Stage	Disc	Field (MD in dB)	Z
Suspect	—	—	0
Mild (early)	Minimal cupping	Nasal step or paracentral scotoma; MD –6 dB or better	1
Moderate	NRR thinning	Arcuate scotoma; MD –6 to –12 dB	3
Severe	Marked cupping	Within the central 5°; MD –12 dB or worse	5
End-stage	Gross cupping	Small residual field	7

Pathogenesis

- **Risk factors:** raised IOP; age (prevalence doubles each decade over 40); Black race; first-degree relatives; myopia; thin CCT; diastolic perfusion pressure < 55 mm Hg; diabetes; vasospasm.
- **Outflow:** the meshwork has uveal, corneoscleral and juxtacanalicular parts. The juxtacanalicular part gives most resistance.
- **Mechanical:** raised IOP bows the lamina cribrosa back. Axoplasmic transport is blocked; ganglion cells die by apoptosis.
- **Vascular** (key in NTG): vasospasm, nocturnal hypotension, blood loss.
- **Neurodegenerative:** released glutamate raises intracellular calcium, causing more apoptosis.
- **Poles first:** larger laminar pores. Cupping often starts inferotemporally, giving an upper arcuate scotoma.

Complications

- **Disease:** field loss to a central and a temporal island, then blindness; sudden loss (acute pressure rise, CRVO, snuff-out).
- **Treatment:** surface toxicity, drug side effects, lower success of later surgery, surgical complications.

Prognosis

- **Good:** IOP kept at target with good compliance; central vision lasts until late. In the young, cupping can reverse once IOP falls.
- **Poor:** advanced damage at presentation (needs a 40% fall); disc haemorrhages; wide pressure swings; poor compliance; pseudoexfoliation.

Recent advances

- (extra) Non-penetrating surgery (deep sclerectomy); Rho kinase inhibitors (under study); neuroprotection (memantine, brimonidine, calcium channel blockers).

Viva questions

Why did you ask...?

Q Why did you ask about night blindness?

Defective dark adaptation is a feature of open-angle glaucoma. Miotics such as pilocarpine make it worse.

Q Why did you ask about frequent change of glasses?

Constant **high pressure on the zonules and ciliary body** weakens accommodation, so near glasses change often. Field loss is also mistaken for poor vision.

Q Why did you ask about sudden loss of vision?

Sudden loss points to an acute pressure rise with corneal oedema, or a **central retinal vein occlusion**. In advanced glaucoma, **snuff-out** after trabeculectomy is another cause.

Q Why did you ask about steroids?

Steroid-induced glaucoma mimics POAG. Any form counts: drops, nasal spray, tablets, after transplant or keratoplasty; also Cushing's syndrome.

Q Why did you ask about asthma, heart and kidney disease and sulpha allergy?

They decide the drug. **Beta-blockers** are barred in asthma, COPD, heart failure, heart block and bradycardia. **CAIs** are barred in sulpha allergy, renal failure and chronic liver disease.

Q Why did you ask about migraine, Raynaud's, blood loss and head injury?

Vasospasm and haemodynamic crises damage the optic nerve, above all in NTG. Past head injury or shock can mimic NTG.

Diagnosis and differentials

Q Which findings make you investigate for open-angle glaucoma?

Raised IOP. A vertical CDR ≥ 0.7 or asymmetry > 0.2 . Rim thinning that breaks the **ISNT rule** (rim thickest inferiorly, then superiorly, nasally, temporally). An RNFL defect in red-free light. A field defect.

Q How does NTG differ from POAG?

NTG is a diagnosis of exclusion: IOP stays below 22 mm Hg on diurnal testing, with an open angle. Disc haemorrhages, notches and beta-zone atrophy are commoner.

Q When will you suspect a lesion other than glaucoma?

Pallor more than cupping, asymmetric dyschromatopsia, a vertical-meridian field, or an RAPD with mild cupping. Mimics: chiasmal compression; ischaemic, toxic or shock optic neuropathy. I order **neuroimaging**.

Q Where is aqueous made, how does it leave the eye, and what sets the IOP?

The **ciliary epithelium** of the pars plicata makes it. About 90% leaves by the **trabecular meshwork** into Schlemm's canal, 10% by the **uveoscleral** route. IOP depends on aqueous production, outflow resistance and episcleral venous pressure (normally 8–10 mm Hg).

Investigations

Q What is diurnal variation, and why do phasing?

IOP peaks in the morning, falls by day and rises at night, tracking aqueous production. The normal swing is 3–6 mm Hg; 8 mm Hg or more is significant. Phasing finds the peak and explains progression despite good readings.

Q How many ganglion cells are lost before a field defect appears?

25–30% are lost before a defect shows on white-on-white perimetry. So the disc and RNFL often change first.

Q Which tools detect pre-perimetric glaucoma?

OCT, scanning laser polarimetry (GDx), Heidelberg retinal tomography, **short-wavelength automated perimetry** and frequency-doubling perimetry.

Q In what order do glaucomatous field defects appear?

Generalised depression and baring of the blind spot come first but are non-specific. Then a **paracentral scotoma**, Seidel's scotoma, **arcuate (Bjerrum's) scotoma**, double arcuate (ring) scotoma and nasal (Roenne) step. Last, a central and a temporal island remain.

Q How will you know that his glaucoma is progressing?

Progression is judged against a baseline. Glaucoma progression analysis compares each follow-up field with the baseline. A reproducible deepening, enlarging or new scotoma means progression. On OCT, event analysis flags change beyond test-retest variability; trend analysis gives the rate of thinning. A new disc haemorrhage or more nasalisation also warns of it.

Management — drugs

Q How do you calculate the target pressure?

$TP = IP (1 - IP/100) - Z \pm 2$; Z is 0 suspect, 1 early, 3 moderate, 5 severe, 7 end-stage. For 30 mm Hg, moderate: $30 \times 0.7 - 3 = 18$, so 16–20 mm Hg.

Q Why is a prostaglandin analogue first line, and why at bedtime?

A **prostaglandin analogue** lowers IOP most, without the systemic effects of beta-blockers. Its effect peaks 10–14 hours later, and bedtime use lessens hyperaemia symptoms.

Q Why is timolol best given in the morning?

Timolol cuts aqueous production more in the morning; production falls at night anyway. It also blunts the early rise and avoids hypotension in sleep.

Q What are the systemic side effects of timolol, and how do you reduce them?

Bradycardia, hypotension, heart failure, bronchospasm and depression. I advise lacrimal occlusion and closing the eyes for 3 minutes after each drop.

Q Why is brimonidine avoided in young children?

Brimonidine crosses the blood–brain barrier. In infants it causes somnolence, hypotension, seizures and apnoea, so it is barred under 2 years.

Q What are the advantages and disadvantages of fixed combinations?

Fixed combinations give simpler dosing, better adherence, less preservative, lower cost. But dosing times clash (timolol morning, prostaglandin bedtime), and each drug's benefit is unproven.

Q How would you treat NTG?

I lower IOP by 30%, including betaxolol for optic nerve blood flow. I treat systemic risks and repeat fields 4–6-monthly.

Q How would you manage ocular hypertension?

Ocular hypertension can often be watched carefully, with baseline fields and nerve fibre layer or disc imaging. Treatment is with drops or laser. The risk rises with higher IOP, older age, a larger cup–disc ratio and a CCT below 555 μm .

Q How will you manage a glaucoma suspect?

A **glaucoma suspect** needs a baseline: applanation pressure, CCT, disc photograph, nerve fibre analysis and fields. I observe him if there are no risk factors. I review after 2 months and start drops if progression is documented.

Management — laser and surgery

Q How does laser trabeculoplasty lower IOP?

Burns shrink the **trabecular meshwork** and stretch the areas between. Released cytokines (interleukin-1 β , tumour necrosis factor- α) induce matrix metalloproteinases that open the meshwork.

Q Why use mitomycin C in trabeculectomy, and when is it essential?

Mitomycin C prevents bleb scarring and failure. It is essential after failed trabeculectomy or past conjunctival or cataract surgery. Also in secondary glaucoma, the young, and after 3 years of drops.

Q Why is a peripheral iridectomy done with trabeculectomy?

The **peripheral iridectomy** stops the iris blocking the ostium and prevents pupillary block. It must be larger than the internal ostium.

Q When will you choose a glaucoma drainage device?

A **glaucoma drainage device** suits failed filtering surgery with antimetabolites, or a badly scarred conjunctiva. It also suits neovascular, uveitic and post-keratoplasty glaucoma.

Complications

Q What are the complications of mitomycin C?

A **cystic, overfiltering bleb** (commonest), late bleb leak, corneal decompensation, scleral or corneal melt, hypotony, endophthalmitis and cataract.

Recent advances

Q What is microinvasive glaucoma surgery? (extra)

Microinvasive glaucoma surgery (MIGS) lowers IOP with minimal tissue handling, in mild to moderate glaucoma. Types: Schlemm's canal stents, goniotomy, canal dilatation, suprachoroidal stents.

Examiner traps

- IOP without method, time and CCT, or measured after gonioscopy or dilatation.
- Pupils checked after dilating: an RAPD can no longer be judged.
- "Open angle" without gonioscopy; mixing up Shaffer 4 (wide open) and Scheie 4 (only Schwalbe's line seen).
- "CDR 0.8" alone: give the vertical ratio, rim (ISNT), notch, haemorrhage, vessels, PPA, RNFL.
- Missing PXF at the pupil or lens: it changes the diagnosis and the cataract plan.
- CCT in the clinical table: it is an investigation.
- Pallor more than cupping called glaucoma: it needs neuroimaging.
- Describing the disc alone: start with red glow and media; end with macula, periphery and "+90 D confirmed".

Short-case version

Glaucomatous optic disc

The instruction you may get: "Examine the fundus of this patient", or "Describe this optic disc."

Spot diagnosis: "Glaucomatous optic disc: a vertically enlarged cup with neuroretinal rim thinning, suggestive of glaucomatous optic neuropathy."

Focused examination:

1. Pupils for an RAPD, before dilating.
2. Direct ophthalmoscope: red glow at arm's length, then media, disc, vessels and macula close up.
3. Slit lamp, +90 D lens: cup, rim (ISNT), notch, haemorrhage, laminar dots; red-free light for RNFL defects. Indirect ophthalmoscope for the periphery.
4. Describe in the department's order, as below.

SAY IT

On distant direct ophthalmoscopy at arm's length, the right eye shows a good red glow. Close direct ophthalmoscopy reveals clear media. The optic disc is normal in size and vertically oval. The cup is vertically oval. The margins are well defined. The disc is pale. The cup-disc ratio is 0.9, with circumferential thinning of the neuroretinal rim. The vessels arise centrally and branch dichotomously, with bayoneting and nasalisation. The arteriovenous ratio is 2 to 3. There is peripapillary atrophy. Red-free light shows superior and inferior wedge-shaped nerve fibre layer defects. The background retina and the macula are normal, with a foveal reflex. Indirect ophthalmoscopy shows the peripheries to be normal. Slit-lamp biomicroscopy with a plus 90 dioptre lens confirmed these findings. My diagnosis is right eye glaucomatous optic neuropathy, with a cup-disc ratio of 0.9 and circumferential rim thinning.

Three signs to show:

1. Vertical cup-disc ratio ≥ 0.7 , or asymmetry > 0.2 between the eyes.
2. Rim thinning or a **notch** that breaks the ISNT rule.
3. **Bayoneting** and **nasalisation** of the vessels.

Q What is the ISNT rule?

The normal **neuroretinal rim** is thickest Inferiorly, then Superiorly, Nasally and Temporally. Glaucoma thins the inferior and superior rim first.

Q Why is the vertical cup-disc ratio more important?

Early rim loss is at the **upper and lower poles**, so the cup elongates vertically. A normal cup is horizontally oval.

Q What is the significance of a disc haemorrhage?

A **disc haemorrhage** (Drance haemorrhage) is flame-shaped, usually inferotemporal. It signals active disease and precedes field loss by months; commoner in NTG.

Q What are bayoneting, nasalisation and baring of circumlinear vessels?

Bayoneting: double angulation of a vessel. **Nasalisation:** vessels pushed nasally as the cup enlarges. **Baring:** a circumlinear vessel exposed as the rim recedes.

Q What are the laminar dot sign and beta-zone atrophy?

Laminar dots are lamina cribrosa fenestrations, seen in advanced cases. **Beta-zone atrophy** lies next to the disc, with sclera visible; the outer alpha zone is irregularly hypopigmented.

Q How do you tell a large physiological cup from a glaucomatous cup?

A **physiological cup** (the atrophied Bergmeister's papilla) is round, with a rim obeying ISNT. A glaucomatous cup is vertically elongated, with notching, haemorrhage or an RNFL defect.

Say-it script

Opening (about 1 minute)

SAY IT

My patient, Mr X, is a 70-year-old man who has come for regular follow-up of glaucoma. He was apparently normal two years ago. Then he noticed defective vision in both eyes. It was gradual in onset, progressive in nature and not associated with pain or headache. An ophthalmologist near his home found raised pressure in both eyes. He was started on eye drops at bedtime, and has come to us for follow-up. He has no headache with vomiting, coloured haloes, redness, watering or pain. There is no history of injury to the eyes or frequent change of glasses. He had cataract surgery with lens implants in both eyes. He is not a known diabetic or hypertensive. He has no asthma, heart disease, sulpha allergy, migraine, Raynaud's phenomenon, stroke or head injury. He has never used steroids. He has used bimatoprost drops at bedtime in both eyes regularly for two years. He does not smoke or drink alcohol. There is no family history of glaucoma.

Closing (2 minutes)

SAY IT

To summarise, my patient is a 70-year-old man on bimatoprost drops for glaucoma in both eyes for two years. He has gradual, painless, progressive defective vision, with no haloes, trauma or steroid use. Visual acuity is 6/18 in the right eye and 6/9 in the left. Intraocular pressure by Goldmann applanation tonometry is 18 and 16 millimetres of mercury at ten in the morning. The anterior chambers are Van Herick grade four, with no relative afferent pupillary defect. Both eyes are pseudophakic. Gonioscopy shows the scleral spur in all four quadrants. The right disc has a vertical cup-disc ratio of 0.8, with superior and inferior rim thinning. It shows nasalisation, bayoneting and peripapillary atrophy. The left disc has a ratio of 0.6, with an inferior notch. Confrontation fields are full.

My diagnosis is both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management.

I will do pachymetry, diurnal phasing, Humphrey fields and optical coherence tomography of the retinal nerve fibre layer. Then I will set a target pressure. If it is met, I will continue the prostaglandin analogue. If not, I will switch or add a drug, then consider laser trabeculoplasty or trabeculectomy. I will counsel him on lifelong treatment and compliance.

Quick recall

- Normal IOP **10–21 mm Hg**; asymmetry ≥ 4 mm Hg and a diurnal swing ≥ 8 mm Hg are significant.
- Order: acuity → pupils → slit lamp → tonometry → gonioscopy → dilate → disc → fields.
- GAT is calibrated for a CCT of **520 μm** . Thinner reads low, thicker high: about 0.7 mm Hg per 10 μm .
- Vertical CDR ≥ 0.7 or asymmetry > 0.2 : suspect glaucoma.
- **25–30%** of ganglion cells are lost before a white-on-white field defect.
- OCT: RNFL thins first at **7 o'clock RE, 5 o'clock LE**; eyes differ by $> 9 \mu\text{m}$.
- **TP = IP (1 - IP/100) - Z $\pm$ 2**; Z = 0, 1, 3, 5, 7.
- PG analogue first, at bedtime; timolol in the morning.

Mnemonic: PART B before dilating: Pupillary reflex, Angle, Rubeosis, Tension, BCVA (best-corrected visual acuity).

Read more: Baidya p.149–166, 181–205, 367, 442, 717, 739–740 · Namrata p.171–178, 183, 200, 352 · Aravind 4.1, 4.2, 4.4, 4.6, 4.8, 4.15–4.20, 6.6, glaucoma model case sheet p.661–667 · Glaucoma case format p.1–2

G4 Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy and the acute attack

★ Kept 4 times at JEH last year (once as the long case) · Long case

At a glance

- **Primary angle closure:** the peripheral iris closes the drainage angle, usually from **pupillary block**. The ISGEO classification stages it as primary angle-closure suspect (**PACS**), primary angle closure (**PAC**) or primary angle-closure glaucoma (**PACG**).
- Why it is kept: common, bilateral, stable and full of signs. Gonioscopy was the task in all four cases last year.
- Examiners test: the Van Herick grade, gonioscopy with indentation, the stage, the mechanism, imaging, laser iridotomy and the acute attack.

Keywords the examiner listens for: pupillary block · iridotrabecular contact · peripheral anterior synechiae · Van Herick · indentation gonioscopy · occludable angle · plateau iris · glaukomflecken · laser peripheral iridotomy · ultrasound biomicroscopy · anterior segment optical coherence tomography · mannitol · Fincham test · combined mechanism

History — positive and negative, with reasons

Chief complaint: “Mrs X, a 58-year-old woman, has come for regular follow-up for glaucoma. Laser was done to both eyes 3 years ago.”

History of present illness (house wording): “Patient was apparently normal ___ years before. Then she had episodes of blurred vision with coloured haloes and brow ache in both eyes. They came on in dim light and while reading. An ophthalmologist found raised pressure and narrow angles. Laser was done to both eyes, and she was started on timolol 0.5% eye drops twice daily. Then she came to our institution for further follow-up.”

Positive history

Question	Why I ask — what the answer tells me
Onset and progression of the blurring?	Gradual, slow loss: chronic closure , mostly symptom-free. Sudden painful loss: an acute attack .
One eye or both?	Primary closure tends to be bilateral; after an attack in one eye, the fellow eye is at risk.
Is the blurring on and off, like a smoke-filled room?	Intermittent closure with corneal oedema.
In the dark, or watching television in a dark room?	Dim light mid-dilates the pupil, where pupillary block is greatest.
While reading?	Near work is a physiological trigger.
After dilating drops?	Pharmacological mydriasis can start an attack.
With emotional stress?	Acute emotional stress is a listed trigger.
When lying face down?	The prone position is a trigger.
Similar episodes before, in either eye?	Repeated attacks; an attack in one eye makes laser in the other urgent.
Laser or eye surgery: which eye, when?	Status after laser peripheral iridotomy (LPI); earlier surgery changes the plan.
Old records of pressure and fields?	Baseline intraocular pressure (IOP) and fields show progression .
Glasses for distance?	Hypermetropia goes with PACG: a short, crowded eye.

Negative history

Question	A “yes” would point to	How to tell them apart
Any coloured haloes?	Corneal epithelial oedema from a pressure rise: angle closure. Other causes of haloes: mucus on the cornea in conjunctivitis, incipient cataract, vitreous opacities, snow blindness, a tilted intraocular lens	Fincham (stenopaeic slit) test: a slit is moved across the pupil. The glaucoma halo stays intact; the halo of incipient cataract breaks into its component colours
Any headache?	Severe one-sided headache: an acute attack	Circumciliary congestion, corneal oedema, mid-dilated oval pupil, IOP 50–100 mm Hg
Nausea or vomiting?	With pain and headache: an acute congestive attack	The same signs, with a very high IOP
Eye pain?	A rapid rise of IOP; pain follows the speed of the rise, not the level	Gonioscopy shows the closed angle
Redness?	An attack, or toxicity from long-term drops	Attack: congestion round the limbus. Drug toxicity: inferior bulbar congestion

Question	A "yes" would point to	How to tell them apart
Watering?	Drop side effects: lacrimation from pilocarpine, allergy to brimonidine, or surface toxicity of long-term drops	Fluorescein 1% staining shows surface toxicity; lid and conjunctival allergy points to the drug
Injury to the eye?	Traumatic glaucoma or traumatic angle closure	Gonioscopy: angle recession, trabecular dialysis, cyclodialysis
Recent eye operation?	Malignant glaucoma	Diffuse shallowing, central and peripheral
Frequent change of near glasses?	Primary open-angle glaucoma	Open angle on gonioscopy
Poor vision from cataract before the pain?	Lens-induced glaucoma	Phacomorphic: one swollen lens, one shallow chamber. Phacolytic: deep chamber, open angle

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Diabetes?	Frequent dilated retinal checks can start an attack: a reason for LPI. No oral glycerol.
High blood pressure?	The same risk from repeated dilatation.
Asthma or chronic airway disease?	Beta-blockers are contraindicated.
Heart failure, heart block or slow pulse?	No beta-blockers; hyperosmotics avoided in cardiac decompensation.
Kidney disease?	No acetazolamide in renal failure; no hyperosmotics in anuria.
Sulpha allergy?	Acetazolamide is avoided.
Dilating drops (atropine, cyclopentolate, tropicamide, phenylephrine)?	Anticholinergic and adrenergic drops can precipitate closure .
Antihistamines for allergy or colds?	Systemic anticholinergic effect.
Antidepressants or antipsychotics?	Anticholinergic; phenothiazines also narrow the angle.
Drugs for Parkinson's disease or stomach cramps?	Anticholinergic.
Inhalers, decongestants (vasoconstrictors) or appetite suppressants?	Systemic adrenergic drugs can precipitate closure.
Topiramate or a sulpha drug?	Ciliary effusion closes the angle without pupillary block.
Present drops: strength, frequency, duration, compliance, side effects?	Shows control and decides the next drug.
Glaucoma in the family?	First-degree relatives are at high risk; I will ask them to be examined.

Examination — must-do steps and their viva

General and systemic (this case only)

- Wheeze and pulse: asthma, bradycardia and heart block rule out beta-blockers.
- Blood sugar before oral glycerol; weight, because osmotic agents are dosed per kilogram.
- Heart failure or dehydration: both contraindicate hyperosmotics.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity	Pinhole and glasses; plus correction	6/12; +2.00 D sphere: 6/9
Extraocular movements	Seniors' addition	Full, free
Tn by GAT	Tension by Goldmann applanation tonometer (GAT); note the time	16 mm Hg at 10 am
Lids	Prostaglandin lid pigmentation, long lashes	Normal
Conjunctiva	Attack: circumciliary congestion . On drops: inferior congestion	Quiet
Cornea	Attack: epithelial and stromal oedema. Chronic: Krukenberg spindle	Clear
Anterior chamber	Central and peripheral depth; Van Herick grade; cells and flare after an attack	Shallow, VH (Van Herick) grade 2
Iris	Iridotomy; sectoral atrophy; posterior synechiae; iris bombé in an attack	Patent LPI at 11 o'clock

Structure	Look for in this case	How to record (example)
Pupil	After an attack: mid-dilated, vertically oval. Relative afferent pupillary defect (RAPD) when advanced	RTL (round, reacting to light) 3 mm, no RAPD
Lens	Glaukomflecken ; thick or swollen lens	Clear
Gonioscopy	Iridotrabecular contact (ITC), peripheral anterior synechiae (PAS)	Cross diagram (step 7)
Fundus	Undilated until iridotomy. Attack: hyperaemic, swollen disc. Chronic: cupping	+90 D: cup-disc ratio (CDR) 0.7
Fields	Confrontation (seniors' addition)	Full

Step 1 — Visual acuity and glasses

Do: Snellen acuity in each eye, then with pinhole and with her glasses. Note a plus (hypermetropic) prescription.

Record: "RE 6/12, 6/9 with +2.00 D sphere; LE 6/9."

Q Why does vision blur in an attack?

Corneal changes. The pressure first distorts the corneal lamellae. Later the epithelium becomes oedematous, which also gives the haloes.

Q Why note the glasses?

Hypermetropia goes with a short eye. A short axial length, small cornea and thick lens crowd the front segment and raise the risk of PACG.

Step 2 — Pupils, before any drop

Do: Note size and shape. Test direct and consensual reactions, then the swinging torch-light test.

Record: "Pupils round, 3 mm, reacting to light, no RAPD." After an attack: "RE pupil mid-dilated, vertically oval, sluggish."

Q Why is the pupil mid-dilated, vertically oval and non-reacting in an acute attack?

Sphincter ischaemia. Above 40 mm Hg the sphincter is ischaemic and cannot constrict. Severe pain also drives sympathetic overactivity, which dilates the pupil vertically.

Q What does a relative afferent pupillary defect mean here?

Advanced glaucoma in that eye. It points to PACG rather than PAC.

Step 3 — Anterior chamber depth: pen-torch, then Van Herick

Do: Shine a pen-torch from the temporal side and note how much iris is lit. At the slit lamp, put a slit-thin beam 1 mm inside the temporal limbus at 60°. Compare the peripheral depth with the corneal thickness, and note the central depth.

Record: "Pen-torch from the temporal side: two-thirds of iris lit, grade 2, shallow chamber. Van Herick ratio 1:1/4, grade 2: shallow chamber with occludable angles."

Q How do you grade the chamber by Van Herick?

Van Herick compares peripheral depth with corneal thickness. Grade 4: one thickness or more (normal). Grade 3: ¼ to ½. Grade 2: ¼. Grade 1: under ¼. Grade 0: iridocorneal contact. Grades 0–2 are suspicious of angle closure.

Q Can Van Herick replace gonioscopy?

No. It is a guide to who needs gonioscopy. Peripheral depth under ¼ corneal thickness means the angle may be occludable.

Q The centre is deep but the periphery shallow. What do you suspect?

Plateau iris: a deep axial chamber, a narrow peripheral chamber and a flat iris.

Step 4 — Slit lamp: signs of a past attack

Do: Before any drop, examine cornea, chamber, iris and lens in order for signs of an old attack.

Record: "RE: sectoral iris atrophy superiorly, glaukomflecken, Krukenberg spindle. LE: no signs of a previous attack."

Q What are the signs of a previous attack?

Iris pigment on the endothelium with endothelial loss, and PAS. The iris shows sectoral atrophy, or whorling from sectoral infarction of the sphincter. Also look for posterior synechiae, a mid-dilated sluggish pupil and glaukomflecken. Field loss and cupping may follow.

Q What are glaukomflecken?

Glaukomflecken are small grey-white anterior subcapsular or capsular opacities in the pupillary zone. They come from infarction of lens fibres in an attack.

Q What is Vogt's triad?

Glaukomflecken, sectoral iris atrophy and a Krukenberg spindle. Together they are characteristic of PACG after an attack.

Step 5 — The peripheral iridotomy

Do: Find the opening with diffuse light. Describe, in the department's order: margin, site, surrounding iris, then patency by retro-illumination.

Record: "Iridotomy at 11 o'clock, ragged irregular edges, hypopigmented surrounding iris. Light reflex seen on retro-illumination: suggestive of a patent laser peripheral iridotomy."

Q How do you tell a laser iridotomy from a surgical iridectomy?

Laser: ragged edges, any clock hour, pigment dispersion around it; it can close. **Surgical:** clean-cut triangular edges near the limbal incision, usually at 12 o'clock; it cannot close.

Q Does a light reflex on retro-illumination prove patency?

Not on its own. Seeing the **anterior lens capsule** through the opening confirms it.

Q Why at 11–1 o'clock?

The upper lid covers it. Some prefer the upper nasal iris, to avoid diplopia and macular burns.

Step 6 — Applanation tonometry

Do: Topical anaesthetic and fluorescein. Goldmann applanation before gonioscopy and before any dilatation. Note the time.

Record: "Tn by GAT: RE 16 mm Hg, LE 18 mm Hg at 10 am."

Q Why record the time?

Diurnal variation. Pressure is higher in the morning. The normal swing is 3–6 mm Hg; in glaucoma it is larger, and 8 mm Hg or more is significant.

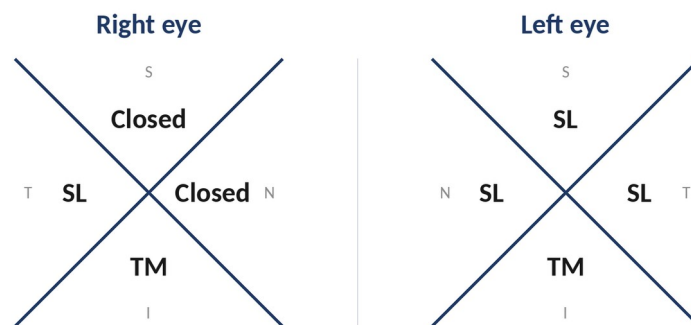
Q When will you measure central corneal thickness?

After any attack settles. Goldmann applanation is calibrated for a mean central corneal thickness of 520 μm . A thinner cornea reads falsely low, a thicker one falsely high, by about 0.7 mm Hg per 10 μm . Thin is under 500 μm ; thick is over 570 μm .

Step 7 — Gonioscopy: static, then indentation

Do: Both eyes, before dilatation. First a Goldmann two-mirror lens, without pressure, which would distort the angle. Ask the patient to look towards the mirror to see into a narrow angle. Name the structures from behind forward: ciliary body band, scleral spur, trabecular meshwork, Schwalbe's line. Count the quadrants where the posterior meshwork is hidden. Then press with a Zeiss four-mirror lens (indentation or dynamic gonioscopy). Clear an oedematous cornea with 1–2 drops of anhydrous glycerin. Write the deepest structure seen in each quadrant (Shaffer grades under Classification).

Record:



Deepest structure before indentation (SL Schwalbe's line, TM trabecular meshwork). ITC in 3 quadrants per eye

"RE: ITC in 3 quadrants; indentation opens the temporal angle to the scleral spur; PAS superiorly and nasally. LE: ITC in 3 quadrants, all opening on indentation."

Q What is an occludable angle?

An **occludable angle** shows no pigmented trabecular meshwork in at least three quadrants without indentation. That is Shaffer grade 1 or 0.

Q What does indentation add?

Indentation gonioscopy forces aqueous into the angle. If it opens, closure is appositional; if not, there are PAS, whose extent I note. A double hump means plateau iris. Indentation can even break an attack.

Q How do you tell PAS from iris processes, and what causes PAS?

Iris processes are lacy and fenestrated; the structures behind stay visible. **PAS** are solid and hide them. Causes of PAS: primary angle closure, anterior uveitis, iridocorneal endothelial syndrome, intraocular surgery (a wound leak, for example) and trauma.

Q What do you expect in acute and chronic closure?

Acute: an occludable angle. **Chronic:** PAS, which appear late. After an attack the meshwork may be pigmented.

Step 8 — Fundus, when it is safe

Do: No dilatation until iridotomies are done. View the undilated disc with a direct ophthalmoscope or a +78 D or +90 D lens. Dilate after a patent iridotomy, usually about 2 weeks after laser.

Record: "Undilated, +90 D: disc vertically oval, CDR 0.7, inferior rim thinning."

Q Why did you not dilate?

Dilatation is contraindicated until iridotomies are done; it can start an attack. In plateau iris syndrome it can do so even after a patent iridotomy.

Q What does the disc look like after an attack?

Hyperaemic and oedematous, sometimes with haemorrhages. It becomes pale and cupped after 9–10 days. Chronic closure gives cupping like open-angle glaucoma.

Step 9 – Fields by confrontation

Do: Confrontation in each eye, all four quadrants, before the diagnosis. Confirm with automated perimetry.

Record: “Fields full to confrontation in both eyes.”

Q What happens to the field during an attack?

Generalised constriction.

Q Why do the disc and fields decide the stage?

Optic neuropathy is the only difference between PAC and PACG.

Summary and complete diagnosis

“Mrs ___, a ___-year-old hypermetropic woman, presented with ___. Visual acuity is ___ / ___. Chambers are shallow, Van Herick grade ___ / ___. Iridotomies are ___. Tn by GAT is ___ / ___ mm Hg at ___. Gonioscopy shows ITC in ___ quadrants and PAS ___. CDR is ___ / ___. Fields are ___.”

Eye → stage → mechanism → acute or chronic → iridotomy status → IOP control → other eye.

- “Both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies, intraocular pressure controlled on one medication.”
- “Both eyes primary angle-closure suspect: iridotrabecular contact in 3 quadrants, no synechiae, normal pressure, discs and fields.”
- “Right eye primary angle closure with an acute congestive attack, pressure 54 mm Hg; left eye primary angle-closure suspect.”

Differential diagnosis

Condition	Points for	Points against	How to tell apart
Primary open-angle glaucoma	Raised IOP, cupping	Shallow chambers; haloes in dim light	Gonioscopy: open angle in all quadrants
Plateau iris syndrome	Narrow angle, attacks	Shallow central chamber, convex iris	Deep central chamber, flat iris; indentation: double hump ; stays occludable after a patent iridotomy
Phacomorphic glaucoma	Shallow chamber, high IOP	Both chambers shallow; clear lenses; hypermetrope	Asymmetric shallowing with a swollen (intumescent) cataract in one eye
Malignant glaucoma	Shallow chamber, high IOP	No recent intraocular surgery	Diffuse shallowing; ultrasound shows forward-rotated ciliary processes
Neovascular glaucoma	Closed angle, high IOP	No diabetes or vein occlusion	New vessels on iris or angle, ectropion uveae
Drug-induced ciliary effusion	Shallow chamber, high IOP	No topiramate or sulpha drug	Drug history; ultrasound shows supraciliary effusion

Exclude first: secondary closure — look at the fellow eye. Primary closure tends to be bilateral; a wide-open fellow angle suggests another diagnosis.

Open-angle mimics of an attack: glaucomatocyclitic crisis, herpes simplex keratouveitis, herpes zoster ophthalmicus, and pigmentary, exfoliative and phacolytic glaucoma. Gonioscopy shows an open angle.

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Gonioscopy with indentation , both eyes, before dilatation	Confirm and stage; the gold standard	ITC in 3 or more quadrants; PAS in clock hours	Decides PACS, PAC or PACG; synechial closure over 270° favours trabeculectomy
Anterior segment optical coherence tomography (AS-OCT): non-contact, seated	Image of the angle	An occludable angle	Shows occludability; limited view behind the iris
Ultrasound biomicroscopy (UBM): 50–100 MHz, supine, eye cup with methylcellulose	Find the mechanism	Iris bowing (block); double hump (plateau); cysts	Iridotomy for block, iridoplasty for plateau; images through corneal oedema
Biometry (A-scan or optical): axial length, chamber depth, lens thickness	Confirm the short, crowded eye	Axial length under the normal 22.5–24.5 mm	Documents risk; lens power if lens extraction is planned

Test — what I do	Why	Expected finding	Significance — what it changes
Automated perimetry	Stage PAC versus PACG	Glaucomatous defects in PACG	Baseline for progression
Optical coherence tomography (OCT) of the nerve fibre layer	Structural baseline	Thinning in PACG	Detects progression
Blood sugar, blood pressure, kidney function	Fitness for drugs and surgery	—	Glycerol avoided in diabetes; acetazolamide and mannitol need kidney function

Say: “I will proceed with gonioscopy with indentation to confirm and stage. Ultrasound biomicroscopy will show the mechanism, and anterior segment optical coherence tomography the occludability. Perimetry and nerve fibre scans give a baseline.”

How I will manage

Aim

To relieve pupillary block, open the angle and protect the optic nerve in both eyes. Management of angle closure is essentially surgical; drugs control the pressure first.

The ladder

- Acute attack, IOP above 50 mm Hg:** miotics fail on an ischaemic iris. I give **mannitol 20% intravenously, 1–2 g/kg over 20–30 minutes**. Without nausea or vomiting, I add oral **acetazolamide 250–500 mg stat**, then 250 mg four times a day. Oral **glycerol 50%, 1–1.5 g/kg**, suits non-diabetics. **Apraclonidine 0.5%** lowers the pressure quickly.
- Trigger — IOP below 40 mm Hg:** **pilocarpine 2%** four times over 30 minutes, then 6-hourly. I add **timolol 0.5%** twice daily, **brimonidine 0.2%** three times daily and **prednisolone acetate 1%** four times daily. Ibuprofen 500 mg twice daily and an antiemetic help. I indent the central cornea, review hourly, and do gonioscopy of both eyes once the cornea clears.
- Trigger — quiet eye, clear cornea:** **Nd:YAG (neodymium:yttrium-aluminium-garnet) iridotomy in both eyes**. Hazy cornea and closure under two-thirds: **surgical iridectomy**, with YAG iridotomy in the fellow eye. Closure over two-thirds, or persistent inflammation with high IOP: **primary trabeculectomy**.
- PACS:** LPI, or observation with IOP, gonioscopy and disc checks. I advise LPI with frequent dilatation (diabetes, hypertension, macular degeneration) or an attack in the other eye. Hypermetropia, poor follow-up, intermittent symptoms and predisposing drugs also favour it.
- PAC and PACG:** LPI, plus drugs or surgery to control the IOP.
- Trigger — gonioscopy 1 week after LPI:** still occludable despite a patent iridotomy means **plateau iris syndrome**. I then do **argon laser peripheral iridoplasty**. An open angle with high IOP means a **combined mechanism**: timolol plus a prostaglandin, dorzolamide, brimonidine or pilocarpine.
- Trigger — failure of laser and drugs:** **trabeculectomy with mitomycin C or 5-fluorouracil**. Indications: PACG not controlled by iridotomy and/or iridoplasty, progression on maximal medical therapy, or synechial closure over 270°.
- Lens extraction**, when the lens drives the closure (approved passage below).

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Mannitol 20% intravenous, 1–2 g/kg over 20–30 min	Raises plasma osmolarity; water leaves the vitreous	Diuresis, dehydration; heart failure, pulmonary oedema; rebound rise	Anuria, severe dehydration, cardiac decompensation, pulmonary oedema
Glycerol 50% oral, 1–1.5 g/kg	The same osmotic effect, by mouth	Nausea, vomiting; hyperglycaemia	Diabetes (use oral isosorbide)
Acetazolamide 250 mg two to four times a day	Blocks carbonic anhydrase II in the ciliary epithelium	Paraesthesia, nausea, acidosis, low potassium, renal stones, aplastic anaemia	Sulpha allergy, renal failure, chronic liver disease
Pilocarpine 2% once IOP is below 40	Constricts the pupil, pulling the peripheral iris off the meshwork	Brow ache, dim vision, induced myopia, retinal detachment	Neovascular, uveitic and phacolytic glaucoma
Timolol 0.5% twice daily	Beta-blocker: less cyclic adenosine monophosphate in the ciliary epithelium, less aqueous	Bradycardia, hypotension, bronchospasm, depression	Asthma, airway disease, heart failure, heart block
Brimonidine 0.2% three times daily	Alpha-2 agonist: less aqueous, more uveoscleral outflow	Allergic blepharoconjunctivitis, dry mouth, somnolence	Children under 2 years
Apraclonidine 0.5–1%	Alpha-2 agonist: less aqueous, better trabecular outflow	Hyperaemia, mydriasis; bradycardia	Children under 2 years
Prednisolone acetate 1% four times daily	Controls inflammation after an attack or laser	Pressure rise in steroid responders	Caution with a family history of glaucoma

Full profiles: card GX.

Laser and surgery in this case

Procedure (indication)	Key steps	Advantages	Disadvantages and complications
Nd:YAG LPI: PACS at risk, PAC, PACG, fellow eye, attack once the cornea clears	Pilocarpine 1%; apraclonidine or brimonidine, one drop before and one after; Abraham lens; outer third at 11–1 o'clock, in a crypt; 2–5 mJ, 1–3 pulses; 150–200 µm	Works on all iris colours; fewer late closures than argon	IOP spike, iritis, hyphaema, focal cataract, corneal burn, glare; rarely malignant glaucoma
Argon laser peripheral iridoplasty: plateau iris syndrome, nanophthalmos	200–400 mW, 0.1 s, 20–24 spots over 360° on the peripheral iris	Shrinks and flattens the peripheral iris; opens an angle that iridotomy alone leaves closed	Not a substitute for iridotomy: most such eyes also have pupillary block, so iridotomy comes first
Surgical peripheral iridectomy: attack, hazy cornea, closure under two-thirds	Iridectomy through a limbal incision, usually at 12 o'clock	Possible when the cornea is too hazy for laser; cannot close	An open intraocular operation
Trabeculectomy with mitomycin C or 5-fluorouracil: ladder step 7	Mitomycin C 0.2–0.5 mg/ml for 1–5 min, or 5-fluorouracil 50 mg/ml for 5 min. Peripheral iridectomy, so the iris cannot block the ostium or cause pupillary block. Scleral flap sutured more tightly	Lowers IOP more than drugs	Shallow chamber (commonest), malignant glaucoma, bleb infection, cataract

Lens extraction

- **Phacoemulsification with intraocular lens implantation** corrects hypermetropia, deepens the anterior chamber and opens the angle: IOP control in almost all patients with normal preoperative IOP, and in up to **80%** with raised preoperative IOP. (*Kanski — not in your practical books*)
- **EAGLE study:** clear lens extraction is more effective and more cost-effective than LPI in PAC with IOP > **29 mm Hg**, or in PACG. (*Kanski — not in your practical books*)
- **PACS — ZAP trial:** LPI has a modest prophylactic effect; offer it only to those at highest risk of PACG. If significant ITC persists after LPI and there is symptomatic cataract, lens extraction usually opens the angle. (*Kanski — not in your practical books*)

Follow-up and counselling

- After laser: IOP at 1 hour. A rise of 8 mm Hg or more, or IOP over 30 mm Hg, is a spike. With advanced damage or a high IOP, I also give oral acetazolamide. Prednisolone acetate 1% four times daily for about a week.
- At 1 week: IOP, patency, inflammation and gonioscopy. Then gonioscopy at every review, because the lens keeps thickening.
- Counsel: no dilating drops, antihistamines or antidepressants without telling the doctor. Come at once with haloes, pain or redness.
- After intravenous mannitol, do not get up at once: hypotension.
- Bring first-degree relatives for examination.

What you must know

Definition

“**Primary angle closure (PAC)** is appositional or synechial closure of the anterior-chamber angle caused by pupillary block. The angle closure may or may not be associated with elevated intraocular pressure (IOP) or glaucomatous optic neuropathy, and may occur in either an acute or chronic form. This entity does not include secondary forms of angle closure induced by other causes, for example subluxed lens.”

In plain words: aqueous is held back at the pupil, and the peripheral iris blocks the angle. **Gonioscopy, IOP, disc and field** make the diagnosis.

- **Relative pupillary block:** lens-iris touch blocks flow through the pupil. Aqueous collects behind the iris and bows it forward.
- **Plateau iris configuration:** the plateau iris shape without angle closure.
- **Plateau iris syndrome:** an anteriorly placed ciliary body pushes the peripheral iris forward. The angle stays occludable despite a patent iridotomy.
- **Combined mechanism:** angle-closure and open-angle elements together.
- **Acute congestive attack:** circumcorneal congestion, corneal oedema, pain and vomiting. The chamber is shallow, the pupil vertically oval, the IOP very high.

Classification

The ISGEO classification uses IOP, gonioscopy, disc and fields:

Stage	Gonioscopy	IOP and PAS	Disc and field
PACS	ITC in 3 or more quadrants (almost 270°)	Normal IOP, no PAS	Normal
PAC	ITC in 3 or more quadrants	Raised IOP and/or PAS	Normal

Stage	Gonioscopy	IOP and PAS	Disc and field
PACG	ITC in 3 or more quadrants	Raised IOP and/or PAS	Glaucomatous optic neuropathy: cupping with field loss

By mechanism

- **Primary:** with pupillary block (acute, subacute or chronic closure); without block (plateau iris).
- **Secondary with block:** swollen lens, mobile lens (ectopia lentis, microspherophakia), miotics. In spherophakia, miotics slacken the zonules and the lens moves forward: **inverse glaucoma**.
- **Secondary without block:** anterior “pulling” (neovascular glaucoma, iridocorneal endothelial syndrome). Posterior “pushing” (ciliary block, cysts, nanophthalmos, tumours, suprachoroidal haemorrhage).
- **Four sites:** iris (pupillary block), ciliary body (plateau iris), lens (phacomorphic), behind the iris (malignant glaucoma).

Clinical stages: prodromal (haloes, white eye, IOP 40–60 mm Hg). Then constant instability, then the acute congestive attack. Chronic closed-angle: PAS, high IOP between attacks. Absolute glaucoma: no light perception, stony hard.

Gonioscopic grading — Shaffer:

Grade	Width	Deepest structure seen	Closure
4 wide open	35–40°	Up to ciliary body	Impossible
3 open	25–35°	Up to ciliary body	Impossible
2 moderately narrow	20°	Pigmented trabecular meshwork	Possible but unlikely
1 extremely narrow	10°	Only Schwalbe's line; maybe anterior meshwork	Possible
Slit	< 10°	No structures; no obvious contact	Portions appear closed
0 closed	—	Iridocorneal contact	Complete

- **Scheie**, by structures hidden: wide open, all seen. Grade 1: iris root hard to see. Grade 2: ciliary body band hidden. Grade 3: posterior meshwork hidden. Grade 4: only Schwalbe's line seen.
- **Spaeth**, four parameters: iris-root insertion **A–E** and angle width **10–40°**. Iris contour is **S** (steep, convex), **R** (regular, flat) or **Q** (queer, deeply concave). Meshwork pigmentation is graded **0–4**.

Pathogenesis

- **Pupillary block** traps aqueous behind the iris, and the peripheral iris bows onto the meshwork.
- Contact is first **appositional**, then **synechial**; outflow falls and IOP rises.
- **Age** thickens the lens, moves it forward and narrows the pupil.
- **Risk factors:** older age, women, family history, Asian or Inuit descent. Also hypermetropia, short axial length, small cornea, shallow chamber and a thick lens. An anteriorly placed ciliary body adds risk.

Complications

- **Of an attack:** glaukomflecken, iris atrophy, PAS, posterior synechiae, endothelial loss, cupping; finally **absolute glaucoma**.
- **Of trabeculectomy in angle closure:** a shallow chamber and malignant glaucoma.

Prognosis

- PACS progresses to acute closure in **6–10%**, and to PAC or PACG in **17–35%**. The fellow eye closes in about **50% within 5 years**.
- **Good:** appositional closure, which iridotomy usually relieves.
- **Poor:** synechial closure over 270°, and plateau iris or a combined mechanism, which persist after a patent iridotomy.

Recent advances

- (extra) Your thesis angle parameters are not defined in your practical books; use your thesis definitions.

Viva questions

Why did you ask...?

Q Why did you ask about coloured haloes?

Corneal epithelial oedema acts as a diffraction grating: blue-green in the centre, yellow-red outside. Haloes point to intermittent closure.

Q Why did you ask about diabetes and hypertension?

Frequent dilated examinations can start an attack in a shallow chamber. This is an indication for LPI in a suspect.

Q Why did you ask about drugs?

Anticholinergic and adrenergic drugs dilate the pupil and precipitate closure. Topiramate and sulpha drugs cause ciliary effusion, which closes the angle without block.

Diagnosis and differentials

Q Why does a mid-dilated pupil precipitate an attack?

Maximum pupillary block. At mid-dilatation the iris–lens contact is greatest, and the slack peripheral iris bunches onto the meshwork. Full dilatation lifts the iris off the lens.

Q Why is pain worse than in open-angle glaucoma at the same pressure?

Rate of rise. Pain follows how fast the pressure rises, not the level.

Q What is the Fincham test?

The **Fincham test** moves a stenopaic slit across the pupil. A glaucoma halo stays intact; a cataract halo breaks into colours.

Q How do you diagnose and treat plateau iris syndrome?

Plateau iris: deep axial chamber, shallow periphery, flat iris. Indentation shows a sine-wave, double-hump iris, because the iris hangs over the anterior ciliary processes. Ultrasound biomicroscopy confirms it. The angle stays occludable after a patent iridotomy. Most eyes also have pupillary block, so iridotomy comes first, then argon laser iridoplasty.

Q What is the combined mechanism?

Combined mechanism: after iridotomy for an attack, the angle opens but the IOP stays high. I treat the open-angle component.

Investigations

Q What is ultrasound biomicroscopy?

Ultrasound biomicroscopy is 50–100 MHz ultrasound of structures in front of the pars plana. Axial resolution is 25 µm, lateral 50 µm. Penetration is about 4 mm.

Q What does ultrasound biomicroscopy show in each mechanism?

Pupillary block: forward iris bowing. **Plateau iris:** double hump on indentation. **Malignant glaucoma:** everything pressed against the cornea, with or without supraciliary fluid.

Q How does UBM differ from AS-OCT?

UBM needs contact, fluid, a skilled operator and lying down; the supine position may falsely widen the chamber. It sees behind the iris pigment epithelium and through opaque media. **AS-OCT** works by low-coherence interferometry. It is non-contact and faster, with higher axial resolution and a wider field, but needs a clear cornea.

Q What angle measurements can UBM give?

Angle width as a linear distance or an angle, and the iris–meshwork area. Also iris thickness and contour, chamber depth and the iris–ciliary body relation, followed after treatment.

Q What are the provocative tests?

Mydriatic, dark room, prone, prone dark room, phenylephrine–pilocarpine and triple tests. In the dark-room test the patient sits awake in the dark for 90 minutes. A rise over 8 mm Hg, or closure, is positive. Done with ultrasound biomicroscopy, dark-room testing gives objective results.

Management — drugs

Q Why mannitol first in an attack?

Mannitol lowers the pressure fast by drawing water from the vitreous. Above 50 mm Hg miotics fail. It is the intravenous drug of choice: safe in diabetes and renal failure.

Q How does pilocarpine work here, and when do you start it?

Pilocarpine constricts the pupil and pulls the peripheral iris off the meshwork, relieving pupillary block. I start it once IOP is below 40 mm Hg.

Q Why give apraclonidine or brimonidine with laser iridotomy?

To prevent the pressure spike. One drop goes before and one after laser. Apraclonidine 1% brings down short-term rises after iridotomy quickly.

Management — laser and surgery

Q Why does iridotomy work?

Laser peripheral iridotomy lets aqueous bypass the pupil. The pressure gradient goes, the iris flattens and the chamber deepens.

Q What are the indications for laser iridotomy?

Angle closure at any stage, and the fellow eye. Also pseudophakic pupillary block, malignant glaucoma, combined mechanism, microspherophakia and phacomorphic glaucoma.

Q What are the contraindications?

Corneal oedema or opacity, a flat chamber and a totally synechial angle. Also uveitis, neovascular glaucoma and iridocorneal endothelial syndrome.

Q Why Nd:YAG rather than argon?

Nd:YAG (1064 nm) uses photodisruption and works on all iris colours. Argon is thermal, needs pigment and more energy, and closes more often. Argon suits anticoagulated patients.

Q What does the Abraham lens do?

The Abraham lens (+55 D button) steadies the eye and keeps the lids apart. It magnifies the periphery and concentrates energy. Its pressure stops bleeding.

Q What are the signs of penetration?

A sudden gush of aqueous with a plume of pigment, and sudden deepening of the chamber.

Q What if the iridotomy does not penetrate?

Retreat after a few days, or choose another site. Thick dark irides often need thermal (argon) pre-treatment first.

Q Why does lens extraction help in angle closure?

The lens thickens and moves forward with age. Phacoemulsification with intraocular lens implantation corrects hypermetropia, deepens the anterior chamber and opens the angle; in the EAGLE study, clear lens extraction was more effective than LPI in PAC with IOP > 29 mm Hg, or PACG. (*Kanski — not in your practical books*)

Q What problems do you expect at phacoemulsification in this short, crowded eye?

A shallow anterior chamber raises the risk of a posterior capsule rent. A hypermetropic eye with a shallow chamber is also a cause of corneal oedema after cataract surgery. I choose the lens-power formula by axial length. Hoffer Q suits 20–22 mm; Holladay II or Hoffer Q suits under 20 mm.

Q When would you do trabeculectomy in angle closure?

Trabeculectomy: PACG uncontrolled by iridotomy and drugs, progression, or synechial closure over 270°. Non-penetrating surgery is relatively contraindicated: the meshwork lies close to the iris root, so filtration may fail.

Complications

Q What are the complications of LPI?

Common: IOP spike, uveitis, hyphaema, focal cataract, posterior synechiae, glare, corneal damage. **Rare:** aqueous misdirection, herpetic keratouveitis, retinal haemorrhage, macular hole.

Q How do you recognise and treat malignant glaucoma?

Malignant glaucoma follows surgery, with diffuse shallowing and high or normal IOP. Ultrasound shows forward-rotated ciliary processes. Hyperosmotics help; Nd:YAG hyaloidotomy opens the anterior hyaloid.

Q How do you manage absolute glaucoma? (extra)

Comfort first: antiglaucoma drops, steroids and atropine. Then cyclodestruction or retrobulbar alcohol, which lasts 3–6 months. Last, enucleation or enucleation.

Recent advances

Q Can imaging replace gonioscopy?

No. Anterior segment optical coherence tomography may detect occludability better. But gonioscopy remains the gold standard; it shows PAS directly.

Examiner traps

- Dilating before gonioscopy and iridotomy.
- Calling an iridotomy patent on retro-illumination alone.
- Pressing during static grading, or skipping indentation.
- Staging without the disc and fields.
- Calling every attack PACG: without glaucomatous optic neuropathy, it is primary angle closure.
- Pilocarpine while the pressure is above 40 mm Hg.
- Forgetting the fellow eye and the relatives.
- Saying imaging replaces gonioscopy.

Say-it script

Opening (about 1 minute)

SAY IT

My patient, Mrs X, is a fifty-eight-year-old woman. She has come for regular follow-up for glaucoma, after laser to both eyes three years ago.

She was apparently normal four years ago. Then she had repeated episodes of blurred vision with coloured haloes around lights and brow ache, in both eyes. These came on in dim light, while watching television in a dark room, and while reading. She has worn plus glasses for distance since the age of forty. An ophthalmologist found raised pressure and narrow angles. Laser was done to both eyes, and she was started on timolol 0.5% eye drops twice daily. Since then she has had no haloes.

There is no history of a sudden painful red eye with headache and vomiting. There is no trauma or eye surgery. She has not used dilating drops, antihistamines or antidepressants. She is not diabetic or asthmatic and has no sulpha allergy. Her mother was treated for glaucoma. She uses timolol regularly, without side effects.

Closing (2 minutes)

SAY IT

To summarise, Mrs X is a fifty-eight-year-old hypermetropic woman with intermittent haloes in dim light. Laser was done to both eyes three years ago, and she uses timolol twice daily.

Best-corrected visual acuity is 6/9 in each eye. The anterior chambers are shallow, Van Herick grade two. Each eye has an iridotomy at eleven o'clock with ragged edges, hypopigmented surrounding iris and a light reflex on retro-illumination. These are suggestive of patent laser peripheral iridotomies. Intraocular pressure by Goldmann applanation is 16 and 18 millimetres of mercury at ten in the morning. Gonioscopy shows iridotrabecular contact in three quadrants in each eye. On indentation gonioscopy the angles open, except superiorly and nasally in the right eye, where there are peripheral anterior synechiae. The undilated discs show cup-disc ratios of 0.7 and 0.6, with inferior rim thinning. Confrontation fields are full.

My diagnosis is both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies. Intraocular pressure is under control on one medication.

I would confirm the stage with automated perimetry and nerve fibre layer scans. I would check occludability with anterior segment optical coherence tomography, and look for plateau iris with ultrasound biomicroscopy. I will continue timolol, repeat gonioscopy at every visit, and consider trabeculectomy if the disease progresses.

Quick recall

- ISGEO: ITC in 3 or more quadrants (almost 270°). PAC adds raised IOP and/or PAS. PACG adds optic neuropathy.
- Van Herick grade 2 = $\frac{1}{4}$ corneal thickness; under $\frac{1}{4}$ may be occludable.
- Attack IOP 50–100 mm Hg. Miotics fail above 50; pilocarpine 2% only below 40.
- Mannitol 20% 1–2 g/kg intravenously over 20–30 min. Glycerol 50% 1–1.5 g/kg, not in diabetes.
- Nd:YAG LPI: 2–5 mJ, 1–3 pulses, 150–200 μ m, 11–1 o'clock.
- Spike after LPI: rise of 8 mm Hg or more, or IOP over 30 mm Hg.
- Trabeculectomy: synechial closure over 270°; after an attack, closure over two-thirds.
- PACS: acute closure 6–10%, PAC or PACG 17–35%. Fellow eye 50% in 5 years.
- Vogt's triad: glaukomflecken, sectoral iris atrophy, Krukenberg spindle.

Mnemonic: I CAN SEE TILL SCHWALBE'S LINE: Iris root, Ciliary body, Scleral spur, Trabecular meshwork, Schwalbe's line. These are the angle structures from the iris root forwards.

Read more: Baidya p.72, 129, 145–146, 148–152, 154, 156–158, 164, 172, 209, 441, 715–716, 721 · Namrata p.173, 175, 178–184, 352 · Aravind 3.1, 4.2, 4.5, 4.6, 4.7, 4.13, 4.14, 4.15, 4.17, 4.18 · Kanski p.377

G5 Neovascular glaucoma, including rubeosis iridis

★ Kept once at JEH last year · Long case and short case (rubeosis iridis)

At a glance

- **Neovascular glaucoma (NVG)** is a severe secondary glaucoma: a fibrovascular membrane covers the iris and angle after long-standing retinal ischaemia. The commonest causes are **proliferative diabetic retinopathy (PDR)**, **ischaemic central retinal vein occlusion (CRVO)** and **ocular ischaemic syndrome**.
- Its signs: neovascularisation of the iris (NVI, **rubeosis iridis**) and angle (NVA), **ectropion uveae** and very high intraocular pressure (IOP). Last year's case had applanation tonometry as its task.
- Examiners test: iris and angle **before any drop**; the **stage**; the **cause**; the ladder; the drugs to avoid.

Keywords the examiner listens for: rubeosis iridis · neovascularisation of the angle · ectropion uveae · fibrovascular membrane · peripheral anterior synechiae · ischaemic central retinal vein occlusion · proliferative diabetic retinopathy · ocular ischaemic syndrome · 100-day glaucoma · Wand's classification · panretinal photocoagulation · bevacizumab · glaucoma drainage device · cyclophotocoagulation

History — positive and negative, with reasons

Chief complaint: "Mr X, 62 years, came with pain, redness and defective vision in the right eye for 10 days."

History of present illness (house wording): "Patient was apparently normal 3 months ago. Then he had a sudden, painless loss of vision in the right eye. For 10 days the eye has been painful and red, with coloured haloes, headache and vomiting."

Positive history

Question	Why I ask — what the answer tells me
When did the pain start?	An acute rise of IOP: the angle-closure stage , when most patients present.
Coloured haloes around lights?	Corneal epithelial oedema from the pressure rise.
Headache, nausea or vomiting?	A severe acute rise of IOP. An NVG attack may have no precipitating factor.
Did the vision fall before the pain — suddenly or gradually?	Dimness often precedes pain by weeks to months. Sudden: ischaemic CRVO or central retinal artery occlusion (CRAO). Gradual: diabetic retinopathy, chronic uveitis, long-standing retinal detachment, radiation retinopathy.
How long ago was the sudden loss?	After ischaemic CRVO, NVI comes at 2–4 months and NVG at 3–5 months (" 100-day glaucoma "). After CRAO, rubeosis comes within 4–5 weeks in 16–18%.
Floaters, or a sudden total blackout?	Vitreous haemorrhage from retinal new vessels; it hides the fundus, so I will need a B-scan.
Blood seen in the front of the eye?	Hyphaema from fragile new vessels, even after a minor knock.
Retinal laser or injections into the eye?	Earlier panretinal photocoagulation (PRP) or anti-vascular endothelial growth factor (anti-VEGF) injection changes the plan and the operation.
Known glaucoma — on which drops?	Pre-existing primary open-angle glaucoma (POAG) gives a higher IOP rise; old records show progression.
Glasses for long sight?	Hypermetropia goes with angle closure. Chronic angle closure can cause vein occlusion and NVG — important in India.
Similar attacks before?	NVG can be associated with recurrent angle-closure attacks .
Diabetes — how long, what treatment, how controlled?	The commonest cause: a third of rubeotic eyes have diabetic retinopathy. Longer duration and poor control raise the risk.
High blood pressure?	It raises the risk in a diabetic eye and causes retinal vein occlusion.
Brief blackouts of vision in one eye?	Amaurosis fugax: carotid disease and ocular ischaemic syndrome , which also gives a dull brow ache (ocular angina).
Stroke or brief weakness of a limb?	A transient ischaemic attack: carotid disease again.
Heart disease?	A cause of ocular ischaemia; it also bars intravenous mannitol.
Eye surgery — cataract, vitrectomy, detachment surgery, laser capsulotomy?	Lens, capsule and vitreous are barriers to angiogenic factors. Capsule rupture, aphakia and vitrectomy remove them.
Radiation to the head or eye?	Radiation retinopathy is a cause.

Negative history

Question	A "yes" would point to	How to tell them apart
Did the attack follow dim light, reading or a dilating drop?	Primary acute angle closure	Angle closure has no iris or angle new vessels, no fibrovascular membrane and no ectropion uveae.
Recurrent red eye treated with steroid drops?	Uveitic glaucoma , or a steroid-induced rise	Keratic precipitates, flare and cells. Iris vessels may be dilated, but there is no true rubeosis.
Any injury to this eye, even years ago?	Angle-recession glaucoma	Gonioscopy: a recessed angle and pigment clumps in the meshwork; no iris new vessels.
Has one eye become a lighter colour?	Fuchs' heterochromic iridocyclitis	White, quiet eye; stellate keratic precipitates; heterochromia. NVI and NVG are rare.
Jaw pain on chewing, or scalp tenderness?	Giant cell arteritis , a cause of ocular ischaemia	Scalp tenderness is most specific; erythrocyte sedimentation rate (ESR) is often above 60 mm/hr.
A bulging red eye with swollen conjunctiva?	Carotid-cavernous fistula , which causes ocular ischaemia	Arterialised episcleral veins (the hallmark), chemosis and exophthalmos.

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Asthma or chronic obstructive airway disease?	No timolol or other beta-blocker.
Allergy to sulpha or other drugs?	No carbonic anhydrase inhibitor (dorzolamide, acetazolamide).
Kidney or liver disease?	Carbonic anhydrase inhibitors are avoided; no mannitol in anuria.
Heart block, slow pulse or low blood pressure?	Timolol worsens them; intravenous mannitol needs a sound heart.
On diabetic treatment?	Oral glycerol is not given to a diabetic.
Aspirin or other blood thinners?	They worsen hyphaema and vitreous haemorrhage; warfarin stops before device surgery.
Each eye drop: strength, frequency, duration, compliance, side effects?	Shows what he uses and what he could not tolerate.
Earlier trabeculectomy, valve or vitreoretinal surgery?	Changes the choice and site of the next operation.
Clots in legs or lungs (patient under 60)?	A hypercoagulable state: I will send a thrombophilia screen.
Smoking?	A risk factor for diabetic retinopathy.
Diabetes, hypertension, clotting disorders or glaucoma in the family?	First-degree relatives are at high risk of the same illnesses.

Examination — must-do steps and their viva

General and systemic (this case only)

- **Blood pressure and pulse:** hypertension is a cause; low pressure bars mannitol.
- **Heart and chest:** heart failure is a cause and bars mannitol; wheeze bars beta-blockers.
- **Diabetic complications:** neuropathy and nephropathy; nephropathy is a risk factor for diabetic retinopathy.

Ocular examination — RE | LE template

Right eye (RE) and left eye (LE) columns. Pupil, iris, angle and tension come **before any drop**.

Structure	Look for in this case	How to record (example)
Visual acuity	Poor: hand movements (HM); perception (PL) and projection (PR) of light	HM, PL+, PR accurate
Extraocular movements	Full; proptosis suggests a fistula	Full
Lids and conjunctiva	Lid oedema; circumciliary congestion	Circumciliary congestion +
Cornea	Epithelial and stromal oedema . Long-standing: pigment on the inferior endothelium, Descemet's folds, pigmented keratic precipitates	Epithelial oedema +
Anterior chamber	Van Herick (VH) grade; flare , cells, hyphaema	VH grade 3; flare +
Iris	NVI in slide order; flat, glistening surface. Chronic: segmental atrophy, posterior synechiae	NVI from 11 to 8 o'clock
Pupil	Mildly dilated, oval, fixed; ectropion uveae ; relative afferent pupillary defect (RAPD)	Oval, fixed; ectropion uveae +; RAPD +

Structure	Look for in this case	How to record (example)
Lens	Cataract; pigment on the anterior capsule; pseudophakia with capsule dehiscence; aphakia	PCIOL (posterior chamber intraocular lens)
Tension	Goldmann applanation tonometry (GAT)	Tn by GAT 48 mm Hg at 10 am
Gonioscopy	NVA; extent of peripheral anterior synechiae (PAS)	Cross diagram
Fundus	The cause; disc; vitreous haemorrhage	Disc pale, CDR (cup-disc ratio) 0.8; suggestive of old CRVO

Step 1 — Visual acuity

Do: Snellen vision in each eye, with pinhole. Below 1/60: counting fingers, hand movements, or light perception with projection.

Record: "VA RE: HM, PL+, PR accurate. LE: 6/9, with pinhole 6/6."

Q Why is the vision so poor?

Corneal oedema from the acute pressure rise, and **glaucomatous optic atrophy**, which can leave no perception of light. Macular ischaemia or vitreous haemorrhage adds to it.

Q Can an NVG eye keep useful vision?

Yes. Vision depends on the stage. A well-treated ischaemic retinal event can leave working visual acuity.

Step 2 — Pupils, before any drop

Do: Before any drop: direct and consensual reactions, then the swinging flashlight test. Then the pupil margin on the slit lamp.

Record: "RE pupil mildly dilated, oval, fixed; RAPD +; ectropion uveae +. LE round, reacting."

Q What is ectropion uveae, and why does it occur?

Ectropion uveae is out-rolling of the posterior pigment epithelium at the pupil margin. The contracting **fibrovascular membrane** pulls the posterior iris layer round onto the front. Repeated uveitic attacks with the rubeosis add to it.

Q What does a relative afferent pupillary defect mean here?

The **RAPD** shows the extent of optic-nerve damage. A marked RAPD (over 0.7 log unit) also marks an ischaemic CRVO.

Step 3 — Slit lamp, before dilating

Do: Dim light, high magnification, no drops. Cornea, anterior chamber by Van Herick, then the pupil margin, iris surface and any iridotomy margin.

Record: "RE: corneal epithelial oedema; VH grade 3, flare +; thin, friable, arborising, irregular new vessels from 11 to 8 o'clock."

Q Why must the iris be seen before dilatation?

Early **new vessels** are very fine and sit at the pupil margin. After dilatation they may not be found.

Q How do you tell new vessels from normal iris vessels?

New vessels are superficial, irregular, varying in size, branching and leaky. Normal vessels are stromal, radial and uniform. Only new vessels cross the scleral spur.

Q What do flare and hyphaema mean here?

Flare and cells go with the florid open-angle stage or a recent attack. **Hyphaema** comes from the fragile new vessels.

Step 4 — Tonometry

Do: Goldmann applanation in both eyes before dilatation, noting the time. With corneal oedema, take the true IOP as higher; repeat once it clears.

Record: "Tn by GAT: RE 48 mm Hg, LE 14 mm Hg at 10 am."

Q What pressure do you expect at each stage?

Normal in prerubeosis and pre-glaucoma, unless POAG or angle closure co-exists. **Raised** in the open-angle stage. **Persistently high**, and it can reach 60 mm Hg, in the angle-closure stage.

Q Why does the exact reading matter?

It sets the first treatment: topical aqueous suppressants below 40 mm Hg; intravenous mannitol added above **40 mm Hg**.

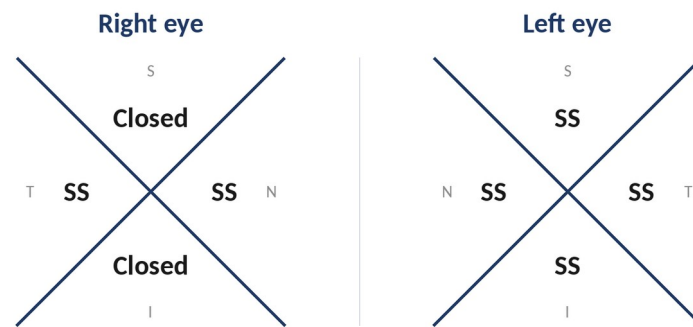
Q The cornea is oedematous. Which tonometer helps?

The **Tonopen** is useful on oedematous and scarred corneas, and it is portable. With Goldmann, I take the true pressure as higher than the reading. I repeat applanation once the oedema clears.

Step 5 — Gonioscopy

Do: Goldmann 2-mirror lens, both eyes, before dilatation; defer it while the cornea is oedematous. Note new vessels and PAS in degrees. Indent with a 4-mirror lens: appositional versus synechial closure.

Record: the cross diagram for each eye.



Deepest structure seen (SS = scleral spur). RE: closed by PAS above and below; new vessels cross the spur elsewhere. LE: open in all quadrants

Q How do you identify new vessels in the angle?

Neovascularisation of the angle crosses the ciliary body band and scleral spur, then arborises over the meshwork. In about 12%, it occurs without iris new vessels.

Q What does gonioscopy show at each stage?

Pre-glaucoma: open angle, early NVA. Open-angle stage: florid NVA under the membrane. Angle-closure stage: **peripheral anterior synechiae**, sometimes complete closure.

Q Why measure the extent of PAS?

PRP still reverses the IOP rise only while synechial closure is **not more than 270°**. PAS also mark where a drainage tube must not go.

Q The cornea is too hazy for gonioscopy. What will you do?

Anhydrous glycerine, one or two drops, clears the oedema enough for gonioscopy if it cannot wait. **Ultrasound biomicroscopy** images the angle even through corneal oedema or opacity.

Step 6 – Dilated fundus for the cause

Do: Dilate with a cycloplegic, which also treats NVG. Expect a festooned pupil from multifocal posterior synechiae. Then +90 dioptre (D) slit-lamp biomicroscopy and indirect ophthalmoscopy.

Record: “RE: media hazy; disc pale, CDR 0.8, optociliary shunts; scattered haemorrhages, suggestive of old ischaemic CRVO.”

Q Why dilate with a cycloplegic when the pressure is high?

In NVG, **cycloplegia is a treatment**, even in the angle-closure stage with corneal oedema. It is both therapeutic and diagnostic.

Q What will you look for in the fundus?

The disc: hyperaemia and oedema if acute; pallor and cupping if chronic. Then the cause: **ischaemic central retinal vein occlusion, PDR, CRAO**, or the mid-peripheral haemorrhages of ocular ischaemia.

Q You recorded optociliary shunts. What are they?

Optociliary shunt vessels are abnormal vessels on the disc. They carry blood from the retinal to the choroidal circulation, a sign of good compensatory circulation after an old CRVO.

Step 7 – The fellow eye

Do: Repeat the undilated pupil margin, tonometry and gonioscopy in the other eye, then its fundus.

Record: “LE: no NVI; scleral spur seen in all quadrants; Tn 14 mm Hg; mild non-proliferative diabetic retinopathy.”

Q Why examine the other eye with the same care?

Diabetes and hypertension affect both eyes. CRVO is also associated with open-angle glaucoma, so I check its pressure and angle. The **fellow eye** has a CRVO risk of 1% a year, 7% within 5 years.

Q What would make you treat the fellow eye now?

Proliferative diabetic retinopathy: it is an indication for prophylactic PRP before rubeosis appears.

Summary and complete diagnosis

Mr ___, ___ years, diabetic / hypertensive, has pain, redness and poor vision in the ___ eye for ___. This followed a sudden / gradual loss of vision ___ months ago. That eye shows vision ___, congestion, corneal oedema and flare. New vessels run from ___ to ___ o’clock, with ectropion uveae and RAPD. Tension by GAT is ___ mm Hg at ___. Gonioscopy shows NVA with PAS over ___°. The fundus shows ___. The other eye shows ___.

eye → disease → stage → aetiology → complications or status → other eye

- “**Right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled intraocular pressure; left eye normal.**”
- “**Left eye neovascular glaucoma, open-angle stage, secondary to proliferative diabetic retinopathy, with pseudophakia; right eye post panretinal photocoagulation.**”
- Book formula: “Secondary closed (or open) angle glaucoma of the right eye due to neovascularisation of the angle due to ___.”

Differential diagnosis

Condition	Points for	Points against	How to tell apart
Primary acute angle closure	Painful red eye, haloes, very high IOP	A trigger; hypermetropia	No iris or angle new vessels, no fibrovascular membrane, no ectropion uveae
Uveitic glaucoma	Flare, cells, raised IOP; dilated iris vessels	Recurrent red eye on steroids	Keratic precipitates; complicated cataract and band keratopathy if chronic; no true rubeosis
Fuchs' heterochromic iridocyclitis	Fine angle vessels, spontaneous hyphaema	White, quiet eye; NVI rare	Stellate keratic precipitates; iris heterochromia
Old trauma (angle recession)	PAS, hyphaema, raised IOP	History of injury; no NVI	Gonioscopy: recessed angle, pigment clumps
Iridocorneal endothelial (ICE) syndrome	PAS, distorted iris, raised IOP	No new vessels	Corneal decompensation, corectopia, pseudopolyopia
Normal iris vessels (light iris)	Visible iris vessels	Stromal, radial, uniform	No fluorescein leak; do not cross the scleral spur

Exclude first: primary acute angle closure, because its miotics harm an NVG eye. The undilated pupil margin and gonioscopy decide.

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Iris fluorescein angiography or fluorophotometry	Earliest sign of rubeosis	Leakage at the pupil margin	In a diabetic before lensectomy or vitrectomy, it calls for prophylactic PRP
Fundus fluorescein angiography (FFA) , once haemorrhages clear	The cause; extent of ischaemia	Non-perfusion over 10 disc areas = ischaemic CRVO; ocular ischaemia: arm-to-retina time over 20 s	Confirms and maps the retina that needs PRP
B-scan ultrasonography , when the fundus is hidden	Retina attached? A mass?	Vitreous haemorrhage, traction or detachment, tumour	Haemorrhage or traction: vitrectomy with PRP; mass: tumour work-up
Carotid Doppler	Ocular ischaemic syndrome	Internal carotid stenosis	Vascular referral; carotid angiography is the gold standard
Blood pressure ; fasting and post-meal sugar, glycated haemoglobin (HbA1c), lipids, urea, creatinine, electrocardiogram	Diabetic control; risk factors; fitness for drugs and surgery	Poor control; hypertension	Physician care; kidney and heart checked before acetazolamide and mannitol
Thrombophilia screen, ESR (under 60 years)	Hypercoagulable state; giant cell arteritis	Abnormal clotting; ESR over 60 mm/hr	Treat the systemic cause
Optical coherence tomography (OCT) of macula and nerve fibre layer; fields when vision allows	Macular oedema; glaucoma baseline	Cystoid macular oedema; thin nerve fibre layer	Macular oedema adds anti-VEGF; baseline for progression

Say: "I will do fluorescein angiography to confirm the cause, and B-scan if the media are hazy. Carotid Doppler and blood tests will find the systemic cause."

How I will manage

Aim

Control the intraocular pressure, stop further new vessels, treat the cause and relieve pain.

The ladder

- Lower the pressure now.** IOP below 40 mm Hg: timolol 0.5% twice daily, brimonidine 0.2% and dorzolamide 2% three times daily. Above 40 mm Hg, I add intravenous mannitol 20%, 1–2 g/kg over 20–30 minutes, if heart and blood pressure allow. Oral acetazolamide 250 mg two to four times a day if there is no sulpha allergy or vomiting.
- Settle the eye.** Atropine 1% (or homatropine 2%) three times daily and prednisolone acetate 1% 6–8 times a day ease inflammation and pain. I avoid miotics, prostaglandin analogues and adrenaline.
- Find the cause** — trigger: the cornea clears. Gonioscopy and a dilated fundus are then a must.
- Anti-VEGF.** Intravitreal **bevacizumab 1.25 mg in 0.05 ml** through the pars plana. Iris new vessels regress in 24–48 hours, but the effect lasts only weeks. So it is an adjunct to PRP, and I give it **2–3 days before surgery**.

5. **PRP — the treatment of choice.** Trigger: ischaemic retina and synechial closure not more than 270°. Settings: 532 nm, 500 µm and 0.1 s. Burns are mild white, one burn-width apart, out to the equator. I give **1200–1600 burns in 3–4 sittings**, 3–4 weeks apart.
6. **Hazy media.** Indirect laser, trans-scleral panretinal cryotherapy, or bevacizumab alone. With vitreous haemorrhage: anti-VEGF, then vitrectomy with PRP and the glaucoma operation.
7. **Surgery** — trigger: IOP stays high after retinal ablation, from extensive PAS.
 - Useful vision (rare): anti-VEGF, then trabeculectomy with mitomycin C (MMC) 0.2–0.5 mg/ml for 1–5 minutes. The alternative is 5-fluorouracil (5-FU) 50 mg/ml for 5 minutes, or 5 mg subconjunctival injections after surgery. A **glaucoma drainage device** (Ahmed valve) is often the primary operation when filtering surgery is likely to fail.
 - Poor vision with pain: **cyclophotocoagulation** (diode, or neodymium:yttrium-aluminium-garnet (Nd:YAG) laser) or cyclocryotherapy.
 - Blind, painful eye: retrobulbar alcohol injection, or enucleation after counselling.
8. **Treat the systemic cause** with the physician. Severe symptomatic carotid stenosis goes for endarterectomy.

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Timolol 0.5% · twice daily	Beta-blocker: less aqueous production	Ocular allergy, punctate erosions; bradycardia, hypotension, bronchospasm	Asthma, chronic airway disease, heart failure, heart block
Brimonidine 0.2% · three times daily	Alpha-2 agonist: less aqueous, more uveoscleral outflow	Allergic blepharoconjunctivitis, dry mouth, somnolence	Infants and children
Dorzolamide 2% · three times daily	Inhibits carbonic anhydrase II in the ciliary epithelium	Burning, stinging, punctate keratitis	Sulpha allergy, renal failure, chronic liver disease
Acetazolamide 250 mg · two to four times a day, oral	Carbonic anhydrase inhibitor	Paraesthesia, acidosis, low potassium, renal stones, aplastic anaemia	As dorzolamide; vomiting
Mannitol 20% · 1–2 g/kg intravenously over 20–30 minutes	Raises plasma osmolarity; draws water from the vitreous	Diuresis, dehydration, heart failure, pulmonary oedema; rebound IOP	Anuria, dehydration, cardiac decompensation, hypotension
Atropine 1% · three times daily	Cycloplegic: rests the iris and ciliary body; the strongest, longest-acting cycloplegic	Mydriasis 7–10 days, cycloplegia 6–12 days; flushing, tachycardia, fever, delirium	None here: it treats NVG
Prednisolone acetate 1% · 6–8 times a day	Steroid: controls inflammation	Raised IOP, cataract, infection	Infection; watch IOP

Avoid: miotics (more inflammation and pain); prostaglandin analogues (the membrane blocks uveoscleral outflow); adrenaline. Full profiles: card GX.

Laser and surgery in this case

Procedure (indication)	Key steps	Advantages	Disadvantages and complications
Intravitreal anti-VEGF (any stage; before PRP or surgery)	Bevacizumab 1.25 mg/0.05 ml; 30 gauge needle 4 mm from the limbus (phakic)	Regression in 24–48 hours; less bleeding at PRP and surgery	Lasts only weeks; an adjunct
PRP (ischaemic retina; closure not more than 270°)	532 nm, 500 µm, 0.1 s; 1200–1600 burns in 3–4 sittings	Treats the cause; vessels regress in 4–6 weeks	Night-vision and field loss, worse macular oedema; one sitting risks angle closure
Trabeculectomy with MMC (useful vision; after anti-VEGF)	Wet-field cautery; MMC 0.2–0.5 mg/ml for 1–5 minutes	Larger IOP fall than a device; success up to 95% with pre-operative bevacizumab or PRP	Poor alone; hyphaema; hypotony, bleb leak and infection; fibrosis
Glaucoma drainage device — Ahmed, Molteno, Baerveldt (failure-prone eyes)	Plate on the sclera; tube into the chamber, away from PAS	Comparable to trabeculectomy in NVG; valved device is one step	Smaller IOP fall; hyphaema, hypotony, tube block, corneal oedema, tube erosion
Cyclophotocoagulation — diode or Nd:YAG (pain, poor vision)	Transscleral diode: 1500–2000 mW for 1.5–2 s; 12–24 burns behind the limbus over 360°, sparing 3 and 9 o'clock	Quick, no incision; repeatable after a month	Uveitis, hypotony, phthisis, sympathetic ophthalmia

Follow-up and counselling

- At every visit: the undilated pupil margin and angle, the IOP and the disc, in both eyes.
- After PRP, new vessels should regress by 4–6 weeks; I add laser if they persist.
- NVG is potentially blinding; useful vision is rarely regained once the angle closes.
- Control sugar and blood pressure; return at once if pain or blurring recurs.

What you must know

Definition

“**Neovascular glaucoma** is a severe form of secondary glaucoma characterised by the proliferation of fibrovascular tissue in the anterior chamber angle, giving rise to highly elevated IOP. NVG develops as a result of aggressive iris neovascularisation and angle neovascularisation due to long-standing severe retinal ischaemia.”

In plain words: a starved retina makes growth factors, and new vessels close the drain.

- **Rubeosis iridis**: new vessels on the iris surface, regardless of the angle or the presence of glaucoma.
- **Neovascularisation of the angle**: new vessels from the iris root, across the scleral spur, onto the meshwork.
- **Ectropion uveae**: out-rolling of the posterior pigment epithelium of the iris at the pupillary margin.
- **Ocular ischaemic syndrome**: the ophthalmic features of chronic hypoperfusion of the entire arterial supply to the eye. The block lies in the carotid or ophthalmic artery.
- **100-day glaucoma**: NVG about 3 months after a CRVO. Other names: haemorrhagic, thrombotic, rubeotic, congestive glaucoma.

Causes — most common first. The top three are diabetic retinopathy, ischaemic CRVO and ocular ischaemic syndrome; in India, chronic angle closure too.

- **Retinal vascular**: diabetic retinopathy, ischaemic CRVO, CRAO, branch occlusions, sickle cell, Eales', Coats', retinopathy of prematurity.
- **Extraocular vascular**: carotid obstruction, giant cell arteritis, Takayasu disease, carotid-cavernous fistula, heart failure.
- **Inflammatory**: chronic uveitis, chronic retinal detachment, endophthalmitis, Vogt-Koyanagi-Harada disease.
- **Tumours**: malignant melanoma, retinoblastoma, metastases.
- **After surgery**: cataract extraction in diabetics, vitrectomy, detachment surgery, Nd:YAG capsulotomy, radiation.

Classification

Clinicopathological stages:

Stage	NVI and gonioscopy	IOP	Other signs
Prerubeosis	No NVI or NVA; open angle	Normal	Iris angiography shows early leakage
Pre-glaucoma (rubeosis iridis)	NVI at the pupil margin; open angle, early NVA	Normal	Usually none
Open-angle glaucoma	Florid NVI; open angle, florid NVA under a fibrovascular membrane	Raised	Cells, flare, hyphaema
Angle-closure glaucoma	Vessels less apparent; PAS up to total closure	Persistently high; can reach 60 mm Hg	Ectropion uveae, glistening iris, corneal oedema, disc damage

Some add a **burnt-out stage**: total synechial closure, few visible vessels.

Wand's classification: stage 1 pupillary margin · 2 collarette · 3 angle · 4 crossing the scleral spur.

Pathogenesis

- **Retinal hypoxia** releases **VEGF**, fibroblast growth factor, tumour necrosis factor- α and angiogenin.
- Lens, capsule and vitreous are barriers and give vaso-inhibitory factors; surgery removes them.
- Vessels bud from capillaries of the **minor arterial circle** at the pupil margin. Thin-walled and fenestrated, they leak and bleed.
- An invisible **fibrovascular membrane** covers the meshwork: open-angle (pretrabecular) glaucoma.
- Its myofibroblasts contract: PAS, flat iris, ectropion uveae, then **synechial angle closure**.

Complications

- **Disease**: hyphaema, vitreous haemorrhage, optic atrophy. Late: **painful bullous keratopathy**, complete synechial closure, intractable glaucoma.
- **Treatment**: PRP (field loss, macular oedema); surgery (hyphaema, hypotony, failure); cyclodestruction (phthisis, sympathetic ophthalmia).

Prognosis

- **Good**: early stage; synechial closure not more than 270°; prompt PRP with anti-VEGF; well-treated ischaemia, which can leave working vision.
- **Poor**: delayed diagnosis; extensive PAS; optic atrophy. Useful vision is then rare, and trabeculectomy alone does poorly.

Recent advances

- (extra) Intracameral bevacizumab alone can stabilise NVI and control IOP in the early open-angle stage.

Viva questions

Why did you ask...?

- Q Why did you ask about a sudden loss of vision three months ago?**
Ischaemic central retinal vein occlusion: NVI appears in about half, usually at 2–4 months. NVG follows at 3–5 months, the 100-day glaucoma.
- Q Why did you ask how long he has had diabetes?**
Diabetic retinopathy is the commonest cause. Longer duration, hypertension and high cholesterol raise the NVG risk in a diabetic eye.
- Q Why did you ask about cataract surgery or vitrectomy?**
Lens, capsule and vitreous are **mechanical barriers** to angiogenic factors. Capsule rupture, aphakia and vitrectomy remove them.
- Q Why did you ask about blackouts of vision?**
Amaurosis fugax suggests carotid disease and **ocular ischaemic syndrome**, one of the three commonest causes.

Diagnosis and differentials

- Q What is the first sign of NVG?**
Clinically, **new-vessel tufts at the pupillary margin**. Fluorescein angiography shows leakage there even earlier.
- Q What are the theories of new-vessel formation in NVG?**
Retinal hypoxia releases angiogenic factors. Tumours release tumour angiogenesis factor. Hypoxic iris vessels dilate chronically. Vitrectomy and lensectomy remove vaso-inhibitory factors.
- Q Where do the new vessels arise?**
At the **pupillary margin**, from capillaries of the **minor arterial circle**, as endothelial buds that become tufts.
- Q What is Wand's classification?**
Wand's classification stages vessels by reach: 1 pupillary margin, 2 collarette, 3 angle, 4 crossing the scleral spur.
- Q Which types of glaucoma occur in NVG, and what closes the angle?**
First **open-angle (pretrabecular)**, as the **fibrovascular membrane** covers the meshwork. Later, its **myofibroblasts** contract and pull the iris into **peripheral anterior synechiae**. This is **secondary angle closure without pupillary block**.
- Q Will a laser iridotomy open this angle?**
No. There is **no pupillary block**: the contracting membrane pulls the iris forward. Neovascular glaucoma is a contraindication to laser iridotomy and to laser trabeculoplasty.
- Q What symptoms suggest ocular ischaemic syndrome?**
Ocular angina, a dull brow ache (40%), with visual loss over weeks to months. Also slow recovery after bright light, amaurosis fugax (10%) and transient neurological deficits.
- Q Why are iris new vessels commoner than disc new vessels after CRVO?**
The retina is too ischaemic to grow vessels. The **anterior-segment capillaries** are still viable, so they respond.
- Q How often does ischaemic CRVO lead to NVG?**
Within 6 months: NVI 49%, NVA 37%, **NVG 29%**. The NVG risk reaches 45% at 3 years.
- Q How do you tell ischaemic from non-ischaemic CRVO?**
Ischaemic CRVO has vision of counting fingers or worse and a marked RAPD, over 0.7 log unit. Haemorrhages and disc and macular oedema are more marked, and cotton-wool spots more numerous. Fluorescein angiography shows non-perfusion over **10 disc areas**, and the electroretinogram is always depressed.
- Q How do you tell ocular ischaemic syndrome from CRVO and diabetic retinopathy on the fundus?**
In **ocular ischaemic syndrome**, the veins are dilated but not tortuous, with dot and blot haemorrhages in the mid-periphery. The disc is normal; angiography shows delayed choroidal filling. CRVO has tortuous veins, flame haemorrhages throughout and a swollen disc. Non-proliferative diabetic retinopathy is always bilateral, with hard exudates and posterior-pole haemorrhages.

Investigations

- Q Why do new vessels leak fluorescein?**
They have **gap junctions** between endothelial cells and **fenestrations**. Normal vessels have three coats and an intact blood-aqueous barrier.
- Q How will you confirm ocular ischaemic syndrome?**
Carotid colour Doppler; carotid angiography is the gold standard. FFA shows an arm-to-retina time over 20 seconds.

Management — drugs

- Q Which drugs are contraindicated, and why?**
Miotics increase inflammation and pain. **Prostaglandin analogues**: the membrane blocks uveoscleral outflow. **Adrenaline** is also contraindicated.
- Q Why atropine and a steroid in a glaucoma?**
They reduce **inflammation and pain**. Cycloplegia treats NVG even in the angle-closure stage.

Q Why not oral glycerol?

Glycerol is not given to a diabetic, and diabetes is the commonest cause of NVG. I use **intravenous mannitol**.

Q What do you tell the patient after intravenous mannitol?

Not to get up at once: it can cause hypotension and, rarely, coning of the brain.

Q What is the role of anti-VEGF?

Bevacizumab 1.25 mg in 0.05 ml, intravitreal or intracameral, regresses NVI in 24–48 hours. It cuts bleeding at PRP and surgery, but lasts only weeks. It cannot replace PRP: when it stops, new vessels return more strongly.

Q Which anti-VEGF agents and doses, and how do they differ?

Bevacizumab 1.25 mg/0.05 ml is a complete antibody, off-label and much cheaper; it is the most reported in NVG.

Ranibizumab 0.5 mg/0.05 ml is a humanised antibody fragment against all isoforms of VEGF-A. **Aflibercept** 2 mg/0.05 ml is a fusion protein that also binds VEGF-B and placental growth factor.

Management — laser and surgery

Q What is the treatment of choice, and how does it work?

Panretinal photocoagulation. Destroying retinal pigment epithelium and photoreceptors cuts oxygen demand, so less VEGF is released.

Q Why 3–4 sittings, not one?

One sitting can worsen **macular oedema** and cause pain, exudative or choroidal detachment and **angle-closure glaucoma**.

Q When will you do prophylactic PRP?

In PDR, and in diabetics with peripupillary leakage before lensectomy or vitrectomy. In ischaemic CRVO, the **Central Vein Occlusion Study** advised prompt PRP once NVI or NVA appears, not prophylactic PRP. I give it early only when follow-up is unreliable or ischaemia is extensive.

Q Which operation would you choose?

Useful vision: trabeculectomy with MMC after anti-VEGF, or a **glaucoma drainage device**. Poor vision with pain: **cyclophotocoagulation**. Blind: retrobulbar alcohol or enucleation.

Q Why does trabeculectomy do poorly in NVG, and how do you help it?

NVG is a poor-prognosis eye for filtering surgery and a high-risk indication for an antimetabolite. Alone, trabeculectomy does poorly. Pre-operative anti-VEGF cuts failure, bleeding and inflammation; with MMC and PRP, success may reach 95%.

Q When is cyclodestruction indicated?

Pain with poor visual potential, with rubeosis and glaucoma that cannot be controlled.

Q How do you give a retrobulbar alcohol injection?

Retrobulbar alcohol relieves pain for 3–6 months. I first inject 2–3 ml of lignocaine behind the globe. Keeping the needle in place, I change to a 1 ml syringe of 95–100% alcohol. Expect transient ptosis, lid swelling and restricted eye movements.

Complications

Q What are the late complications of NVG?

Painful bullous keratopathy, complete synechial angle closure and intractable glaucoma.

Q What are the complications of cyclophotocoagulation?

Uveitis, **hypotony**, phthisis, malignant glaucoma and **sympathetic ophthalmia**.

Recent advances

Q What newer treatments exist? (extra)

Anti-VEGF antibodies such as bevacizumab, alpha-interferon, troxerutin and pigment epithelium-derived factor gene transfer.

Examiner traps

- Any drop before seeing the pupil margin and angle: early NVI is missed.
- No gonioscopy because the iris is clean: NVA occurs alone in 12%.
- Treating it as primary angle closure: miotics and prostaglandins are contraindicated.
- Lowering pressure without naming the cause: PRP is the treatment of choice.
- No stage or PAS extent: they decide whether PRP can still work (closure not more than 270°).
- Forgetting the other eye: it may need prophylactic PRP.
- Calling radial stromal vessels in a light iris “rubeosis”.
- Offering laser iridotomy for the closed angle: there is no pupillary block.

Short-case version

Rubeosis iridis

The instruction you may get: “Examine this patient’s right iris.”

Spot diagnosis: rubeosis iridis — fine new vessels at the pupillary margin and on the iris surface.

Focused examination:

1. No drops. Torch: pupil reactions and RAPD.
2. Slit lamp, high magnification, undilated: the **pupillary margin first**, for tufts and ectropion uveae.
3. Then the **iris surface** to the root, and any iridotomy margin; count the clock hours.
4. **Gonioscopy**: new vessels crossing the scleral spur; PAS.
5. **IOP** by Goldmann applanation: normal means the pre-glaucoma stage.
6. Ask to dilate and examine the fundus for the cause.

SAY IT

On slit-lamp examination of the right eye, before dilatation, the iris shows new vessels. They are thin in appearance and friable in character. Their course is arborising. They are irregularly arranged, not radial like normal vessels. They start at the pupillary margin and extend from 11 o'clock to 8 o'clock. There is no ectropion uveae. On gonioscopy, new vessels cross the scleral spur, without peripheral anterior synechiae. The pressure is 16 millimetres of mercury by Goldmann applanation. These features are suggestive of neovascularisation of the iris. My diagnosis is right eye rubeosis iridis, pre-glaucoma stage, probably due to proliferative diabetic retinopathy.

Three signs to show:

1. New-vessel **tufts at the pupillary margin**.
2. **Irregular, branching, non-radial** surface vessels, with their extent in clock hours.
3. **New vessels crossing the scleral spur** on gonioscopy.

Q Define rubeosis iridis.

Rubeosis iridis means new vessels on the iris surface, regardless of the state of the angle or the presence of glaucoma.

Q Is rubeosis iridis the same as NVG?

No. **Rubeosis iridis** with normal pressure is the pre-glaucoma stage. It becomes NVG when the pressure rises.

Q What does "PART B" remind you to do?

Before dilating, check the **Pupillary reflex, Angle, Rubeosis, Tension** and **Best-corrected visual acuity**.

Q What else can cause iris new vessels?

Chronic uveitis, chronic retinal detachment and intraocular tumours such as melanoma and retinoblastoma.

Q How will you manage this patient?

I will find the cause on fundus examination and FFA, then give **PRP**, with anti-VEGF as an adjunct.

Say-it script

Opening (about 1 minute)

SAY IT

My patient, Mr X, is a sixty-two-year-old man. He came with pain, redness and defective vision in the right eye for ten days. He was apparently normal three months ago, when he had a sudden, painless loss of vision in the right eye. Ten days ago the eye became painful and red, with coloured haloes, headache and vomiting. There was no precipitating factor such as dim light or dilating drops.

There is no history of injury or of recurrent red eye treated with steroid drops. He has had no eye surgery, retinal laser or injections. He has had no brief blackouts of vision and no stroke. There have been no similar attacks in either eye.

He is a known diabetic for twelve years, poorly controlled on tablets, and hypertensive for eight years, on treatment. He has no heart disease, no kidney disease, no asthma and no sulpha allergy. He was started on timolol eye drops twice daily five days ago.

Closing (2 minutes)

SAY IT

To summarise, a sixty-two-year-old diabetic and hypertensive man has a painful, red right eye with poor vision for ten days. This followed a sudden loss of vision in that eye three months ago.

The right eye has hand-movements vision, circumciliary congestion, mild corneal oedema and flare. Before dilatation, I saw thin, friable, arborising, irregular new vessels from eleven to eight o'clock. There is ectropion uveae and a relative afferent pupillary defect. By Goldmann applanation at ten in the morning, the pressure is forty-eight millimetres of mercury in the right eye. It is fourteen in the left. Gonioscopy shows peripheral anterior synechiae above and below, with new vessels elsewhere. The fundus shows an old ischaemic central retinal vein occlusion.

My diagnosis is right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled pressure. The left eye has mild non-proliferative diabetic retinopathy.

I will lower the pressure with timolol, brimonidine, dorzolamide, oral acetazolamide and intravenous mannitol. I will add prednisolone acetate and atropine, and avoid miotics. Once the cornea clears, I will do fluorescein angiography and carotid Doppler. I will give intravitreal bevacizumab, then panretinal photocoagulation. If the pressure stays high, I will advise a glaucoma drainage device or trabeculectomy with mitomycin C. If vision is poor and the eye painful, I will offer cyclophotocoagulation.

Quick recall

- Top three causes: **diabetic retinopathy, ischaemic CRVO, ocular ischaemic syndrome**.
- Ischaemic CRVO: NVI at 2–4 months; NVG at 3–5 months, the “**100-day glaucoma**”.
- Vessels start at the **pupillary margin**; NVA without NVI in **12%**.
- Stages: prerubeosis → pre-glaucoma → open-angle → angle-closure.
- Above **40 mm Hg**: mannitol 20%, 1–2 g/kg intravenously over 20–30 minutes.
- **Avoid** miotics, prostaglandin analogues, adrenaline.
- **Bevacizumab 1.25 mg/0.05 ml**: regression in **24–48 hours**; **2–3 days** before surgery.
- **PRP**: 500 µm, 0.1 s, **1200–1600 burns in 3–4 sittings**; closure **not more than 270°**.

Mnemonic: RUBEOTIC (causes) — Retinopathy, Retinal vein occlusion, Retinal detachment • Uveitis • Branch retinal vein occlusion • Eales' disease • Ocular ischaemic syndrome • Trauma • Intraocular tumours • Carotid-cavernous fistula.

Mnemonic: ABCD (medical treatment) — Atropine • Beta-blockers • Cidamex (oral acetazolamide) • Dexamethasone (topical steroid).

Mnemonic: PART B (before dilating) — Pupillary reflex • Angle • Rubeosis • Tension • Best-corrected visual acuity.

Read more: Baidya p.157–159, 166, 175–180, 231, 234–235, 237, 251, 264–273, 277, 286, 437–438, 441, 448, 578, 626–627 • Namrata p.192–197, 219–222, 230, 235, 283 • Aravind 3.1, 3.7, 4.1, 4.5, 4.6, 4.9, 4.15, 4.17, 4.18, 4.19, 4.20, 4.21, 7.15, 7.16

G6 Pseudoexfoliation syndrome and pseudoexfoliation glaucoma

★ Kept once at JEH last year (with POAG) · Long case and short case

At a glance

- **Pseudoexfoliation syndrome (PXF)** is an age-related systemic disease. White, flaky material coats the pupillary margin, lens capsule, angle and cornea. **40%** develop glaucoma or ocular hypertension.
- **Pseudoexfoliation glaucoma (PXG)** is a secondary open-angle glaucoma, more aggressive than primary open-angle glaucoma (POAG). Kept last year as “POAG with pseudoexfoliation glaucoma”; also a short case.
- Examiners test the **pupillary margin before dilatation**, the four points, the **three zones**, **Sampaolesi’s line** and the **cataract-surgery precautions**.

Keywords the examiner listens for: pupillary margin · moth-eaten · transillumination defects · poor mydriasis · three zones · phacodonesis · zonular dialysis · Sampaolesi’s line · blotchy pigmentation · diffuse neuroretinal rim · diurnal variation · laser trabeculoplasty · capsular tension ring · true exfoliation

History — positive and negative, with reasons

Chief complaint: “Mr X, 68-year-old male, has come for regular follow-up.” Many patients have no symptoms.

History of present illness (house wording): “Patient was apparently normal ___ years before. Then he had defective vision in the ___ eye, gradual in onset, progressive, not associated with pain or headache. An ophthalmologist found increased pressure and started eye drops. Then he came here for follow-up.”

Positive history

Question	Why I ask — what the answer tells me
How old are you?	PXF starts after 50, mostly in the 7th decade. PXG usually starts after 60; POAG at 40–50.
Which eye? Is the other eye affected?	PXG is usually unilateral; POAG is bilateral. So glaucoma in one eye favours PXG.
Did the vision fall gradually or suddenly?	Gradual: nuclear sclerosis (commoner in PXF) or glaucoma. Sudden: an acute pressure rise, or a central retinal vein occlusion.
Which drops, how many, since when? Used regularly?	One drug is usually not enough in PXG. Drops for over 3 years call for mitomycin C at trabeculectomy.
Any side effects from the drops?	Breathlessness on timolol, or a dry mouth on brimonidine, changes my choice of drug.
Any laser or operation for glaucoma?	Laser trabeculoplasty helps in PXF but wears off sooner. A trabeculectomy means I assess the bleb (card G2).
Any cataract operation in either eye? Any complication?	PXF raises the risk of zonular dialysis, vitreous loss and posterior capsular rent. The intraocular lens (IOL) can dislocate late.

Negative history

Question	A “yes” would point to	How to tell them apart
No headache with vomiting and pain?	Acute angle closure: primary, or a lens pushed forward by weak zonules	Gonioscopy shows a closed angle.
No coloured haloes?	Corneal epithelial oedema from a pressure rise; also incipient cataract, mucus or vitreous opacities	Fincham (stenopaic slit) test: the glaucoma halo stays intact; the cataract halo breaks up.
No redness?	An acute rise of intraocular pressure (IOP), seen in a few PXF patients; or uveitic glaucoma	Uveitic glaucoma shows an anterior chamber reaction and other signs of uveitis. PXF shows flakes on the pupillary margin and capsule.
No watering?	Drop side effects: lacrimation from pilocarpine, allergy to brimonidine, or surface toxicity of long-term drops	Fluorescein 1% staining shows surface toxicity; lid and conjunctival allergy points to the drug.
No pain?	An acute pressure rise	Tonometry shows a very high pressure.
No injury to the eye?	Angle-recession glaucoma; traumatic subluxation or iris defects	Gonioscopy shows a recessed angle.
No herpes of the eye or forehead?	Post-herpetic iris transillumination defects	PXF defects lie at the pupillary margin, beside flakes.
No glass blowing or infrared exposure?	True exfoliation from heat or infrared	Clear scrolls split off the capsule; glaucoma is infrequent.
No short sight since youth?	Pigment dispersion syndrome	Mid-peripheral defects, a deep chamber and a uniform dark band.

Question	A “yes” would point to	How to tell them apart
No frequent change of glasses?	Accommodation impaired by high pressure, or field loss taken for poor vision	Perimetry shows the field defect.

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Heart attack?	Myocardial infarction is linked with PXF.
Heart failure, heart block or a slow pulse?	Each rules out a beta-blocker.
Stroke?	Cerebrovascular events are linked with PXF.
High blood pressure?	Hypertension is linked with PXF, and is weighed before surgery.
Diabetes?	It is weighed before surgery.
Aortic aneurysm?	Linked with PXF through abnormal basement membrane.
Asthma or chronic obstructive pulmonary disease (COPD)?	No beta-blocker: it causes bronchospasm.
Sulpha allergy?	No carbonic anhydrase inhibitor (dorzolamide, acetazolamide).
Kidney or liver disease?	Renal failure and chronic liver disease also rule out carbonic anhydrase inhibitors.
Steroid use: which, and how long?	Steroids can raise the pressure.
Treatment: “E/D (eye drops) timolol 0.5% BD (twice daily) for 3 years”	I record drug, strength, frequency, duration, compliance and side effects.
Glaucoma or PXF in the family?	A genetic predisposition is often present.

Examination — must-do steps and their viva

General and systemic (this case only)

- **Pulse and blood pressure:** bradycardia, heart block or hypertension change the drug and the fitness for surgery.
- **Heart, nervous system, chest:** old infarction or stroke, as PXF is systemic. Heart failure, or a wheeze of asthma or COPD, rules out beta-blockers.

Ocular examination — RE | LE template

Columns: right eye (RE), left eye (LE); extraocular movements (EOM) follow visual acuity.

Structure	Look for in this case	How to record (example)
Visual acuity, EOM	Nuclear sclerosis; pinhole (PH)	6/24, PH 6/12; full
Tension (Tn)	Goldmann applanation tonometry (GAT), timed; asymmetry ≥ 4 mm Hg is significant	Tn by GAT 18 mm Hg at 10 am
Lids, conjunctiva	Lids normal for age; whitish powdery deposits	N; clear
Cornea	Flakes and pigment on the endothelium; guttata	Flakes on endothelium
Anterior chamber	Van Herick (VH) grade; unequal depth; flare	VH grade III
Iris	Flakes; defects at the margin; iridodonesis	PxF at pupillary margin
Pupil	Flecks on the margin; moth-eaten margin	RTL 3 mm (round, reacting to light)
Lens	Three zones (dilated); nuclear sclerosis (NS); phacodonesis	NS grade 2; PxF in ALC (anterior lens capsule)
Gonioscopy	Blotchy pigment; Sampaolesi's line	Cross diagram (step 5)
Fundus	Cup–disc ratio (CDR); diffuse neuroretinal rim (NRR) loss	CDR 0.8, diffuse NRR thinning

Step 1 — Visual acuity

Do: Distance acuity in each eye, with pinhole, then near. Tell him: “Cover one eye and read the smallest line you can.”

Record: “RE 6/24, PH 6/12; LE 6/9.”

Q Why is vision often worse in PXG than in POAG?

Nuclear sclerosis is more common in PXF, so the loss of vision is more marked. The glaucoma is also more severe, with greater field loss.

Q What can cause sudden loss of vision in this patient?

An **acute rise of pressure** with corneal oedema, or a central retinal vein occlusion, which is linked with PXF. After trabeculectomy in advanced glaucoma, a sudden loss of vision called **snuff-out** can occur.

Step 2 — Pupils, before any drop

Do: In dim light, note size, shape and symmetry (pupils may be unequal in PXF). Test direct and consensual reactions, and look for a relative afferent pupillary defect (RAPD).

Record: "RE: 3 mm, round, sluggish, RAPD present. LE: 3 mm, RTL."

Q Why examine the pupils before any drop?

Dilating drops abolish the reactions and hide the margin signs. Mnemonic PART B: before dilating, check the pupillary reflex, angle, rubeosis, tension and best-corrected acuity.

Q Why look for a relative afferent pupillary defect?

A **relative afferent pupillary defect** can be seen in advanced glaucoma. The test needs a normal fellow eye for comparison.

Step 3 — Slit lamp before dilatation

Do: Examine the cornea, chamber, iris and pupillary margin at high magnification. Grade Van Herick in both eyes. Retro-illuminate the iris. Watch the iris and lens for quivering.

Record: "RE: white flaky material at the pupillary margin as tufts; moth-eaten margin; transillumination defects at the margin; VH grade III."

Q What is the hallmark sign of PXF?

Flecks of material on the pupillary margin are the hallmark. The whitish powdery ring on the anterior capsule is the more consistent and diagnostic sign.

Q What is a moth-eaten margin, and where are the transillumination defects?

A **moth-eaten margin** is loss of the pupillary ruff with transillumination defects at the margin, from sphincter atrophy. In pigment dispersion the defects lie in the mid-periphery.

Q Why compare the chamber depth of the two eyes?

The chamber may be shallow in PXF. Unequal depth is evidence of **zonular dialysis**, which I confirm by ultrasound biomicroscopy before cataract surgery.

Step 4 — Tonometry

Do: Goldmann applanation in both eyes, after anaesthetic and fluorescein, before gonioscopy and dilatation. Note the time (card G8).

Record: "Tn by GAT: RE 18 mm Hg, LE 15 mm Hg at 10 am."

Q Why measure before gonioscopy and dilatation, and why note the time?

Gonioscopy presses on the cornea and opens the angle, so a later reading is falsely low. Dilatation raises the pressure by 4–5 mm Hg. The time is recorded because the pressure varies through the day.

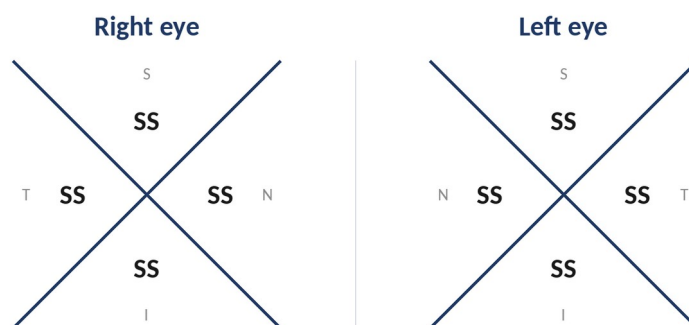
Q How does the pressure in PXG differ from POAG?

It is **higher**, often above 40 mm Hg, and harder to control. Its greater **diurnal variation** is kept in mind when setting the target pressure.

Step 5 — Gonioscopy, before dilating

Do: In a dim room, name the deepest structure seen in each quadrant (card G9). Note pigment, Sampaolesi's line and flakes. Indent if the angle looks narrow.

Record: the cross below, then "Blotchy trabecular pigmentation; Sampaolesi's line present."



Deepest structure seen in each quadrant (SS = scleral spur): open angles in all quadrants

Q What is Sampaolesi's line? Is it specific?

Sampaolesi's line is a dark, dense, scalloped band of pigment on or anterior to Schwalbe's line. It is not specific: pigment dispersion and chronic inflammation also show it.

Q How do you tell Sampaolesi's line from a pigmented meshwork?

Sampaolesi's line is dark, granular and discontinuous ("salt and pepper"). A pigmented meshwork is fine, continuous and "brown sugar" in colour.

Q Why are 9–18% of PXF angles occludable?

Weak zonules let the lens-iris diaphragm move forward, and the cataract thickens the lens. So I do gonioscopy before dilating.

Step 6 – Dilatation and the anterior lens capsule

Do: Dilate only after gonioscopy shows the angle is not occludable. Note how wide the pupil goes. Examine the anterior capsule, the opacity and lens stability.

Record: “RE: pupil dilates poorly to 5 mm; PxF in ALC with three zones; NS grade 2; no phacodonesis.”

Q Describe the three zones.

The **three zones** form the target sign. A translucent central disc, sometimes with curled edges, is absent in 20%. A clear intermediate zone is rubbed clean by the moving iris. A peripheral granular zone shows radial striations.

Q Why must you dilate before you call the capsule clear?

The peripheral granular zone is **seen only after dilatation**, and the central disc is absent in 20%. An undilated look can miss PXF.

Q Why note how well the pupil dilates?

Poor mydriasis is a sign of PXF, from atrophy of the iris sphincter. It warns of a difficult cataract operation needing hooks or a ring.

Step 7 – Disc

Do: Examine the disc with a +90 D lens, in the department’s disc order. If a mature cataract hides it, order a B-scan.

Record: “RE: CDR 0.8, diffuse NRR thinning, bayoneting present. LE: CDR 0.3, healthy NRR.”

Q How does the disc in PXG differ from POAG?

Cupping is usually unilateral, with **diffuse neuroretinal rim loss**; POAG prefers a sector. PXG also shows more glaucomatous optic atrophy.

Q What other fundus findings do you look for?

Central retinal vein occlusion and retinal detachment, both complications of PXF.

Step 8 – Fields

Do: Confrontation at 60 cm, one eye at a time, a finger brought in from each quadrant. Then automated perimetry (card G10).

Record: “Fields by confrontation: RE superior defect; LE full.”

Q Why is confrontation not enough?

Confrontation gives only a rough assessment. I need baseline automated perimetry, because PXG causes greater field loss than POAG.

Q How do the cataract and small pupil of PXF affect the Humphrey field?

They cause a **uniform generalised depression**. The pattern deviation probability plot stays normal in it, and reveals scotomas masked under an irregular depression.

Summary and complete diagnosis

Mr X, ___ years, on ___ drops for glaucoma in the ___ eye for ___ years. Visual acuity ___ / ___. The ___ eye shows pseudoexfoliation (four points), ___ mydriasis, ___ cataract. Tn by GAT ___ / ___ mm Hg at ___. Gonioscopy: ___ angles, Sampaolesi’s line ___. CDR ___, ___ rim thinning. Other eye: ___.

Formula: eye → disease → type / stage / grade → aetiology → complications or status → other eye.

- “**Right eye: pseudoexfoliation glaucoma with immature senile cataract, intraocular pressure controlled on two medications. Left eye: pseudoexfoliation syndrome.**”
- House style: “(RE) pseudoexfoliation glaucoma with NS grade 3 with intraocular pressure under control with medical management.”

Differential diagnosis

Condition	Points for	Points against	How to tell apart
POAG	Open angle; raised pressure; cupping	One eye; age over 60; flakes	Flakes on the pupillary margin; three zones; Sampaolesi’s line
Pigment dispersion syndrome, pigmentary glaucoma	Angle pigment; Krukenberg spindle (a spindle of pigment on the endothelium); transillumination defects	Older patient; flakes; defects at the margin	Mid-peripheral defects; posterior iris bowing; uniform dark band
True exfoliation (capsular delamination)	Material on the anterior capsule	No heat or infrared exposure; glaucoma present	A thin, clear scroll split off the capsule
Primary amyloidosis	Whitish powdery deposits; glaucoma	Flakes on the pupillary margin; target sign	Fine whitish deposits throughout the eye, both eyes
Uveitic glaucoma	Flare; pigment; posterior synechiae	No cells or keratic precipitates	Slit lamp: cells and keratic precipitates in uveitis
Angle-closure glaucoma	Shallow chamber; pressure spikes	Open angle on gonioscopy	Gonioscopy: closed angle or peripheral anterior synechiae

Exclude first: angle closure — by gonioscopy, with indentation, before dilating, because 9–18% of PXF angles are occludable.

PXG, POAG and pigment dispersion compared

Feature	PXG	POAG	Pigment dispersion
Age, sex	Over 60 (PXF after 50); both sexes	40–50 years	20–30 years; men; myopes
Laterality	Usually unilateral	Bilateral	Usually bilateral
Pressure	Often above 40 mm Hg; wide diurnal swings	Above 21 mm Hg	Rises with pigment release
Anterior segment	Flakes; defects at the margin; weak zonules	Normal	Krukenberg spindle; mid-peripheral defects
Gonioscopy	Blotchy pigment; Sampaolesi's line	Open	Wide open; uniform dark band
Treatment	Poor drug response; surgery often	Good drug response	Medical in most

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Diurnal variation (phasing): GAT every 3 hours for 24 hours	True pressure range	Wider swings than in POAG	Sets the target; 8 mm Hg or more between readings is significant
Central corneal thickness (pachymetry)	Correct the applanation reading	Thin < 500 µm; thick > 570 µm	GAT is calibrated for 520 µm. A thinner cornea reads falsely low, a thicker one falsely high, by about 0.7 mm Hg per 10 µm
Humphrey perimetry; optical coherence tomography (OCT) of the nerve fibre layer	Stage; baseline	Field loss; diffuse thinning	Grades severity; shows progression
Blood homocysteine	A risk factor	May be raised	I lower it if high
Specular microscopy	Endothelium before surgery	Low cell density; altered size and shape	Endothelium-friendly viscoelastic; warns of decompensation
Ultrasound biomicroscopy (UBM)	Phacodonesis or unequal chamber depth	Extent of zonular dialysis	Plans a capsular tension ring
Biometry, keratometry; B-scan if media hazy	IOL power; retina behind a cataract	Axial length; attached retina	IOL choice; visual prognosis
Blood pressure, blood sugar, heart	Fitness for surgery	Hypertension, diabetes or heart disease	Weighed before surgery, with the physician

Say: “I will proceed with diurnal variation of intraocular pressure and central corneal thickness to confirm the pressure. Perimetry and optical coherence tomography will stage it. Before cataract surgery I need specular microscopy, ultrasound biomicroscopy and biometry.”

How I will manage

Aim

I aim for a **target pressure** at which further damage is unlikely, allowing for the wider diurnal swing.

The ladder

- Drops.** I start latanoprost 0.005% at bedtime; it is preferred to a beta-blocker, which has systemic side effects. I review after 2 months.
- Add early.** One drug is usually not enough. Target not reached: I add timolol 0.5% twice daily, or brimonidine or dorzolamide. A fixed combination simplifies dosing, helps adherence and cuts preservative toxicity. Its drawback: the drugs cannot be timed apart.
- Laser trabeculoplasty.** If the target is still not reached, or drops cannot be used reliably, I offer selective or argon laser trabeculoplasty. It works better in PXG than in POAG, but wears off sooner.
- Trabeculectomy with mitomycin C.** If drops and laser fail to reach the target, or the glaucoma progresses (card G2).
- Cataract with glaucoma.** Intractable pressure or major field loss: combined phacoemulsification and trabeculectomy. Pressure controlled on one or two drugs: cataract surgery alone. Both need the PXF precautions.
- General.** I lower a high homocysteine and advise antioxidants. Before surgery, I weigh his diabetes and hypertension with his physician.

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Latanoprost 0.005% · at bedtime	More uveoscleral outflow (ciliary muscle relaxes)	Hyperaemia, long lashes, iris pigmentation, uveitis, cystoid macular oedema, reactivated herpes keratitis; no systemic effects	Uveitis: it breaks down the blood–aqueous barrier and increases inflammation
Timolol 0.5% · twice daily (morning dose matters most)	Beta-blocker: less aqueous production	Bradycardia, hypotension, heart failure, bronchospasm	Asthma, COPD, heart failure, heart block, bradycardia
Brimonidine 0.15% · three times, sometimes twice	Alpha-2 agonist: less aqueous, more uveoscleral outflow	Allergy, dry mouth, somnolence	Children under 2 years
Dorzolamide 2% · three times daily	Inhibits carbonic anhydrase in the ciliary epithelium	Burning, stinging	Sulpha allergy, renal failure, chronic liver disease
Acetazolamide 250 mg tablet · two to four times a day	Systemic carbonic anhydrase inhibitor	Paraesthesia, renal stones, acidosis	As dorzolamide

Full profiles: card GX.

Laser and surgery in this case

Procedure (indication)	Key steps	Advantages	Disadvantages and complications
Selective laser trabeculoplasty: open-angle PXG above target	Q-switched frequency-doubled Nd:YAG (neodymium-doped yttrium aluminium garnet) laser, 532 nm; 400 µm, 3 ns, 0.5–1.5 mJ; 50–100 spots over 180°; end point tiny bubbles	Higher success in PXG than in POAG; targets only pigmented cells, with no heat spread	Wears off sooner in PXF; pressure rise; mild uveitis. Not in uveitic glaucoma
Argon laser trabeculoplasty: the same	200–1200 mW, 50 µm, 0.1 s; 40–50 spots over 180°; end point blanching or a bubble	Control for 3–5 years; best results in PXF	Pressure rise, peripheral anterior synechiae, hyphaema
Trabeculectomy with mitomycin C: target not reached, or progression	Scleral flap; mitomycin C 0.2–0.5 mg/ml for 1–5 minutes; limbal fistula; peripheral iridectomy; aqueous drains to a bleb	Plain trabeculectomy lowers pressure by about 41%; the flap gives uniform control and less hypotony	Shallow chamber, hypotony, bleb leak, infection, cataract; scarring
Phacotrabeculectomy: intractable pressure or major field loss, with cataract	Both operations in one sitting, with PXF precautions	One operation; prompt vision; lasting pressure control	More complications (shallow chamber, choroidal effusion, hypotony); slower recovery
Cataract surgery alone: pressure controlled on 1–2 drugs	Phacoemulsification with PXF precautions	Easier, shorter, safer	Early pressure spike; later filtering surgery fails more

Follow-up and counselling

- **Review:** every 3 months with glaucoma; yearly for PXF without glaucoma. Each visit: pressure, disc and fields.
- **Drops:** after each drop, lacrimal occlusion (pressing the inner corner) or closing the eyes for 3 minutes lessens systemic absorption.
- **Counsel:** cataract surgery is riskier in PXF. The other eye needs regular review. PXF is systemic, linked with heart attack, stroke and hypertension.

What you must know

Definition

Pseudoexfoliation syndrome (PXF or PEX) “is an age-related systemic syndrome where mainly ocular tissues are affected through the gradual deposition of fibrillary white flaky material from the lens, mainly on the lens capsule, ciliary body, zonules, corneal endothelium, iris and pupillary margin resulting in secondary glaucoma.”

In plain words: with age, a white, dandruff-like material coats the front of the eye and clogs its drain. The diagnosis rests on **flakes on the pupillary margin**, the **three zones** and **Sampaolesi’s line**.

Pseudoexfoliation glaucoma is the secondary open-angle glaucoma caused by PXF. It has a high pressure at onset, progresses faster and responds poorly to medical therapy.

- **Glaucoma capsulare:** glaucoma in an eye with PXF; usually unilateral.
- **True exfoliation:** lamellar delamination of the anterior capsule from heat (glass blowing) or infrared. Clear scrolls split off and may float in the chamber (glassblower’s cataract). Goggles prevent it. Trauma, intraocular inflammation and advanced age can also delaminate the capsule.
- **Sampaolesi’s line:** a dark, scalloped band of pigment on or anterior to Schwalbe’s line.

- **Target sign:** the three zones on the anterior capsule: central disc, clear zone and peripheral granular zone.
- **Iridodonesis and phacodonesis:** quivering of the iris and of the lens, from weak zonules.
- **Target pressure:** the pressure at which further glaucomatous optic neuropathy is unlikely.
- **Ocular hypertension:** pressure above normal (often above 21 mm Hg), with open angles and no detectable glaucomatous damage.

Who gets it: over 50, mostly the 7th decade; both sexes equally; commonest in Scandinavians.

Classification

Clinical classification by the morphology of the anterior lens capsule:

Stage	Anterior lens capsule
Preclinical	Clinically invisible
Suspected PXF	A precapsular layer
Mini PXF	A focal defect that starts superonasally
Classic PXF	The three zones (target sign)

This patient has classic PXF. The glaucoma may be **open-angle** (most) or **angle-closure**.

Pathogenesis

- **The material** is a grey-white fibrillogranular matrix: fibrillin, α -elastin and laminin, coated with hyaluronic acid. Abnormal basement membranes of ageing epithelial cells are thought to make it.
- **Open angle:** material made in the meshwork swells the juxtacanalicular meshwork and narrows Schlemm's canal. Trabecular cells fail.
- **Pigment** adds to the block. The iris pigment epithelium rubs against PXF material on the lens capsule and sheds pigment. It settles on the endothelium and in the meshwork.
- **Angle closure:** weak zonules let the lens move forward. The cataract thickens the lens, and the iris sticks to it.

Systemic associations

- **Diseases:** myocardial infarction, cerebrovascular events, systemic hypertension, aortic aneurysm; a raised homocysteine.
- **Deposits:** skin, heart, lungs, liver, kidneys, gall bladder and cerebral meninges.

Why PXG is more aggressive than POAG

- The pressure is higher at onset, fluctuates more and is harder to control. Drops work less well.
- Damage progresses faster, with diffuse rim loss and greater field loss. Surgery is often needed.
- The meshwork is blocked from within, by material made in it and by shed pigment.

Cataract surgery in pseudoexfoliation

Risks: small pupil; zonular dialysis and vitreous loss; friable posterior capsule; IOL decentration or late dislocation; corneal decompensation; inflammation; pressure rise.

Problem	Precaution
Poorly dilating pupil	Atropine; stretch the pupil; iris or Kuglen hooks; a pupil expansion ring; mini-sphincterotomies; sector iridectomy
Endotheliopathy	Dispersive (Viscoat, chondroitin sulphate) or visco-adaptive (Healon 5) viscoelastic; soft-shell technique
Weak zonules	Phaco chop or prolapse of the nucleus out of the bag; capsular tension ring; 3-piece IOL; or extracapsular surgery
Inflammation after surgery	Aggressive, possibly longer, steroids; a heparin-surface-modified IOL
Pressure spike after surgery	Postoperative Diamox (oral acetazolamide)
Capsular phimosis	Nd:YAG laser relaxing incisions in the anterior capsule at the 4 cardinal positions

Complications

- **Anterior segment:** open-angle and angle-closure glaucoma; nuclear cataract, zonular dialysis, subluxation; posterior synechiae, uveitis; early endothelial decompensation.
- **Retina:** central retinal vein occlusion, retinal detachment.

Prognosis

- **Good:** your practical books give no list of good factors. They state that PXF without glaucoma needs yearly review, and that laser trabeculoplasty works better in PXG than in POAG.
- **Poor:** PXG progresses faster and responds poorly to drops (see above). Weak zonules make cataract surgery riskier.

Recent advances

- (extra) **Trabecular aspiration:** surgery that removes intertrabecular and pretrabecular debris.

- (extra) **Genetics**: a polymorphism in the LOXL1 gene (viva below). Nothing else in your practical books.

Viva questions

Why did you ask...?

Q Why did you ask about heart attack, stroke and blood pressure?

PXF is systemic: it is linked with myocardial infarction, cerebrovascular events, hypertension and aortic aneurysm.

Q Why did you ask about glass blowing?

To exclude **true exfoliation**: heat or infrared splits the anterior capsule into clear scrolls. Glaucoma is infrequent.

Q Why did you ask about previous cataract surgery?

Zonular weakness makes surgery riskier: zonular dialysis, vitreous loss and posterior capsular rent. The IOL can decentre or dislocate years later, from loss of zonular support.

Q Why did you ask about asthma and sulpha allergy?

Asthma rules out beta-blockers, which cause bronchospasm. **Sulpha allergy** rules out carbonic anhydrase inhibitors.

Diagnosis and differentials

Q How common is PXF, and how often does glaucoma follow?

Incidence is **1.5–27%**, rising with age. **40%** develop glaucoma or ocular hypertension; PXG occurs in up to 50% of eyes.

Q What are the risk factors for PXF?

Age above 70, Scandinavian origin, a genetic predisposition and a high blood homocysteine.

Q What are the corneal signs of PXF?

Flakes and pigment on the endothelium, sometimes a Krukenberg spindle, and guttata. Early endothelial decompensation is common.

Q What is the differential diagnosis of PXF?

True exfoliation, pigment dispersion, primary amyloidosis, uveitis, and melanosis or melanoma. Trauma or herpes can mimic its iris defects.

Q What are the causes of trabecular meshwork pigmentation?

PIGMENT: Pseudoexfoliation and Pigment dispersion, Iritis, Glaucoma after angle closure, Melanosis, Endocrine (diabetes, Addison's), Naevus (Cogan-Reese), Trauma.

Q How do you tell PXF from pigment dispersion syndrome?

Pigment dispersion affects young myopic men, 20–30 years, in both eyes, with mid-peripheral defects and a uniform dark band. PXF affects older people, with flakes, the target sign and **blotchy pigmentation**.

Q Why are the zonules weak in PXF?

PXF material invades the zonules at their origin and insertion; its proteolytic enzymes help them disintegrate. PXF zonules are also modified fibres.

Investigations

Q Why do you do diurnal variation of pressure in PXG?

PXG shows a **greater diurnal variation**. Phasing shows the true size of the pressure fall and guides the target pressure.

Q What does specular microscopy show?

Reduced endothelial cell density, with altered cell size (polymegathism) and shape.

Q When will you order ultrasound biomicroscopy?

When there is **phacodonesis** or unequal chamber depth before cataract surgery. It confirms the extent of zonular dialysis.

Management — drugs

Q What do you keep in mind when you treat PXG medically?

It responds less well than POAG, so I **treat aggressively** with frequent follow-up. I allow for the higher diurnal variation and consider laser.

Q Which drug will you start, and why?

A **prostaglandin analogue**, latanoprost 0.005% at bedtime. It is preferred to a beta-blocker, which causes bronchospasm and bradycardia.

Q The target is not reached on latanoprost. What will you add?

Combination therapy, because one drug is usually not enough in PXG. I add timolol 0.5% twice daily, unless asthma, heart failure or heart block forbids it. Otherwise I add brimonidine 0.15% or dorzolamide 2%.

Management — laser and surgery

Q Is laser trabeculoplasty useful in PXG, and how does it work?

Yes. **Laser trabeculoplasty** succeeds more often in PXG than in POAG, but wears off sooner. Burns shrink the meshwork and stretch the adjacent areas; cytokines raise matrix metalloproteinases.

Q When is laser trabeculoplasty contraindicated?

In **uveitic glaucoma**, neovascular glaucoma and iridocorneal endothelial (ICE) syndrome. Also in synechial angle closure and angle recession.

Q When will you do a trabeculectomy?

When drops and laser fail to reach the **target pressure**, or the glaucoma progresses. I use mitomycin C, especially after more than 3 years of drops.

Q When will you choose combined surgery over cataract surgery alone?

Combined surgery when the pressure needs many drugs, field loss is significant, or exfoliation is present. Cataract alone when one or two drugs control the pressure.

Q What will you check before cataract surgery?

The **endothelium**; zonular dialysis (phacodonesis, unequal chamber depth, ultrasound biomicroscopy); pupil dilatation; posterior synechiae; and subluxation.

Q How does a capsular tension ring help?

A **capsular tension ring** in the bag spreads any force over the whole zonular apparatus, not the weak part. It manages zonular dialysis and stabilises the bag.

Q How will you enlarge a small pupil at surgery?

I stretch it, or use **iris hooks**, Kuglen hooks or a pupil expansion ring such as the Malyugin ring. Mini-sphincterotomies also help.

Q Why chop, and why the soft-shell technique?

Phaco chop puts less stress on the zonules. In the **soft-shell** method a dispersive viscoelastic goes in first and coats the weak endothelium; a cohesive one follows.

Q How will you operate on a subluxated lens?

By its **extent in clock hours**. Up to 3: phacoemulsification. From 3 to 5: add a capsular tension ring. From 5 to 7: a Cionni ring. From 7: intracapsular extraction and anterior vitrectomy, with a scleral-fixated or anterior chamber IOL.

Complications

Q What complications do you expect during and after cataract surgery?

Poor dilatation, zonular dialysis and vitreous loss, IOL decentration, corneal decompensation, inflammation and a pressure rise. Late IOL prolapse into the posterior segment can follow.

Q How will you handle inflammation after cataract surgery in PXF?

Postoperative iritis is common in PXF, which also disrupts the blood–aqueous barrier (markedly increased flare). I give steroids aggressively and for longer, and may use a heparin-surface-modified IOL.

Recent advances

Q Which gene is linked with exfoliation syndrome? (extra)

The **LOXL1** gene: a polymorphism in exon 1.

Examiner traps

- Dilating before showing the pupillary margin and doing gonioscopy.
- Confusing pseudoexfoliation with true exfoliation (capsular delamination from heat or infrared).
- Swapping the angles: **blotchy pigment with Sampaolesi's line = PXF; uniform dark band = pigment dispersion**.
- Treating PXG like POAG: one drug, one pressure reading, no phasing.
- Offering cataract surgery without checking phacodonesis, chamber depth, dilatation and the endothelium.
- Forgetting the other eye, and that PXF is systemic.
- Calling Sampaolesi's line specific to PXF: pigment dispersion and chronic inflammation also show it.

Short-case version

Pseudoexfoliation

The instruction you may get: "Examine the anterior segment of this patient." Or: "Look at this patient's pupil and lens."

Spot diagnosis: White, flaky, dandruff-like material on the pupillary margin and the anterior lens capsule — suggestive of pseudoexfoliation.

Focused examination:

1. Before any drop: flakes and a moth-eaten pupillary margin; pupil reactions.
2. Retro-illumination: transillumination defects at the margin.
3. Iridodonesis, phacodonesis; Van Herick depth in both eyes.
4. Ask for tonometry and gonioscopy before dilating: blotchy pigment, Sampaolesi's line.
5. After dilatation: how far the pupil dilates; the three zones; the cataract. Offer the disc and other eye.

SAY IT

On slit-lamp examination of the right eye before dilatation, the colour of the material is white. Its nature is flaky, dandruff-like. It is located on the pupillary margin as tufts and on the anterior lens capsule as a disc-shaped deposit. There is associated iris atrophy at the pupil edge.

The pupillary margin looks moth-eaten, with transillumination defects at the margin. The pupil dilates poorly. After dilatation, the anterior capsule shows three zones: a central disc, a clear zone and a peripheral granular zone. There is no phacodonesis. There is nuclear sclerosis. Gonioscopy shows open angles with blotchy pigmentation and Sampaolesi's line.

These findings are suggestive of pseudoexfoliation. My diagnosis is right eye pseudoexfoliation syndrome with immature senile cataract.

Three signs to show:

1. White flecks on the pupillary margin (the hallmark), before dilatation.
2. The three-zone target sign on the anterior capsule, after dilatation.
3. Sampaolesi's line and blotchy pigment on gonioscopy.

Q Why do you look at the pupillary margin before dilating?

The **flecks on the pupillary margin** are the hallmark, and dilatation hides the margin. The angle must be checked first, too.

Q How will you show the iris transillumination defects?

By **retro-illumination**: the beam is set coaxial, then moved to the pupil edge. The fundus glow shines through the defects.

Q What does "with or without iris atrophy" mean in your description?

Sphincter atrophy at the pupil edge gives the moth-eaten margin of PXF. In pigment dispersion the atrophy lies in the iris periphery.

Q You have found PXF. What else will you examine?

The **pressure**, gonioscopy, the disc and the fields, because 40% develop glaucoma or ocular hypertension. I also examine the other eye.

Say-it script

Opening (about 1 minute)

SAY IT

Mr X, a sixty-eight-year-old man, has come for regular follow-up of glaucoma in the right eye.

He was apparently normal three years ago. Then he noticed defective vision in the right eye, gradual in onset and progressive in nature, without pain or headache. An ophthalmologist at his place found raised pressure and started eye drops. He then came to our institution for follow-up.

There is no history of headache with vomiting, coloured haloes, redness, watering or pain. There is no history of injury, herpes, glass blowing, short sight or frequent change of glasses.

He is a known hypertensive for ten years, on regular treatment. He has no asthma, heart disease, stroke or sulpha allergy, and has not used steroids. He has had no eye surgery or laser.

For three years he has used latanoprost 0.005 percent at bedtime and timolol 0.5 percent twice daily. He uses them regularly, without side effects. There is no family history of glaucoma.

Closing (2 minutes)

SAY IT

To summarise, Mr X, sixty-eight, has right eye glaucoma on two drugs for three years.

Visual acuity is 6/24 in the right eye, 6/12 with pinhole, and 6/9 in the left. The right pupillary margin shows white, flaky, dandruff-like tufts and is moth-eaten, with transillumination defects. The pupil dilates poorly; the anterior capsule shows three zones. There is an immature senile cataract, without phacodonesis. At ten in the morning, intraocular pressure by Goldmann applanation is 18 millimetres of mercury on the right. It is 15 on the left. Gonioscopy shows open angles, with blotchy pigmentation and Sampaolesi's line on the right. The right cup-disc ratio is 0.8, with diffuse neuroretinal rim loss. The left eye shows flakes on the margin and a healthy disc.

My diagnosis is right eye pseudoexfoliation glaucoma with immature senile cataract, intraocular pressure controlled on two medications. The left eye has pseudoexfoliation syndrome.

I will proceed with diurnal variation of pressure, central corneal thickness, perimetry, optical coherence tomography and specular microscopy. I will continue both drugs, and offer laser trabeculoplasty if the target is not reached. I will plan phacoemulsification with precautions for poor dilatation and weak zonules, with a capsular tension ring ready. If the field loss is significant, I will combine it with trabeculectomy.

Quick recall

- **Four points:** white · flaky, dandruff-like · tufts on the margin, disc on the capsule · iris atrophy or not.
- **Hallmark:** flecks on the pupillary margin. **Most consistent:** the whitish ring on the anterior capsule.
- **Three zones:** central disc (absent in 20%) · clear zone · peripheral granular zone (only after dilatation).
- **Defects:** at the pupillary margin in PXF; in the mid-periphery in pigment dispersion.
- **Numbers:** incidence 1.5–27% · glaucoma or ocular hypertension in 40% · occludable angle in 9–18%.
- **PXG:** over 60, usually unilateral, pressure often above 40 mm Hg, diffuse rim loss, poor drug response.
- **Laser trabeculoplasty:** works better than in POAG but wears off sooner.
- **Cataract surgery:** hooks or ring · dispersive viscoelastic · chop · capsular tension ring · 3-piece IOL · longer steroids.

Mnemonic: PIGMENT (causes of trabecular pigmentation) is expanded in the viva. **PART B** (before dilating): Pupillary reflex · Angle · Rubeosis · Tension · BCVA (best-corrected visual acuity).

Read more: Baidya p.125–127, 129, 150, 154, 156–158, 161–162, 164–169, 190, 442, 731–732, 736–737, 740 · Namrata p.174–178, 181, 198, 200–207, 342, 344, 418 · Aravind 2.4, 4.1, 4.2, 4.6, 4.8, 4.9, 4.10, 4.11, 4.15, 4.17, 4.18, 4.19, 6.6; glaucoma model case sheet p.661–667

G7 CRVO with POAG, including hemi-CRVO with POAG

★ Kept twice at JEH last year (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case) · Short case

The instruction you may get

- “Examine the fundus of this glaucoma patient and draw it.”
- “Do applanation tonometry and gonioscopy; read his Humphrey field” (card G10).

Spot

- **Superficial flame-shaped and deep dot-blot haemorrhages** radiate from the disc into all four quadrants, with **dilated, tortuous veins**. This is central retinal vein occlusion (CRVO).
- The same picture in one **hemisphere** only: hemi-central retinal vein occlusion (hemi-CRVO).
- **Glaucomatous cupping** in either eye: primary open-angle glaucoma (POAG).
- Old occlusion: **optociliary shunts**, laser scars, **epiretinal membrane**.

Keywords the examiner listens for: blood and thunder · relative afferent pupillary defect · neovascularisation of the iris · undilated gonioscopy · ischaemic CRVO · 10 disc areas · lamina cribrosa · 100-day glaucoma · panretinal photocoagulation · anti-VEGF

Focused examination — steps and their viva

Step 1 — Visual acuity

Do: Snellen distance vision in each eye, then with a pinhole.

Record: “RE (right eye) 6/36, no improvement with pinhole; LE (left eye) 6/9.”

Q Why does the presenting vision matter?

It guides type and prognosis. Counting fingers or worse suggests ischaemia; better than 6/60 is a good sign.

Step 2 — Pupils, before any drop

Do: Swinging flashlight test.

Record: “RE relative afferent pupillary defect (RAPD)” or “No RAPD.”

Q Why test before dilating, and how reliable is it here?

A dilated pupil makes it useless. A marked **relative afferent pupillary defect**, over 0.7 log unit, means ischaemia. Glaucomatous discs or a miotic pupil make it unreliable; the electroretinogram (ERG) is not affected.

Step 3 — Iris, before dilating

Do: Slit lamp, high magnification, pupillary margin first. Vessels: thin, friable, arborising, irregular; extent in clock hours.

Record: “RE neovascularisation of the iris (NVI), 2 clock hours” or “No NVI.”

Q Why look so carefully?

Neovascularisation of the iris starts at the pupillary margin and decides treatment. Over 2 clock hours, or any angle new vessels (NVA), needs prompt laser. It appears at 2–4 months: the **100-day glaucoma**.

Step 4 — Tonometry

Do: Goldmann applanation with anaesthetic and fluorescein, before gonioscopy and dilatation.

Record: “Tn (tension) by GAT (Goldmann applanation tonometry) 18 / 16 mm Hg at 10 am.”

Q Why might the pressure be high?

Over 22 mm Hg needs a cause: the **open-angle glaucoma**, neovascular glaucoma (NVG), angle closure or intravitreal steroids.

Step 5 — Gonioscopy, before dilating

Do: Goldmann gonioscopy lens, each quadrant: new vessels, synechiae, an occludable angle.

Record: “BE (both eyes) scleral spur in all quadrants; no NVA.”

Q Why before dilating?

Undilated gonioscopy is essential, because NVA occurs without NVI in 12% of eyes. It also shows angle closure.

Step 6 — Dilated fundus

Do: In the department’s order: red glow, media, disc, vessels, background, macula, periphery; confirm with a 90 D (dioptre) lens. At the disc: swelling (acute), pallor (late), cupping, shunts, new vessels.

Record: Draw both fundi: red haemorrhages, blue veins, yellow sclerosed vessels, black laser scars, green epiretinal membrane.

Q Is that a shunt or new vessels on the disc (NVD)?

Optociliary shunts are thicker and drain retinal blood to the choroid, a sign of good compensation. NVD are finer and form a network.

Q Why are some haemorrhages flame-shaped and others dot-blot?

Flame-shaped haemorrhages lie in the nerve fibre layer, which runs parallel to the surface. **Dot-blot haemorrhages** lie in the deeper, compact middle layer, which runs perpendicular to it.

Step 7 — The other eye

Do: Pressure, angle and disc; diabetic or hypertensive retinopathy.

Record: "LE cup-disc ratio 0.8; open angles; no retinopathy."

Q Why examine it carefully?

The **fellow eye** risk is 1% a year, 7% in 5 years. Its disc shows the glaucoma; an occludable angle suggests angle closure.

Say-it description

SAY IT

Mr X is a 65-year-old hypertensive man on drops for primary open-angle glaucoma. He has sudden painless defective vision in the right eye for two months.

Vision: right eye 6/36, no better with pinhole; left eye 6/9. No relative afferent pupillary defect. Goldmann applanation tonometry: 18 and 16 millimetres of mercury at 10 in the morning.

Both eyes: anterior segment normal, no new vessels on the iris, grade 2 nuclear sclerosis. Undilated gonioscopy: scleral spur seen in all quadrants, no new vessels.

Right fundus: good red glow, clear media. Disc vertically oval, pink, well-defined margins. Cup-disc ratio 0.7, with inferior rim thinning and nasalisation, suggestive of glaucomatous cupping. Superior veins dilated and tortuous. Red flame-shaped lesions, dots and blots radiate from the disc into the superior hemisphere only, suggestive of retinal haemorrhages. No cotton-wool spots. Foveal reflex absent, retina thickened, suggestive of macular oedema. Confirmed with a 90 dioptre lens.

Left fundus: cup-disc ratio 0.8, inferior rim thinning, suggestive of glaucomatous cupping.

My diagnosis: right eye clinically non-ischaemic hemi-central retinal vein occlusion involving the superior hemisphere, with macular oedema. Both eyes: primary open-angle glaucoma, pressure under control on medical treatment, and grade 2 nuclear sclerosis.

History to ask, if the examiner allows

Question	Why (what it rules in or out)
Since when; was it sudden?	Sudden painless loss is typical; new vessels are likeliest in the first 6 months
Any diabetes?	A major risk factor; diabetic retinopathy is the main look-alike
Any high blood pressure?	A major risk factor: the hardened artery compresses the vein
Which glaucoma drops, how regularly?	A raised pressure slows venous outflow; pilocarpine (a miotic) spoils the pupil test
Any blood disorder or past clot?	Hyperviscosity or a clotting disorder, commoner under 60: blood work-up
Oral contraceptives or diuretics?	Both are drug risk factors for vein occlusion

Differentials

Condition	How to tell apart (sign or test)
Branch retinal vein occlusion (BRVO) versus hemi-CRVO	BRVO: a wedge of haemorrhage, apex at an arteriovenous crossing, one quadrant; a block at the disc covers two. Even a major BRVO has no RAPD; hemiretinal occlusion may
Ocular ischaemic syndrome	Veins dilated but not tortuous; mid-peripheral dot-blot haemorrhages; normal disc; carotid stenosis on imaging
Diabetic retinopathy	Always bilateral; beaded veins; hard exudates (absent in CRVO)

How I will proceed

Test	What it shows	Significance
Fundus fluorescein angiography (FFA), once the haemorrhages clear (about 3 months)	Delayed transit, blocked fluorescence, vessel-wall staining, capillary non-perfusion, leaking new vessels	Over 10 disc areas of non-perfusion = ischaemic; shows macular ischaemia
Optical coherence tomography (OCT) of the macula	Cystoid macular oedema, subretinal fluid; epiretinal membrane when old	Decides and monitors injections
Visual fields (card G10); ERG	Central scotoma is the commonest defect; b-wave under 60% if ischaemic	Grade ischaemia when the pupil test fails
Systemic work-up	Blood pressure, fasting glucose, lipids, blood count, electrocardiogram, renal tests; thrombophilia screen if under 60	Treat the cause

How I will manage

1. **Aim:** save central vision by treating macular oedema, and prevent neovascular glaucoma; a physician treats risk factors.
2. **Macular oedema: intravitreal anti-VEGF first.** Bevacizumab 1.25 mg/0.05 ml, ranibizumab 0.5 mg/0.05 ml or aflibercept 2 mg/0.05 ml. It blocks vascular endothelial growth factor (VEGF), which makes ischaemic capillaries leak. Disadvantage: repeated injections, monthly for three or six months, then as needed.
3. **Second line: intravitreal triamcinolone or the dexamethasone implant.** It reduces leakage but causes cataract and raises the pressure, a real risk here.
4. **Prompt panretinal photocoagulation (PRP)** for NVI over 2 clock hours, any NVA, NVD or new vessels elsewhere (settings: card G5). It is definitive but costs peripheral field and night vision. Not prophylactic, unless follow-up is impossible in ischaemic CRVO.
5. **Pressure:** drops to target; trabeculectomy or a drainage valve if they fail; neovascular glaucoma, card G5. Full profiles: card GX.
6. **Follow-up** monthly for 6 months: undilated gonioscopy, OCT, fellow eye.

Must know

- **Definition.** "Retinal vein occlusion (RVO) is an obstruction of the retinal venous system" that "may involve the central, hemicentral or branch retinal vein." In CRVO, a thrombus blocks the central retinal vein at or just proximal to the **lamina cribrosa**.
- **Hemiretinal vein occlusion** (*Kanski — not in your practical books*): to some a CRVO variant, ischaemic or non-ischaemic. It blocks the **superior or inferior branch** of the central retinal vein. **Hemispheric:** a major branch at or near the disc. **Hemicentral:** one trunk of a dual-trunked vein. It looks like a branch occlusion of that hemisphere. Iris new vessels: commoner than in branch, rarer than in central occlusion. Extensive ischaemia: manage as ischaemic CRVO.
- **Types (Central Vein Occlusion Study, CVOS):** ischaemic, non-ischaemic (about **75%**) and indeterminate (haemorrhage hides perfusion). About **34%** of non-ischaemic eyes turn ischaemic by **3 years**.

Feature	Non-ischaemic	Ischaemic
Vision	Moderate to severe loss	Counting fingers or worse
RAPD	Absent or mild	Marked, over 0.7 log unit
Fundus	Fewer haemorrhages and cotton-wool spots	Extensive haemorrhages, many cotton-wool spots, marked oedema
FFA non-perfusion	Under 10 disc areas	Over 10 disc areas
ERG	Normal or mildly depressed	b-wave under 60% of normal; b-to-a ratio under 1

- **Why POAG is a risk factor.** Raised pressure causes venous stasis (Virchow's triad); a bowed lamina cribrosa and nasalisation compress the vein.
- **CRVO causes glaucoma too:** neovascular, steroid-induced and angle-closure glaucoma.
- **100-day glaucoma.** After ischaemic CRVO, NVI appears in about half the eyes, usually at 2–4 months. Neovascular glaucoma follows in **one-third** of these, usually at 3–5 months.

Viva questions

Q What are the risk factors for CRVO?

Systemic: hypertension, diabetes, hyperlipidaemia, cardiac disease and carotid insufficiency, mainly over 60. Under 60: clotting disorders and vasculitis. Drugs: oral contraceptives and diuretics. **Ocular:** open-angle glaucoma first, then ischaemic optic neuropathy, tilted disc, disc drusen and hypermetropia.

Q What is the pathogenesis, and where is the block in each type?

Artery and vein share a sheath through the rigid **lamina cribrosa**. A hardened artery or a bowed lamina compresses the vein, and a **thrombus** forms. Ischaemic: at or just behind the lamina, few collaterals. Non-ischaemic: 6 mm behind it, many.

Q What is "blood and thunder"?

Blood and thunder: four-quadrant haemorrhages radiating from the disc, with dilated, tortuous veins. Ischaemic CRVO's widespread haemorrhages give a "tomato ketchup" fundus.

Q Why do you call this eye non-ischaemic?

Vision 6/36, no RAPD, no cotton-wool spots (ischaemic infarcts of the nerve fibre layer). Over **10 disc areas** of non-perfusion would make it **ischaemic CRVO**.

Q Why does the type matter?

The type predicts new-vessel risk, visual prognosis, spontaneous improvement and the follow-up interval. New vessels in a non-ischaemic eye suggest diabetes or carotid artery disease.

Q What are the complications?

Macular oedema, the commonest cause of visual loss. Also macular ischaemia, neovascular glaucoma, vitreous haemorrhage, tractional retinal detachment and optic atrophy.

Q What is the ischaemic index?

Ischaemic index = non-perfused area ÷ total retinal area; about 50% (10 disc areas) marks significant neovascular risk.

Q What turns a non-ischaemic CRVO ischaemic?

Vision under 6/60, 5–9 disc areas of non-perfusion, age over 60, cardiovascular disease, blood dyscrasias and nocturnal hypotension.

Q Why is NVI commoner than NVD in CRVO?

The retina is too ischaemic to form them; anterior segment capillaries stay viable.

Q What did the CVOS show?

Grid laser did not improve vision, because leakage is diffuse. Prophylactic laser did not prevent anterior new vessels; delayed laser regressed them better. So: prompt **panretinal photocoagulation** once they appear.

Q Can anti-VEGF replace PRP?

No. **Anti-VEGF** only delays new vessels, which return more strongly when it stops.

Q Why is vision worse on waking?

Nocturnal hypotension slows venous flow, completing a partial thrombosis.

Q How will you investigate a patient under 60?

Thrombophilia screen (protein C and S, factor V Leiden, antiphospholipid antibodies), homocysteine, autoantibodies, treponemal serology.

Q What is the visual prognosis?

Non-ischaemic: final vision better than 6/60 in 50%. Ischaemic: worse than 6/60 in 93%.

Quick recall

- Before dilating: RAPD, iris margin, pressure, undilated gonioscopy.
- Ischaemic: counting fingers or worse, RAPD over 0.7 log unit, over 10 disc areas.
- NVI over 2 clock hours or any NVA: prompt PRP, not prophylactic.
- Macular oedema: anti-VEGF first; steroids raise the pressure.

Read more: Baidya p.251, 259–273 · Namrata p.216–229, 233 · Aravind 6.6, 6.4, 4.8, 4.9 · Kanski p.521

G8 Applanation tonometry — perform and record

★ Done in 8 of the 12 glaucoma cases at JEH last year · Task

Indications

- **Every glaucoma case, both eyes, every visit**, to check the target pressure. Goldmann applanation tonometry (GAT) is the preferred method for intraocular pressure (IOP): **reliable, reproducible, not influenced by scleral rigidity**.
- **When GAT is unsuitable:** **Tono-Pen** for a scarred, oedematous or irregular cornea. **Perkins** (portable, hand-held applanation) for a bed-bound or anaesthetised patient or a child. **Non-contact** for screening. **Digital** for a corneal ulcer.

Keywords the examiner listens for: Imbert-Fick law · 3.06 mm · green semicircular mires · inner edges just touch · dial reading × 10 · central corneal thickness · phasing · before gonioscopy · 10–21 mm Hg

Instrument check

- **Slit lamp** with the Goldmann tonometer; the patient sits.
- **Calibration** checked each session with the manufacturer's **calibration arm**.
- **Dial at 1** (10 mm Hg), never 0: at 0 **microvibrations** can cause **corneal erosions**.
- **Prism:** for **astigmatism over 3 dioptres**, turn it so the **red line** on the housing meets the **minus-cylinder axis**.
- **Ready:** anaesthetic drops, fluorescein (strip or 0.25% drops), disinfectant.

Steps

Step	Why — the viva answer
1. Order: after visual acuity and pupils, before gonioscopy and dilatation	Gonioscopy opens the angle, so IOP falls: a later reading is falsely low. Dilatation raises IOP 4–5 mm Hg: falsely high
2. Introduce yourself, explain, take consent: "Numbing drops and a dye, then a gentle touch on the eye. It does not hurt."	Examiners tick it. Informed patients do not squeeze
3. Disinfect the prism tip: 70% isopropyl alcohol, 3% hydrogen peroxide or 1:10 sodium hypochlorite	The tip can spread adenovirus, herpes simplex type 1, human immunodeficiency virus and hepatitis B
4. Topical anaesthetic drop (for example 0.5% proparacaine), then fluorescein , into the conjunctival sac	Contact is painless. Fluorescein shows the tear meniscus
5. Instruct: chin on the rest, forehead firmly on the headband. "Look straight ahead. Keep both eyes open. Do not squeeze. Breathe normally."	Squeezing raises IOP (orbicularis tone). Breath-holding (Valsalva) raises it through episcleral venous pressure
6. Cobalt blue filter , maximal light, light and microscope about 60° apart	Fluorescein glows yellow-green under cobalt blue
7. Part tight lids gently, without pressing the globe	Pressure on the globe raises the reading
8. Advance the prism until it just touches the corneal apex : two green semicircular mires , above and below the horizontal midline	The mires are the tear meniscus
9. Centre: semicircles equal in size	Unequal semicircles read falsely high
10. Turn the dial until the inner edges just touch	Now exactly 3.06 mm of cornea is flattened
11. Dial × 10 = mm Hg (1.8 → 18). Repeat in the other eye; note the time	IOP varies through the day
12. Disinfect the prism again	Protects the next patient

How to record the result

Eye · method · value · time. Case sheet: "Tn by GAT" (Tn = tension). Right eye (RE) 18 mm Hg, left eye (LE) 16 mm Hg, at 10 am. Write central corneal thickness (CCT) under investigations, not in the clinical table.

SAY IT

Intraocular pressure by Goldmann applanation tonometry is 18 millimetres of mercury in the right eye and 16 in the left. It was measured at 10 in the morning, before gonioscopy.

Normal values and interpretation

- **Normal 10–21 mm Hg.** Asymmetry of **4 mm Hg or more** between the eyes is significant.
- **Diurnal variation:** usually highest in the morning, lowest in the evening. It swings 3–6 mm Hg normally; a swing of 8 mm Hg or more is significant. Record the time.

- **CCT:** GAT is calibrated for a mean CCT of **520 µm**. A thinner cornea reads falsely **low** and a thicker one falsely **high**, by about **0.7 mm Hg per 10 µm**. Thin is under 500 µm (also after refractive surgery); thick is over 570 µm. Ultrasonic **pachymetry** measures it.
- **Corneal oedema** under-reads: assume a higher IOP and repeat after it settles.
- **Ocular hypertension:** over 21 mm Hg on 2 consecutive applanation readings, with a normal disc, field and open angles.

Common errors

Error	Effect on the reading
Mires too thick (wide meniscus) / too thin	Falsely high / falsely low
Unequal semicircles (poor vertical alignment)	Falsely high
Astigmatism (elliptical contact)	With-the-rule low , against-the-rule high , about 1 mm Hg per 4 dioptres. Over 3 dioptres: red-line method, or average two readings 90° apart
Squeezing, breath-holding, pressure on lids or globe	Falsely high
Quick repeated readings	Slight fall
Oedematous, scarred or irregular cornea; after refractive surgery	Low or unreliable. Use a Tono-Pen

Viva questions

Q What is the principle of applanation tonometry?

The **Imbert-Fick law** ($P = F/A$). The pressure inside an ideal dry, thin-walled sphere equals the force needed to flatten it, divided by the area flattened.

Q Why is the flattened area 3.06 mm?

At **3.06 mm**, corneal rigidity (resisting) and the tear meniscus's capillary attraction (pulling) cancel. Then 0.1 g equals 1 mm Hg, hence **dial reading × 10**.

Q What do you see, and what is the end point?

A blue central circle (flattened cornea) between two **green semicircular mires** (tear meniscus). End point: equal semicircles whose **inner edges just touch**.

Q Central corneal thickness 600 µm, GAT 26 mm Hg. Comment.

It is 80 µm above 520 µm, so GAT over-reads by about $8 \times 0.7 = 5.6$ mm Hg. The corrected IOP is about 20 mm Hg. Always record **central corneal thickness**.

Q Why is 21 mm Hg the upper limit?

Normal **10–21 mm Hg** is the mean (15.5 ± 2.57) plus 2 standard deviations, about 20.5. That covers about 95% of people. The curve is skewed right, so 21 is approximate.

Q What factors change the IOP?

Diurnal variation and **posture**: lying down raises IOP by 0.3–6 mm Hg. Valsalva and lid squeezing raise it. Most general anaesthetics lower it, but ketamine raises it. Caffeine raises it; alcohol lowers it.

Q What is phasing?

Phasing records IOP every 3 hours for 24 hours. A difference of 8 mm Hg or more between two readings is significant. It finds the peak, times the drops and explains progression despite good clinic readings.

Q Classify tonometers.

Contact: indentation (Schiotz) or applanation, either variable-force (Goldmann, Perkins, Mackay-Marg) or variable-area (Maklakov). **Non-contact**: air puff.

Q How do you use a Schiotz tonometer? Why is applanation preferred?

Schiotz is an **indentation** tonometer. Its plunger (5.5 g, or 7.5, 10, 15 g) sinks deeper when IOP is lower. I check zero on the test plate, then rest the footplate on the anaesthetised cornea of a supine patient. A **conversion chart** gives mm Hg. **Scleral rigidity** skews it (low in high myopia, high in high hypermetropia); applanation avoids this.

Q What is the Tono-Pen?

A **Mackay-Marg** tonometer. It works on scarred corneas, over a bandage contact lens and in any position. Compared with GAT, it under-reads below 24 mm Hg and over-reads above.

Q How does a non-contact tonometer work?

An **air puff** flattens the cornea; the time taken relates to IOP. It needs no contact or anaesthetic, so it suits screening. Average at least three readings.

Q How is digital tonometry done?

The patient looks down. Two fingertips press through the upper lid, above the tarsal plate: one still, one indenting.

Read more: Baidya p.150, 154, 170, 179, 732, 738–740 · Namrata p.96, 136, 173–178, 180 · Aravind 1.5, 2.3, 4.1, 4.8

G9 Gonioscopy — perform, grade and draw

★ Done in about 8 of the 12 glaucoma cases at JEH last year · Task

Indications

- **Every glaucoma case, both eyes:** open or closed angle, early narrow angles, extent of closure, secondary causes.
- **Yearly in open-angle glaucoma:** the lens thickens with age, so angle closure may develop.
- **Also:** iridotomy patency; new vessels in central retinal vein occlusion (undilated); tumour, foreign body, blood, Kayser–Fleischer ring.

Keywords the examiner listens for: total internal reflection · Goldmann lens · Zeiss four-mirror · indentation · scleral spur · Schwalbe's line · Shaffer grade · occludable angle · peripheral anterior synechiae · angle recession

Instrument check

Lens	Contact and coupling	Best for
Goldmann single-, two- or three-mirror (indirect; angle mirror smallest, dome-shaped)	12 mm; steeper than the cornea, so it needs viscous coupling fluid	Grading without pressure; steadies the globe
Zeiss four-mirror (also Posner, Sussman)	9 mm; no coupling fluid (tears bridge the gap)	Whole angle quickly; indentation
Koepe (direct)	Supine; hand-held microscope	Infants under general anaesthesia

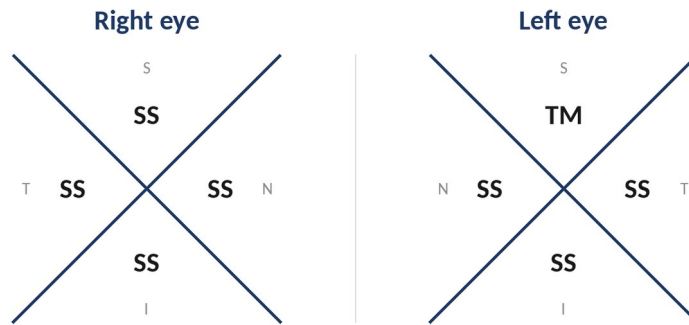
Also ready: anaesthetic drops, disinfectant, anhydrous glycerin for corneal oedema.

Steps

Step	Why — the viva answer
1. Order: Van Herick, then applanation tonometry , then gonioscopy, before dilatation. Defer if the cornea is oedematous	Gonioscopy lowers the pressure, so later tonometry reads falsely low. Dilating a narrow angle can cause closure. Oedema hides the angle
2. Introduce yourself, explain, take consent: “A numbing drop, then a lens on your eye. It does not hurt.”	Examiners tick it. A calm patient keeps still
3. Disinfect the lens before and after each patient	It touches every patient's cornea
4. Topical anaesthetic drop	Lens contact is painless
5. Instruct: chin on the rest, forehead on the band. “Keep both eyes open; look where I ask.”	For a narrow angle, the patient looks towards the mirror in use
6. Semi-dark room; slit beam 2 mm wide, at a right angle to the mirror	Keeps the examiner's eyes partly dark-adapted
7. Coupling fluid in the Goldmann cup; place the lens gently	The lens curve is steeper than the cornea
8. Grade before any pressure	Pressure distorts the angle and opens an appositional closure
9. Rotate through 360°. The mirror at 12 o'clock shows the inferior angle	Each mirror shows a mirror image of the opposite angle
10. From the iris root forwards: ciliary body band, scleral spur , trabecular meshwork, Schwalbe's line	The deepest structure seen grades the angle
11. Indentation with the Zeiss lens	Aqueous pushes the iris back: apposition opens, synechiae stay closed
12. Note iris contour, pigment, synechiae, new vessels, angle recession, pseudoexfoliation material	Secondary causes change the diagnosis

How to record the result

One cross each for the right eye (RE) and left eye (LE): in each quadrant, the **deepest structure seen**. Use scleral spur (SS), ciliary body band (CBB), trabecular meshwork (TM), Schwalbe's line (SL) or “Closed”. Mark a closed quadrant “appositional” (opens on indentation) or “peripheral anterior synechiae (PAS)”.



Deepest structure seen — right eye open all round; left eye narrower superiorly; neither occludable

Written: “RE: scleral spur in all quadrants, open angles. LE: pigmented TM superiorly (Shaffer grade 2), scleral spur elsewhere. No PAS or new vessels.”

Your department writes a grade in each quadrant “by modified Shaffer’s grading” (III in all four on its primary open-angle glaucoma sheet). Confirm with your seniors which structure each grade stands for, and write the structure too.

Normal values and interpretation

Structure	How to recognise it
Ciliary body band	Pink, dull brown or slate-grey band behind the spur; wider in myopes
Scleral spur	Narrow white line; the longitudinal ciliary muscle and iris processes attach here
Trabecular meshwork	The posterior part is pigmented : the main outflow site. Schlemm’s canal may show as a darker line. Gonioscopy itself, raised episcleral venous pressure or hypotony fills it with blood
Schwalbe’s line	Most anterior; whitish or pigmented; the end of Descemet’s membrane

Shaffer	Width	Structures seen	Closure
4 wide open	35–40°	Up to ciliary body	Impossible
3 open	25–35°	Up to ciliary body	Impossible
2 moderately narrow	20°	Pigmented TM	Possible, unlikely
1 extremely narrow	10°	Only SL ± non-pigmented TM	Possible
S slit	< 10°	None; no obvious contact	Parts look closed
0 closed	—	Iridocorneal contact	Complete

System	Grades
Scheie	Wide open all seen · narrow 1 iris root hard to see · 2 CBB obscured · 3 posterior TM obscured · 4 only SL
Spaeth	Four parameters: insertion A anterior to TM, B at TM, C at SS, D deep to SS, E into ciliary body · width 10°/20°/30°/40° · contour S steep, R regular, Q queer (deeply concave) · TM pigment 0–4

Van Herick (peripheral depth ÷ corneal thickness): $4 \geq 1 \cdot 3 \frac{1}{4} - \frac{1}{2} \cdot 2 \frac{1}{4} \cdot 1 < \frac{1}{4}$. Grade 1 may be occludable.

Common errors

Error	Effect
Pressure while grading	Apposition opens: the angle looks wider
Charting the mirror’s own quadrant	Wrong quadrant
Pigmented SL or Sampaolesi’s line taken for TM	Grade 1 called 2: looks wider
Synechial closure called without indentation	PAS over-called
Iris processes taken for PAS	PAS over-called

Viva questions

Q Why can’t you see the angle without a lens?

Light from the angle undergoes **total internal reflection** at the tear–air interface. The gonioslens replaces that interface with one of higher refractive index.

Q Why grade with a Goldmann lens and indent with a Zeiss?

The **Goldmann lens** avoids accidental pressure that distorts the angle. **Indentation** with the **Zeiss four-mirror** shows the extent of PAS. In plateau iris it shows a double hump (iris draped over the ciliary processes).

Q Is this angle occludable? How does the ISGEO classification stage angle closure?

An **occludable angle** shows no pigmented meshwork (**Shaffer grade 1 or 0**) without indentation in at least 3 quadrants. A primary angle-closure suspect has iridotrabecular contact in 3 or more quadrants (almost 270°), normal pressure and no PAS. Raised pressure and/or PAS make it primary angle closure; optic neuropathy makes it primary angle-closure glaucoma.

Q Direct versus indirect gonioscopy?

Direct gives an erect view and sees over a convex iris, but it is slow and supine. **Indirect** is quick and allows compression, but orientation confuses at first.

Q Iris processes versus peripheral anterior synechiae?

Iris processes are lacy and fenestrated, with structures visible between. **Peripheral anterior synechiae** are solid and hide the angle.

Q Causes of PAS?

Primary angle closure, anterior uveitis, iridocorneal endothelial syndrome, after intraocular surgery, trauma.

Q Normal versus new vessels in the angle?

Normal vessels are broad and stop short of the spur. New vessels are fine, cross the **scleral spur** and arborise on the meshwork.

Q What is Sampaolesi's line?

Sampaolesi's line is a dark, scalloped, granular, discontinuous pigment band on or in front of **Schwalbe's line**. Meshwork pigment is fine and continuous. It occurs in pseudoexfoliation, pigment dispersion and chronic inflammation.

Q Causes of meshwork pigmentation?

Pigment dispersion, pseudoexfoliation, trauma, laser iridotomy, acute angle closure, anterior uveitis.

Q Gonioscopy after blunt trauma?

Angle recession: a wide ciliary body band; compare the other eye. Also cyclodialysis (ciliary body off the spur, hypotony), foreign body, blood.

Q Gonioscopy versus imaging?

Gonioscopy is the gold standard: it tells PAS from apposition. Optical coherence tomography and ultrasound biomicroscopy complement it.

Read more: Baidya p.148–152, 154, 169, 171, 715, 741–742 · Namrata p.173, 177–184, 204–205, 354 · Aravind 1.5, 4.2, 4.6, 4.8, 4.10

G10 Humphrey visual field in glaucoma — reading the printout

★ Done in 3 of the 12 glaucoma cases at JEH last year · Chart

What it is

- The **Humphrey field analyser** does **static automated threshold perimetry**. A dim, still light is brightened at each point until seen. Automated perimetry is the **gold standard** for diagnosing and following glaucoma.
- **Threshold**: the intensity seen **50%** of the time. Needing a brighter light means lower sensitivity.
- Stimulus **Goldmann size III, white**, on a **31.5 apostilb (asb)** background. **Decibels (dB)**: **0 dB = 10,000 asb** (brightest), **40 dB = 1 asb**.
- **24-2** (54 points) and **30-2** (76 points) use a **6°** grid. The 24-2 suits suspects and established glaucoma, the 30-2 suspects. **10-2** (68 points, **2°** grid) is for advanced glaucoma.
- **Swedish interactive threshold algorithm (SITA)**: half the time, with little or no loss of accuracy.

Keywords the examiner listens for: reliability indices · pattern deviation · glaucoma hemifield test · mean deviation · pattern standard deviation · visual field index · Bjerrum's area · arcuate scotoma · nasal step · Anderson and Patella criteria

Reading order

Read zone by zone, then conclude. Judge reliability before naming any defect.

1. **Zone 1, patient data:** eye and age. A wrong age compares him with the wrong normals.
2. **Zone 2, test conditions:** programme, strategy, pupil (dilate if under 2 mm), acuity, correcting lens.
3. **Zone 3, reliability indices:** fixation losses, false positives, false negatives, foveal threshold.
4. **Zone 4, grey scale:** a picture of the raw values, without statistics.
5. **Zone 5, raw data:** threshold in dB at each point; 0 dB is an absolute scotoma.
6. **Zone 6, deviation plots** (lower half): numbers above, probability symbols below. **Total deviation** = patient minus age-matched normal. **Pattern deviation** removes overall depression: name the defect here.
7. **Zone 7, glaucoma hemifield test:** glaucomatous loss is asymmetrical. So five sectors above the horizontal raphe are compared with their mirror images below.
8. **Zone 8, global indices:** mean deviation (MD), pattern standard deviation (PSD), visual field index (VFI).
9. **Conclude:** defect, criteria, stage, and the matching disc damage. A superior defect matches the inferior pole.

SAY IT

Humphrey central 30-2 threshold test of the right eye. Age, pupil size and correcting lens are correct. The reliability indices are good: fixation losses 1 of 15, false positives 2%, false negatives 3%. The foveal threshold fits his 6/6 vision. The grey scale shows a dark area above fixation. On the pattern deviation plot, a cluster above the horizontal meridian is depressed, several at p less than 0.5%. The glaucoma hemifield test is outside normal limits. Mean deviation is minus 8 decibels and pattern standard deviation 9 decibels, both at p less than 0.5%. Visual field index is 80%. This is a superior arcuate scotoma in Bjerrum's area. It meets the Anderson and Patella criteria: a moderate glaucomatous defect, matching inferior rim thinning.

Findings to name

Finding	What it means
Fixation losses 20% or more	Unreliable. A light flashed into the blind spot was seen: the eye had moved
False positives over 15%	Unreliable. He pressed when no light was shown ("trigger-happy"); white scotomas on the grey scale
False negatives over 33%	Unreliable. He missed a light brighter than one seen there before: fatigue, inattention or advanced glaucoma. Clover-leaf grey scale
Low foveal threshold, good acuity	Wrong optical correction
Total deviation depressed, pattern deviation normal	Uniform depression: cataract, media opacity, small pupil, uncorrected refraction
Hemifield test outside normal limits (p<1%) or borderline (p<3%)	Upper and lower halves differ, as in glaucoma. Other results: within normal limits, general reduction of sensitivity, abnormally high sensitivity
MD more negative	Overall loss; normal is typically within 2 dB of 0
PSD high	Focal loss; low PSD means no loss or diffuse loss
VFI	Per cent of a normal field for age (100 normal, 0 blind); unaffected by cataract

Glaucomatous defects

They begin in **Bjerrum's area**, the central 25° (first defects 10–25° from fixation). They respect the **horizontal meridian** and appear in this order.

Defect	What it is
Generalised depression	Earliest; not diagnostic
Baring of the blind spot	Early, very non-specific
Paracentral scotoma	Earliest clinically significant; usually superonasal, ending at the horizontal midline
Seidel's scotoma	Paracentral scotoma joined to the blind spot: sickle-shaped
Arcuate (Bjerrum's) scotoma	From the blind spot, arching above or below fixation to the horizontal raphe
Ring (double arcuate) scotoma	Superior and inferior arcuates joined
Nasal (Roenne's) step	A right-angled step at the horizontal meridian, nasally
Tubular vision, temporal island	End stage: only a central and a temporal island remain
Dense hemifield (altitudinal) defect	Not a stage: goes with deep focal notching of the rim. Commonest defect in non-arteritic ischaemic optic neuropathy

Grading or classification

Is it glaucomatous? Anderson and Patella criteria

Shown on at least **2 consecutive reliable fields**:

- **3 or more contiguous non-edge points** on the 30-2 pattern deviation plot at **p<5%**, one at **p<1%**.
- **PSD at p<5%**.
- **Glaucoma hemifield test outside normal limits**.

How severe?

Parameter	Early	Moderate	Severe
MD	-6 dB or better	-6 to -12 dB	-12 dB or worse
Central 5° points under 15 dB	None	One hemifield	Both hemifields
Central 10°	Spared	Spared	Involved

What each finding leads to (management)

- **Unreliable**: do not diagnose; re-instruct and repeat. The criteria need 2 reliable fields.
- **Uniform depression**: correct the refraction, dilate a small pupil, look for cataract; repeat.
- **Criteria met, matching the disc**: glaucomatous defect. Stage it and set the target pressure (card G1).
- **Central 10° involved**: serial **10-2** fields; a **size V** stimulus shows the macular split.
- **Progression** (worsening MD or VFI, deeper or wider scotoma, new defect): seen on trend and progression analyses. Despite maximal medical therapy, it calls for trabeculectomy.
- **Repeat fields** 6-monthly in mild to moderate, 3-monthly in advanced glaucoma.
- **Does not fit** (vertical meridian, no matching disc, pallor more than cupping, rapid progression): order neuroimaging.

Viva questions

Q What is a visual field, and what is a scotoma?

Visual field: Traquair's "island of vision in a sea of darkness". It spans about 60° up, 75° down, 60° nasally and 100° temporally. A **scotoma** is an area of lower sensitivity within greater sensitivity.

Q What is an absolute and a relative scotoma?

In an **absolute scotoma** even the largest, brightest target is not seen. In a **relative scotoma** a brighter light is seen there, but a dimmer one is not.

Q What is a decibel in perimetry?

A **decibel** is 0.1 log unit of attenuation of the brightest stimulus. A higher dB means a dimmer light was seen: more sensitive.

Q Why name the defect on the pattern deviation plot?

Pattern deviation shifts total deviation so its 7th best point reads 0. This removes generalised depression and unmasks localised defects.

Q How does a cataract change the printout?

Total deviation and **mean deviation** fall; pattern deviation stays near normal. **Pattern standard deviation** stays low; the **visual field index** is unaffected.

Q What does p<2% mean?

Under 2% of normal people have such low sensitivity there. It shows significance, not depth.

Q Is the defect glaucomatous or neurological?

Glaucomatous: horizontal meridian (a **nasal step**), relative, incongruous. Neurological: vertical meridian, absolute, mostly congruous.

Q What else causes an arcuate scotoma?

An **arcuate scotoma** also follows myopic peripapillary atrophy, disc drusen, optic pit, ischaemic optic neuropathy and pituitary adenoma.

Q Why is the superior field hit first?

Inferior arcuate fibres are more susceptible. Central fibres arise closest to the disc and exit its centre, so central vision goes last.

Q Which changes mean progression?

A reproducible fall of over 7 dB in a scotoma means deepening. Over 9 dB beside it means enlarging; over 11 dB at a normal point, a new scotoma.

Read more: Baidya p.159, 161, 163–164, 181–194 · Namrata p.176 · Aravind 4.4

G2 POAG with trabeculectomy — bleb assessment and the failed bleb

On the department checklist (not kept last year) · Long case

At a glance

- **Primary open-angle glaucoma (POAG) after trabeculectomy:** the bleb has **failed** and the intraocular pressure (IOP) is above target. For primary open-angle glaucoma itself, see card G1.
- A checklist long case: the bleb, the surgical iridectomy and the cupped disc last for years.
- Tested: the **bleb description in the department's order**, Seidel's test, iridectomy, ostium, antimetabolites, the shallow chamber.

Keywords the examiner listens for: filtering bleb · failed bleb · microcysts · subconjunctival fibrosis · encapsulated bleb (Tenon's cyst) · Seidel's test · peripheral iridectomy · internal ostium · mitomycin C · 5-fluorouracil · digital pressure · laser suture lysis · shallow anterior chamber · malignant glaucoma · bleb-related endophthalmitis

History — positive and negative, with reasons

Chief complaint: "Mr X, a 56-year-old man, came with defective vision in both eyes. Right eye (RE) for 1 year, left eye (LE) for 6 months. The right eye was operated for raised pressure 6 months ago."

History of present illness (house wording): "Patient was apparently normal ___ years before. Then he had defective vision in both eyes, gradual in onset and progressive. He was found to have raised pressure and started on eye drops (___). The ___ eye underwent trabeculectomy (with / without ___) ___ months ago at ___. He is now on ___."

Positive history

Question	Why I ask — what the answer tells me
Which eye was operated, and when?	Early problems (shallow chamber, blocked fistula, failing bleb) differ from late ones (leak, infection, Tenon's cyst, cataract, failure)
Why was the operation advised?	The indication: failed maximal tolerated medical therapy, poor compliance, or progressing disc or field damage
Was a drug put on a sponge during surgery, or were injections given after it?	An antimetabolite : mitomycin C (MMC) on a sponge, or 5-fluorouracil (5-FU) injections. Both raise the risk of hypotony, leak and infection, more so MMC
In the first weeks, was the eye pressed, or were stitches cut with a laser?	Digital pressure and laser suture lysis are used when the early pressure is above target, or the bleb is failing. If they were needed, the pressure was hard to control early
Was the eye soft or the chamber flat after surgery?	A shallow chamber is the commonest problem after trabeculectomy. It can leave peripheral anterior synechiae (PAS), endothelial damage and cataract
Has vision fallen slowly since surgery?	Cataract is a late complication
Which drops now: drug, strength, frequency, since when?	Restarting drops is a step for a failing or failed bleb: the bleb no longer controls the pressure alone
Has the other eye been operated?	A Tenon's cyst in the fellow eye is a risk factor for one here. If 5-FU surgery failed there, MMC is used here
Was laser treatment of the angle done before surgery?	Argon laser trabeculoplasty raises the risk of an encapsulated bleb (Tenon's cyst) up to 3 times

Negative history

Question	A "yes" would point to	How to tell them apart
Any redness of the operated eye?	Bleb infection (blebitis): mild hyperaemia around the bleb is its first sign. Delay risks bleb-related endophthalmitis	Endophthalmitis inflames the vitreous: hypopyon, vitreous cells, loss of the red reflex
Any pain?	Endophthalmitis (pain and redness are its symptoms); an acute pressure rise	High pressure with corneal oedema means an acute rise. Hypopyon and vitreous cells mean endophthalmitis
Any watering?	Endophthalmitis (tearing and photophobia are its symptoms). Drop side effects: lacrimation from pilocarpine, allergy to brimonidine, or corneal surface toxicity	Hypopyon and vitreous cells mean endophthalmitis. Fluorescein 1% staining shows surface toxicity; lid allergy points to the drug
Any sudden loss of vision?	Snuff-out after trabeculectomy in advanced glaucoma; central retinal vein occlusion (CRVO); acute pressure rise	Corneal oedema with high pressure: acute rise. Haemorrhages in all four quadrants with dilated tortuous veins: CRVO (card G7)
Any change in the upper lid since surgery?	Upper eyelid retraction, a late complication	Compare the lid margins of the two eyes in the primary position

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Asthma or chronic obstructive pulmonary disease?	Beta-blockers such as timolol are contraindicated when drops restart
Heart failure or a slow pulse?	Beta-blockers are contraindicated in cardiac failure and bradycardia
Allergy to sulpha drugs?	Carbonic anhydrase inhibitors are avoided; also in renal failure and chronic liver disease
Diabetes or hypertension?	Weighed before any further surgery on a glaucoma eye
Aspirin?	Stopped 5 days before surgery
Other eye surgery: cataract, pterygium, keratoplasty, scleral buckle?	Each worsens the prognosis of filtering surgery. Previous conjunctival or cataract surgery calls for an antimetabolite
Uveitis?	Active uveitis means a poor prognosis; inflammation is treated before surgery
Steroids in any form?	Steroids raise the pressure in some people; POAG patients are at higher risk
Treatment history: drops before surgery, which and how long?	Long-term drops increase conjunctival fibroblasts and inflammatory cells and reduce goblet cells. Over 3 years calls for MMC

Examination — must-do steps and their viva

General and systemic (this case only)

- **Chest and pulse:** a wheeze or bradycardia means no beta-blocker when drops restart.
- **Blood pressure and blood sugar:** weighed before repeat or cataract surgery.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity	Fall since surgery: cataract, snuff-out	6/60, pinhole 6/12
Extraocular movements	Seniors' addition after visual acuity	Full
Lids	Upper lid retraction; punctal occlusion (5-FU)	Normal
Conjunctiva	Bleb (Step 3); hyperaemia around it	See Step 3
Cornea	Dellen (local thinning from tear-film break-up); epithelial (5-FU) or endothelial (MMC) toxicity	Clear
Anterior chamber (AC)	Depth; cells; hyphaema	Van Herick (VH) grade IV (ND, normal depth)
Iris	Peripheral iridectomy (PI) (Step 6)	See Step 6
Pupil	Relative afferent pupillary defect (RAPD)	4 mm, sluggish, grade 3 RAPD
Lens	Cataract	Nuclear sclerosis grade 2
Tension (Tn)	Goldmann applanation tonometry (GAT)	Tn by GAT 26 mm Hg at 10 am
Gonioscopy	Internal ostium, PI, PAS	See Step 8
Fundus	Disc; macula; choroidal detachment	Cup-disc ratio (CDR) 0.9
Fields	Confrontation	"Fields (confrontation test): ____"

Step 1 — Visual acuity

Do: Distance acuity with and without pinhole, each eye; then near vision.

Record: "RE 6/60, pinhole 6/12; LE 6/24, pinhole 6/9."

Q Why can vision fall after trabeculectomy?

Cataract is a late complication, also caused by a shallow chamber after surgery. Early on, hypotony maculopathy and loss of central vision can occur.

Q What is snuff-out?

Snuff-out is sudden loss of vision after trabeculectomy in advanced glaucoma.

Step 2 — Pupils

Do: Before any drop: size, shape and the swinging-flashlight test.

Record: "RE 4 mm, round, sluggish, grade 3 RAPD; LE normal."

Q Why check the pupils before any drop?

A **relative afferent pupillary defect** shows more optic-nerve damage in one eye. Dilating drops would hide it.

Q What does a grade 3 RAPD tell you here?

Advanced damage: a relative afferent pupillary defect is seen in advanced glaucoma. Here it fits the 0.9 cup in the right eye.

Step 3 — Bleb examination in the department's order

Do: Ask the patient to look down; lift the upper lid gently. Use diffuse light, then high magnification. Describe in order:

1. **Height** (flat, medium) and **clarity** (translucent or opaque).
2. **Horizontal extent** (clock hours) and **location** over the limbus.
3. **Vascularity** (avascular, or highly vascular with corkscrew vessels) and **microcysts**.
4. **Subconjunctival fibrosis** and **sutures**.

Record: "Flat, opaque bleb from 11 to 2 o'clock, superiorly above the limbus. Highly vascular with corkscrew vessels, no microcysts, subconjunctival fibrosis present, no sutures — suggestive of a failed bleb."

Feature	Good functioning bleb	Failing or failed bleb
Height	Medium: low and diffuse, or circumscribed and elevated	Flat ; reduced height
Clarity and wall	Translucent , thin	Opaque ; thicker wall
Vascularity	Avascular	Highly vascular , corkscrew vessels
Microcysts	Numerous	Absent
Subconjunctival fibrosis	None	Present

Q What are the features of a good filtering bleb?

A **good filtering bleb** is avascular, with numerous epithelial **microcysts**. It is low and diffuse, or circumscribed and elevated. Such blebs control the IOP.

Q What are the signs of a failing bleb?

A **failing bleb** shows reduced height, a thicker wall, vascularisation and loss of microcysts. The IOP rises.

Q What do the microcysts mean?

Microcysts come from aqueous flowing across the conjunctiva. They mean the bleb is filtering.

Q Why do you note the scleral flap sutures?

Visible flap sutures can be cut by **laser suture lysis** when the target pressure is not reached. This is done from a few days to 6 months after surgery. Gentle pressure on the conjunctiva shows them better. A tenonectomy at surgery also makes them easier to see.

Step 4 — Seidel's test

Do: Put fluorescein over the bleb: 0.25% shows many leaks, 2% the subtle ones. With cobalt blue light, look for aqueous leaking through the dye.

Record: "Seidel's test negative."

Q Why do Seidel's test on every bleb?

A **leak** and **overfiltration** both give a low pressure. **Seidel's test** is positive only in the leak.

Q How is a late leaking bleb managed?

A **late leak** is treated conservatively when minimal, with a deep chamber and a pale, elevated bleb. I use aqueous suppressants, gentamicin drops and patching, and a shell if the chamber is flat. If these fail, I repair it surgically.

Step 5 — Anterior chamber depth

Do: Van Herick: a thin slit beam 1 mm inside the temporal limbus at 60°. Compare the peripheral chamber with corneal thickness. Then judge central depth, cells and hyphaema.

Record: "VH grade IV (ND); AC quiet."

Q What is the commonest problem after trabeculectomy?

A **shallow anterior chamber**. With low IOP: leak, overfiltration or choroidal detachment. With high IOP: pupillary block or delayed suprachoroidal haemorrhage. Malignant glaucoma has a high or normal IOP.

Q Why does a shallow chamber matter?

A **shallow chamber** causes peripheral anterior synechiae, corneal endothelial damage and cataract.

Step 6 — Peripheral iridectomy and its patency

Do: On the slit lamp describe **margin** → **site** → **surrounding iris**. Then **retro-illumination**: a light reflex seen through the defect means patency.

Record: "Sharp, well-defined margin at 12 o'clock near the limbus; iris normal; retro-illumination positive — patent surgical iridectomy."

Q Why is a peripheral iridectomy done with trabeculectomy?

The **peripheral iridectomy** stops the iris blocking the internal ostium and prevents **pupillary block**. It is basal and wider than the ostium.

Q How do you tell a surgical iridectomy from a laser iridotomy?

A **surgical iridectomy** has clean-cut triangular margins near the incision and cannot close. A laser iridotomy has ragged margins at any site, may close, and may show pigment dispersion.

Step 7 — Tonometry

Do: Goldmann applanation after the slit-lamp examination, before gonioscopy and dilatation; note the time.

Record: "Tn by GAT RE 26 mm Hg, LE 24 mm Hg at 10 am."

Q At what pressure would you act soon after surgery?

Above **12 mm Hg**, or whenever the target is missed, I apply **digital pressure**. I do not wait for the bleb to fail.

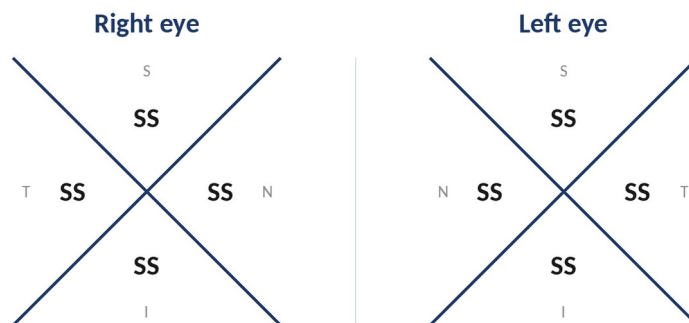
Q What does a soft eye with a flat bleb mean?

A **wound leak**: soft eye, poor bleb, positive Seidel's test. A very low pressure with a good bleb and negative Seidel's test means overfiltration.

Step 8 — Gonioscopy and the internal ostium

Do: After tonometry, before dilatation (technique: card G9). Note the deepest structure in each quadrant; superiorly, the **internal ostium** and iridectomy.

Record: "Scleral spur in all quadrants; patent internal ostium and PI superiorly."



Deepest structure seen in each quadrant — open angles in all quadrants (SS = scleral spur). RE: patent ostium and iridectomy superiorly

Q Why do gonioscopy in an operated eye?

The **internal ostium** can be blocked by clot, iris, ciliary body, lens or vitreous. Gonioscopy shows whether it is open.

Q How do you treat a blocked ostium?

Argon laser at low energy around the obstruction, or internal bleb revision. For fibrin or clot, intracameral tissue plasminogen activator **6–12.5 µg**.

Step 9 — Disc and macula

Do: After dilatation, with a +90 D lens: disc, rim, vessels, nerve fibre layer, macula, then the periphery.

Record: "RE CDR 0.9, diffuse rim loss, bayoneting, laminar dot sign. LE CDR 0.75, superior notch."

Q What do you look for in the fundus after trabeculectomy?

The **optic disc** for progression, the **macula** for hypotony maculopathy, and the periphery for choroidal detachment or suprachoroidal haemorrhage.

Q How is hypotony maculopathy prevented?

Hypotony maculopathy is prevented by a partial-thickness flap, a tighter flap after MMC, and releasable sutures or laser suture lysis. In eyes at risk, 5-FU replaces MMC.

Step 10 — Fields

Do: Confrontation at the bedside; plan Humphrey 24-2 perimetry.

Record: "Fields (confrontation test): ____."

Q How often will you repeat the fields?

Perimetry is repeated every **6 months** in mild to moderate disease and every **3 months** in advanced disease. Progression despite surgery means I must lower the pressure further.

Q How do you stage the damage in this eye?

Severe damage: marked cupping and an extensive field defect including the central 5°. The mean deviation is -12 dB or worse.

End-stage: gross cupping with a small residual field. The 0.9 cup and grade 3 RAPD point to advanced damage; the field confirms the stage (target formula: card G1).

Summary and complete diagnosis

Mr ____, a ____-year-old man with POAG, had trabeculectomy (with ____) in the ____ eye ____ months ago. He now uses _____. Vision ____ / _____. Bleb ____ (department order); Seidel's test _____. Surgical iridectomy at ____ o'clock, patent. AC ____; lens _____. Tn by GAT ____ / ____ mm Hg at _____. Ostium _____. CDR ____ / _____. Fields ____.

Formula: eye → POAG, stage → status post trabeculectomy (± antimetabolite) → bleb → IOP → lens → other eye.

- “**Right eye: primary open-angle glaucoma status post trabeculectomy with failed bleb and grade 2 nuclear sclerosis. Left eye: primary open-angle glaucoma and grade 2 nuclear sclerosis.**”
- “Right eye primary open-angle glaucoma status post trabeculectomy with a good functioning bleb, intraocular pressure under control.”
- “Right eye primary open-angle glaucoma status post trabeculectomy with an encapsulated bleb, intraocular pressure controlled on medical treatment.”

Differential diagnosis

Raised IOP after trabeculectomy:

Condition	Points for	Points against	How to tell apart
Failed bleb (type 3) — this eye	Flat, vascular, no microcysts, fibrosis; IOP raised	—	Flat bleb with engorged vessels, no microcysts; ostium patent
Encapsulated bleb (type 4, Tenon's cyst)	IOP raised; engorged vessels	This bleb is flat	Localised, highly elevated, dome-shaped bleb
Blocked internal ostium	IOP raised with a deep AC	Seen mainly early after surgery	Gonioscopy: clot, iris, ciliary body, lens or vitreous in the ostium
Failing bleb (early)	Rising IOP; reduced height, thicker wall, more vessels	This bleb is flat with no microcysts left	Some height and microcysts remain: still partly filtering
Raised IOP with a flat chamber: malignant glaucoma, pupillary block, delayed suprachoroidal haemorrhage	IOP raised	AC deep here	Flat AC; PI patency on retro-illumination; ultrasound biomicroscopy; B-scan

Exclude first: a blocked internal ostium — by gonioscopy, because laser or revision can reopen it.

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Seidel's test: fluorescein 0.25% (2% for subtle leaks), cobalt blue light	Confirm or exclude a leak	Negative	A leak means patching, a shell or repair
Gonioscopy of the ostium	Find a blocked fistula	Patent ostium and PI	A block is lasered or revised
Humphrey 24-2 with progression analysis	Stage and follow the damage	Advanced defect in the RE	Progression above target pushes me to surgery
Optical coherence tomography (OCT): nerve fibre layer, ganglion cell complex	Baseline for progression	Thin, as the cup suggests	Structural progression supports escalation
Ultrasound biomicroscopy (UBM)	PI patency and how the trabeculectomy works; cause of a flat chamber	Patent PI and ostium	Malignant glaucoma: ciliary processes rotated forward
B-scan ultrasonography if the chamber is flat or the fundus hidden	Choroidal detachment or suprachoroidal haemorrhage	Not expected	Dome-shaped elevation: clear space = serous; echoes = blood
Disc photograph; central corneal thickness (CCT)	Baseline	—	Goldmann is calibrated for a mean CCT of 520 µm. A thinner cornea reads falsely low, a thicker one falsely high, by about 0.7 mm Hg per 10 µm. Thin is under 500 µm; thick is over 570 µm
Blood sugar and blood pressure	Fitness for surgery	Diabetic	Control before repeat surgery

Say: “I will confirm the failed bleb with Seidel's test and gonioscopy of the ostium. I will stage the damage with Humphrey fields and OCT, then check fitness for surgery.”

How I will manage

Aim

Bring the operated eye to its target IOP, protect the remaining field and prevent infection. The left eye is managed as on card G1.

The ladder

1. **Find what has failed.** I examine the bleb, do Seidel's test and look at the ostium by gonioscopy. A blocked ostium gets argon laser or internal revision; fibrin or clot, tissue plasminogen activator.

2. **Early failing bleb** (trigger: IOP above 12 mm Hg or target). I increase steroids to hourly and apply **digital pressure**. Then **laser suture lysis**, or I release a releasable suture.
3. **Encapsulated bleb**. It usually settles in **2–4 months**; I control the IOP with drops meanwhile.
4. **Failed bleb** (trigger: IOP above target with a flat vascular bleb). I restart antiglaucoma drops, avoiding beta-blockers because he is asthmatic.
5. **Above target on maximal drops, or progressing**. I advise repeat trabeculectomy with **mitomycin C** at a fresh superotemporal or superonasal site. With scarred conjunctiva, I choose a **glaucoma drainage device** (card G3).
6. **Refractory after failed filtration**. I use transscleral diode cyclophotocoagulation.
7. **Cataract**. With IOP controlled, cataract surgery through a temporal clear cornea spares the conjunctiva. With poor control or a marginal filter, I combine it with trabeculectomy.

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Mitomycin C 0.2–0.5 mg/ml on a sponge, 1–5 min, at surgery	Alkylating agent; stops fibroblasts in all cell-cycle phases; suppresses vascular ingrowth	Cystic overfiltering bleb (commonest), late leak, hypotony, bleb infection, scleral melt, endothelial toxicity	Use 5-FU after MMC complications before, or with a risk of hypotony maculopathy
5-fluorouracil 50 mg/ml for 5 min at surgery; 5 mg injections after	Pyrimidine analogue; acts in the S phase; stops fibroblast proliferation	Epithelial toxicity (punctate keratitis, ulcer), punctal occlusion, cicatricial ectropion	None stated in your books
Topical steroid , hourly for a failing bleb	Controls inflammation; inhibits phospholipase A2; suppresses fibroplasia	Raised IOP in steroid responders; POAG patients at higher risk	In bleb infection, only 12–24 hours after antibiotics
Atropine 1%	Paralyses the ciliary muscle: deepens the AC, prevents posterior synechiae	Flushing, tachycardia, fever, delirium	None stated in your books for this use
Timolol 0.5% once or twice daily; dorzolamide 2% three times daily; oral acetazolamide 250 mg two to four times a day	Reduce aqueous production: for a leak, overfiltration, a Tenon's cyst or a failed bleb	Timolol: irritation, dry eye. Acetazolamide: paraesthesia, nausea, renal stones	Timolol: asthma, chronic obstructive pulmonary disease, cardiac failure, bradycardia. Carbonic anhydrase inhibitors: sulpha allergy, renal failure, chronic liver disease

Full profiles: card GX.

Laser and surgery in this case

Procedure (indication)	Key steps	Advantages	Disadvantages and complications
Digital pressure (IOP above 12 mm Hg or target missed)	Firm pressure on the lower globe through the lower lid, eye up; each press 10 s at most	Pushes aqueous through; expands the bleb	Subconjunctival haemorrhage, hyphaema, wound rupture or dehiscence
Laser suture lysis (target not reached; a few days to 6 months)	Argon green or neodymium:yttrium-aluminium-garnet (Nd:YAG) laser through a Hoskins lens; 50 µm, 0.02–0.1 s, 250–1000 mW	Tight flap at surgery; less hypotony and choroidal separation	Conjunctival burn, flap leak, flat AC, hypotony maculopathy, hyphaema, blebitis, endophthalmitis
Repeat trabeculectomy with MMC (failed bleb above target)	Fresh superotemporal or superonasal site; MMC; tight flap; watertight closure	Partial-thickness flap: steadier IOP, less hypotony and infection	Previous failure is a relative contraindication; poorer prognosis; MMC toxicity
Glaucoma drainage device (failed surgery, scarred conjunctiva)	Tube in the AC, plate at the equator (card G3)	Works where the conjunctiva cannot be dissected	Tube problems, hypotony, diplopia, corneal decompensation, endophthalmitis
Combined cataract and trabeculectomy (cataract with poor IOP control or a marginal filter)	One sitting; antimetabolite possible	Earlier visual recovery; fewer early IOP spikes; one operation	Long-term IOP control questionable; more hyphaema, uveitis, shallow chamber, hypotony
Transscleral diode cyclophotocoagulation (refractory, after failed filtration)	12–24 burns behind the limbus over 360°, sparing 3 and 9 o'clock (settings: card GX)	Used when filtering has failed or is likely to fail	Pain, inflammation, cystoid macular oedema, hypotony, phthisis

Follow-up and counselling

- Each visit: bleb, Seidel's test, IOP, AC depth, disc and macula. Fields every 3–6 months by stage.
- Surgery protects the remaining vision; lost vision does not return.

- **Come at once if the eye turns red, painful or watery:** bleb infection can become endophthalmitis.

What you must know

Definition

Trabeculectomy is a partial-thickness filtering operation. It has “a partial thickness flap of sclera overlying the opening between the anterior chamber and the subconjunctival space”. In plain words: a guarded hole at the limbus drains aqueous under the conjunctiva into a bleb. It works mainly by **filtration**: aqueous leaves by transconjunctival filtration or through lymphatics and vessels.

- **Failing bleb:** reduced height, thicker wall, vascularisation, loss of microcysts, rising IOP.
- **Failed bleb:** no longer filters (types 3 and 4), so the IOP is elevated.
- **Encapsulated bleb (Tenon's cyst):** a localised, highly elevated, dome-shaped cavity of hypertrophied Tenon's capsule.
- **Malignant (ciliary block) glaucoma:** aqueous is blocked at the ciliary body and forced back into the vitreous.
- **Endophthalmitis:** an inflammation of the vitreous and inner layers of the eyeball by microbial agents. It is characterised by loss of vision with exudation of the vitreous.

Classification

Filtering operations: full thickness, with no scleral flap (sclerectomy, trephination, Scheie's thermal sclerostomy). Partial thickness: **trabeculectomy**. Non-penetrating: viscocanalostomy, canaloplasty, deep sclerectomy.

Bleb type	Appearance	Filters?
1	Thin, polycystic , from transconjunctival flow of aqueous	Yes
2	Flat, thin, diffuse , relatively avascular; microcysts at high magnification	Yes
3	Flat , no microcysts, engorged surface vessels	No: IOP up
4	Encapsulated (Tenon's cyst): localised, highly elevated dome; engorged vessels	No: IOP up

Your books give these four types without a named grading system.

Trabeculectomy and wound-healing modulation

Indications in POAG: as in the viva question below. Also drug side effects and difficult follow-up.

Relative contraindications: likely to respond to less invasive treatment; previously failed trabeculectomy; severely scarred conjunctiva. Also neovascular glaucoma with active new vessels, and uveitic glaucoma.

Key steps

1. Bridle suture; conjunctival flap based at the limbus or the fornix.
2. Scleral flap based at the limbus, **3 mm radially × 4 mm circumferentially**, 2/3 thickness.
3. MMC now, if used.
4. Dissect the flap into clear cornea. Make a **paracentesis**: an entry for fluid into the AC at any stage of the operation.
5. Enter the AC behind the hinge; Kelly Descemet's punch: **internal ostium 1.5 × 2 mm**.
6. **Peripheral iridectomy**, basal and wider than the ostium.
7. Flap: interrupted **10-0 nylon**; reform the AC with balanced salt solution (BSS) through the paracentesis.
8. Conjunctiva: **8-0 vicryl**, watertight.
9. **Atropine 1%**; subconjunctival betamethasone and gentamicin.

Antimetabolites go in the sub-Tenon's space, because subconjunctival fibrosis is the commonest cause of failure.

Feature	Mitomycin C	5-fluorouracil
Dose	0.2–0.5 mg/ml for 1–5 min	50 mg/ml for 5 min; injection 5 mg
Method	Sponge 4.5 × 4.5 mm under conjunctiva and Tenon's; conjunctival edges pulled over it; rinse with BSS; tighter flap	30-gauge needle 90°–180° from the site; twice daily in week 1, daily in week 2: 21 injections (105 mg)
Choice	Almost all cases; essential after failed trabeculectomy, conjunctival or cataract surgery, secondary glaucoma, the young or dark-skinned, drops over 3 years	After MMC complications, or a risk of hypotony maculopathy

Pathogenesis

- Fibroblasts lay down **subconjunctival fibrosis**, the commonest cause of bleb failure.
- Blood under the conjunctival flap scars the bleb, so haemostasis matters.
- Aqueous damps fibroblasts. Older eyes do better: atrophic Tenon's, less fibroblast activity.

Complications

During surgery: conjunctival buttonhole, flap amputation, haemorrhage, lens damage, vitreous loss, choroidal effusion, malignant glaucoma.

Early, after surgery	IOP	Seidel	Bleb and other signs
Wound (conjunctival) leak	Low, soft eye	Positive	Poor bleb
Overfiltration	Very low	Negative	Good bleb ; choroidal detachment possible
Serous choroidal detachment	Low	—	Flat AC
Hypotony maculopathy	Low	—	AC formed
Malignant (ciliary block) glaucoma	High (hard eye) or normal	Negative	No bleb ; diffuse flat AC
Pupillary block (incomplete PI)	High	—	Flat AC; PI not patent
Delayed suprachoroidal haemorrhage	High	—	Flat AC; aphakic and highly myopic eyes at risk
Blocked ostium or failing bleb	High	—	Deep AC

— = not stated in your books. Also early: uveitis, hyphaema, dellen, loss of central vision.

- **Wound leak:** pressure dressing, Simmons shell, soft contact lens, cyanoacrylate glue; a large defect is sutured or grafted.
- **Overfiltration:** firm patching, shell, aqueous suppressants, atropine, steroids. Then reform the AC and drain a very deep choroidal detachment.

Late: bleb infection and endophthalmitis, cataract, leaking bleb, Tenon's cyst, failure, malignant glaucoma, upper lid retraction, sympathetic ophthalmia.

- **Bleb infection:** frequent topical antibiotics, empirically; steroids 12–24 hours later.
- **Bleb-related endophthalmitis:** your books give no bleb-specific regimen. I treat it by the general protocol for postoperative endophthalmitis (viva below).

Prognosis

- Better: older patients (atrophic Tenon's capsule).
- Poorer: age under 40, previous failed filter, aphakia or pseudophakia, neovascular glaucoma, active uveitis, previous eye surgery.
- Plain trabeculectomy lowers IOP by about **41%**.

Recent advances

(extra) **Photodynamic therapy** and **non-penetrating surgery**: see the last two viva questions.

Viva questions

Why did you ask...?

Q Why did you ask whether an antimetabolite was used?

Mitomycin C causes more complications than **5-fluorouracil**: early or late hypotony, bleb leak and bleb infection.

Q Why did you ask how long he used drops before surgery?

Long-term drops increase conjunctival fibroblasts and cut goblet cells, so the bleb fails more. Over 3 years, I would use mitomycin C.

Q Why did you ask about asthma?

Beta-blockers such as timolol are contraindicated in asthma and chronic obstructive pulmonary disease. When drops restart, I choose another class, such as dorzolamide.

Q Why did you ask about the other eye?

An **encapsulated bleb** (Tenon's cyst) in the fellow eye raises the risk here. If 5-FU surgery failed there, I would use MMC here.

Diagnosis and differentials

Q Is "trabeculectomy" a misnomer?

Yes, a misnomer: the block removed need not contain trabecular meshwork. It is more a peripheral posterior keratectomy.

Q What are the causes of failure of filtration?

Subconjunctival fibrosis is the commonest; it is extraocular, like racial and genetic factors. Intraocular causes are a blocked fistula, bleb failure and poor technique.

Q What are the risk factors for a Tenon's cyst?

Tenon's cyst: young age, previous conjunctival surgery, secondary glaucoma, laser trabeculoplasty, sympathomimetic drops, and one in the fellow eye. Argon laser trabeculoplasty raises the risk up to 3 times.

Investigations

Q What does ultrasound biomicroscopy add after trabeculectomy?

Ultrasound biomicroscopy shows PI patency and whether the trabeculectomy works. In malignant glaucoma, the anterior structures press against the cornea.

Q How does B-scan show a choroidal detachment?

Choroidal detachment is a smooth, dome-shaped, highly reflective elevation that spares the optic nerve. A clear space is serous; echoes mean blood. Elevations from both sides form "kissing choroids".

Management — drugs

Q How do mitomycin C and 5-fluorouracil work?

Mitomycin C is an alkylating agent acting in all cell-cycle phases, and far more potent. The pyrimidine analogue **5-fluorouracil** blocks DNA synthesis in the S phase. Both stop fibroblast proliferation.

Q Which eyes need an antimetabolite?

High risk: neovascular glaucoma, failed trabeculectomy, inflammatory or post-traumatic glaucoma. **Intermediate:** drops for over 3 years, previous conjunctival or cataract surgery. **Low:** black patients, age under 40.

Q How do you prevent antimetabolite toxicity?

Thorough irrigation after the set time, watertight closure, and careful dissection to avoid a buttonhole.

Q Why give atropine after trabeculectomy?

Atropine tightens the lens-iris diaphragm and deepens the chamber. It also keeps the blood-aqueous barrier, relieves ciliary spasm and prevents posterior synechiae.

Management — laser and surgery

Q What are the indications for trabeculectomy in POAG?

Failed maximal tolerated medical therapy with progressive or anticipated damage. Also IOP above target on 3 drugs, baseline IOP over 40 mm Hg, one eye lost, poor compliance.

Q What is the preoperative care?

Preoperative care: I lower a very high IOP and treat inflammation. I stop timolol 2 weeks and carbonic anhydrase inhibitors 1–2 days before, to prevent hypotony and let a bleb form. Aspirin stops 5 days before. Pilocarpine 1%, three times in the hour before surgery, constricts the pupil and prevents iris prolapse.

Q How will you improve the success of the repeat operation?

Before surgery, I treat surface infection. I stop pilocarpine drops, aqueous suppressants, carbonic anhydrase inhibitors and aspirin. **During surgery,** I constrict the pupil and avoid a buttonhole or flap disinsertion. I make a 1.5×2 mm ostium, a larger iridectomy and a tight closure. **Wound healing** is modulated with steroids and an antimetabolite, here mitomycin C.

Q When would you choose a drainage device or cyclophotocoagulation instead?

A **glaucoma drainage device** suits failed surgery with conjunctiva too scarred to dissect. **Transscleral diode cyclophotocoagulation** suits refractory glaucoma after failed filtration. It also suits eyes where filtration is likely to fail: aphakic, pseudophakic or neovascular eyes.

Q Which structures are removed in trabeculectomy?

Structures removed, from outside in: a scleral block, Schlemm's canal, trabecular meshwork, Schwalbe's line, peripheral cornea and iris. Removing Tenon's is optional.

Q Why avoid the 12 o'clock position?

Bleeding: the anterior ciliary vessel enters the ciliary body 3–4 mm from the limbus. Scarring there also blocks a repeat operation.

Q Fornix-based or limbus-based conjunctival flap?

Success is equal. Fornix-based: easier, better exposure, but early limbal leak. Limbus-based: safer early suture lysis, 5-FU and digital pressure, and lower late IOP.

Q When do you suture the scleral flap tighter?

Angle-closure glaucoma or past malignant glaucoma, aphakic or highly myopic eyes, nanophthalmos and raised episcleral venous pressure.

Q What are the advantages of releasable sutures and laser suture lysis?

Releasable sutures and suture lysis let the eye leave theatre firm, protected from hypotony. The pressure can then be titrated after surgery.

Q What is the role of digital pressure?

Digital pressure pushes aqueous through the sclerostomy and expands the bleb. It stops an obstruction to filtration becoming permanent.

Q When do you combine cataract surgery with trabeculectomy?

Combined surgery suits a visually significant cataract with poor IOP control, drug intolerance, advanced glaucoma or a marginal filter. Two stages lower the IOP more, with fewer complications.

Q What will you consider before cataract surgery in this eye?

Prior surgery has left a bleb or scarring, so a temporal clear-cornea approach spares the conjunctiva. I expect a miotic pupil, posterior synechiae and bleeding from a congested eye. Other risks are an early pressure rise and suprachoroidal haemorrhage; diabetes and hypertension need control. After cataract surgery alone, later filtering surgery is prone to fail.

Complications

Q How do you treat malignant glaucoma?

Malignant glaucoma: strong mydriatics, then intravenous mannitol, then Nd:YAG disruption of the anterior hyaloid face through the iridectomy. If that fails, pars plana vitrectomy.

Q What is the earliest sign of bleb infection, and how is it treated?

Mild conjunctival hyperaemia around the bleb. I start frequent topical antibiotics empirically, and steroids after 12–24 hours.

Q Which organisms cause bleb-related endophthalmitis?

Bleb-related endophthalmitis: Streptococcus species (most common), Haemophilus influenzae and Staphylococcus.

Q How do you treat bleb-related endophthalmitis?

Bleb-related endophthalmitis is treated by the general protocol for postoperative endophthalmitis; your books give no separate regimen for blebs. After a B-scan, I take an aqueous or vitreous sample. I inject intravitreal **vancomycin 1 mg/0.1 ml** with **ceftazidime 2 mg/0.1 ml**. If there is no improvement, I may repeat the injection after 48 hours, or advise vitrectomy. I advise immediate vitrectomy if vision is only perception of light.

Q What is the commonest side effect of mitomycin C?

A **cystic bleb with overfiltration**. Others: late leak, corneal decompensation, scleral necrosis, hypotony, endophthalmitis and cataract.

Recent advances

Q What is new in wound-healing modulation? (extra)

Photodynamic therapy: diffuse blue light with a photosensitising agent kills fibroblasts and limits the treated area.

Q What is non-penetrating filtration surgery? (extra)

Deep sclerectomy or viscocanalostomy. The AC is not entered, so overfiltration and hypotony are less. It suits POAG, not angle closure.

Examiner traps

- Calling a high dome “good”: an **encapsulated bleb** does not filter.
- “No leak” without **Seidel’s test**: a leak and overfiltration both give a low pressure.
- Not retro-illuminating the **iridectomy**: an incomplete one causes pupillary block.
- Gonioscopy without naming the **internal ostium**.
- A **beta-blocker** for this asthmatic.
- Steroids first in a bleb infection: antibiotics come first.
- “Status post trabeculectomy” without the **bleb status, IOP and lens**.

Say-it script

Opening (about 1 minute)

SAY IT

Mr X is a fifty-six-year-old man. He came with defective vision in the right eye for one year and in the left eye for six months. The vision fell gradually in both eyes. A doctor found raised pressure and started drops twice daily in both eyes. He did not use them regularly. The right eye was operated six months ago to lower the pressure. He has no operation notes, so I do not know whether an antimetabolite was used. He still uses drops in both eyes.

Since surgery he has had no redness, watering or pain in the operated eye. He has had no sudden loss of vision and no change in the upper lid.

He is diabetic for fifteen years on tablets. He is asthmatic and uses an inhaler when needed. He has no sulpha allergy and takes no aspirin or steroids. He has had no other eye surgery. There is no family history of glaucoma.

Closing (2 minutes)

SAY IT

To summarise, Mr X, a fifty-six-year-old asthmatic diabetic man, had trabeculectomy in the right eye six months ago.

Vision is 6/60 on the right and 6/24 on the left, improving to 6/12 and 6/9 with pinhole. The right eye has a flat, opaque bleb from eleven to two o'clock superiorly. It is highly vascular, without microcysts, with subconjunctival fibrosis. Seidel's test is negative. A surgical iridectomy at twelve o'clock is patent. The chambers are of normal depth. There is a grade three relative afferent pupillary defect on the right. Both lenses show grade two nuclear sclerosis. Goldmann applanation pressure is twenty-six millimetres of mercury on the right and twenty-four on the left. Gonioscopy shows open angles and a patent internal ostium on the right. The cup-disc ratio is 0.9 on the right and 0.75 on the left.

My diagnosis is right eye primary open-angle glaucoma status post trabeculectomy with failed bleb and grade two nuclear sclerosis. Left eye: primary open-angle glaucoma and grade two nuclear sclerosis.

I will stage the damage with Humphrey fields and optical coherence tomography. I will restart drops, avoiding beta-blockers because of his asthma. If the pressure stays above target, I will advise repeat trabeculectomy with mitomycin C at a fresh site. A glaucoma drainage device is the alternative. He must come at once if the eye becomes red or painful.

Quick recall

- **Good bleb:** avascular, numerous microcysts. **Failing:** lower, thicker, vascular, microcysts lost, IOP up.
- **Types 3 and 4** do not filter; a Tenon's cyst settles in **2-4 months**.
- Commonest cause of failure: **subconjunctival fibrosis**. Commonest problem after surgery: **shallow AC**.
- **Leak:** soft eye, Seidel positive. **Overfiltration:** good bleb. **Malignant glaucoma:** no bleb, IOP high or normal.
- Fluorescein **0.25%** for many leaks, **2%** for subtle ones.
- **MMC 0.2-0.5 mg/ml for 1-5 min; 5-FU 50 mg/ml for 5 min**, injection **5 mg**, 21 injections.
- Flap **3 × 4 mm**; ostium **1.5 × 2 mm**; PI wider than the ostium.
- Digital pressure above **12 mm Hg**; laser suture lysis up to **6 months**.
- First sign of bleb infection: **hyperaemia around the bleb**; steroids only after **12-24 hours** of antibiotics.
- Bleb-related endophthalmitis: **Streptococcus** most common.

Mnemonic: none in your mnemonics file for this case.

Read more: Baidya p.151, 153-155, 158, 164-166, 173, 177, 193-194, 265, 421, 441, 443-448, 578 · Namrata p.172-178, 200 · Aravind 4.1, 4.5, 4.15, 4.17, 4.18, 4.19, glaucoma model case sheet p.661-667

G3 POAG with a glaucoma drainage device

On the department checklist (not kept last year) · Long case

At a glance

- **Primary open-angle glaucoma (POAG) with a glaucoma drainage device (GDD):** a tube drains aqueous to a plate at the equator. It usually follows a **failed trabeculectomy**.
- Why it is kept: a stable patient, a tube and plate easy to show, and a long surgical viva.
- What examiners test: tube, plate cover, motility, intraocular pressure (IOP) control, the devices, the operation and its complications.

For primary open-angle glaucoma itself, see card G1.

Keywords the examiner listens for: alloplastic tube · equatorial reservoir · fibrous capsule · Ahmed glaucoma valve · Venturi effect · Molteno implant · Baerveldt implant · superotemporal quadrant · scleral patch graft · tube exposure · tube–corneal touch · pars plana tube · hypertensive phase · pseudo-Brown's syndrome · cyclophotocoagulation

History — positive and negative, with reasons

Chief complaint: “Mr X, 65-year-old male, has come for regular follow-up.”

History of present illness (house wording): “Patient was apparently normal ___ years before, after which he had defective vision in both eyes. It was gradual in onset, progressive in nature, not associated with pain or headache. He was found to have increased pressure in both eyes and started on eye drops. He underwent trabeculectomy in the right eye ___ years ago. As the pressure rose again, a glaucoma drainage device was put in the right eye ___ years ago at ___. Then he came to our institution for further follow-up.”

Positive history

Question	Why I ask — what the answer tells me
Which eye has the device, and since when?	It dates the surgery. Hypotony, a tied tube and passing diplopia are early; erosion, migration and corneal decompensation are late.
What was done before the device?	The usual indication is failed filtering surgery : failed trabeculectomies, failure with mitomycin C, or scarred conjunctiva . A device as the first operation suggests a refractory secondary glaucoma.
Which device was used?	A valved Ahmed works at once. A non-valved Molteno or Baerveldt is tied for 2–3 weeks, so the pressure stays high until the tie opens.
Was a laser or a stitch-pull done in the first weeks?	It points to a tied tube, opened by argon laser suturolysis or by pulling a rip cord .
Any re-operation in the first weeks?	Early complications: hypotony with a flat anterior chamber (AC) and choroidal effusion, suprachoroidal haemorrhage, or aqueous misdirection.
Any double vision since surgery?	Early and passing: oedema in the orbit and recti. Lasting: the bleb displaces the globe, or the muscles fibrose.
Any foreign-body feeling?	A dellen (a patch of corneal thinning from drying): raised conjunctiva over the patch graft breaks up the tear film. Or an exposed tube or plate.
How is the vision since surgery?	A fall suggests cataract progression, hypotony maculopathy, corneal oedema or glaucoma progression.
Which drops now: drug, strength, frequency?	The IOP fall is smaller than after trabeculectomy, so drops may still be needed. Miotics and alpha agonists must not be among them.
Does he use them regularly?	Poor compliance, not device failure, may explain a high pressure.

Negative history

Question	A “yes” would point to	How to tell them apart
Headache with vomiting and eye pain?	A sudden pressure rise: angle closure, neovascular glaucoma, or a blocked tube.	Gonioscopy: a closed angle; new vessels in the angle (NVA); or an open angle with a blocked tube tip.
Coloured haloes?	Corneal epithelial oedema from a pressure rise or endothelial failure. Also mucus, incipient cataract, vitreous opacities, snow blindness, a tilted lens implant.	Fincham (stenopaic slit) test : the glaucoma halo stays intact; a cataract halo breaks into colours.
Redness?	Chronic iritis, tube or plate exposure, or endophthalmitis.	Slit lamp: cells and flare in iritis; a bare tube or plate in exposure.
Watering?	Surface toxicity from long-term drops, a dellen, or exposure.	Fluorescein 1% staining for toxicity; slit lamp for a dellen or a bare tube.

Question	A "yes" would point to	How to tell them apart
Pain?	A pressure rise, iritis, or infection through an exposed tube.	Tonometry; cells and flare; inspect the plate quadrant.
Sticky discharge?	Infection through an exposed tube: risk of endophthalmitis.	Lift the upper lid for a bare tube; check the lids for blepharitis.
Injury to the eye?	Post-traumatic (angle-recession) glaucoma.	Gonioscopy: a widened ciliary body band compared with the other eye.
Frequent change of glasses?	High pressure impairs accommodation; field loss is mistaken for poor vision. Hypermetropia goes with angle closure.	Refraction; gonioscopy for an open or closed angle.
Retinal laser or injections into the eye?	Neovascular glaucoma , where a device is often the first operation.	Undilated slit lamp for new iris vessels (NVI); gonioscopy for NVA.
Red eye treated with steroid drops?	Uveitic glaucoma , or steroid-induced glaucoma.	Keratic precipitates, posterior synechiae, cells; peripheral anterior synechiae (PAS).
Corneal transplant?	Post-keratoplasty glaucoma; a pars plana tube suits it.	The graft is seen at the slit lamp.

Past, drug, treatment, personal and family history

Question	Why it matters (diagnosis or treatment)
Diabetes?	New vessels (neovascular glaucoma); diabetic retinopathy adds to the risk of retinal detachment after a device.
Asthma or chronic airway disease?	Beta-blockers are contraindicated.
Heart failure or a slow pulse?	Beta-blockers are contraindicated.
Drug allergy, above all to sulphonamides?	Sulphonamide allergy bars carbonic anhydrase inhibitors.
Warfarin?	It is stopped before any further device surgery.
Steroid use?	Steroid-induced glaucoma; on the department's checklist.
Cataract surgery?	Pseudophakic eyes may take a pars plana tube.

Examination — must-do steps and their viva

General and systemic (this case only)

- Pulse and chest: bradycardia, heart failure or wheeze bars beta-blockers.
- Face: a **port-wine stain** suggests Sturge-Weber syndrome, another glaucoma treated with devices.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity	Useful vision; fall since surgery	6/18, pinhole (PH) 6/12
Extraocular movements (EOM)	Deviation, restriction, diplopia	Orthophoric; EOM full, no diplopia
Lids	Lid over the plate; closure; blepharitis	Normal; complete closure
Conjunctiva	Bleb over the plate; graft; exposure; dellen; old trabeculectomy bleb	Bleb over the plate superotemporally; no exposure; old bleb at 12 o'clock flat and vascularised
Cornea	Oedema near the tube; tube–corneal touch	Clear; no tube touch
Anterior chamber	Van Herick (VH) grade; cells; hyphaema	VH grade IV (ND = normal depth); quiet
Tube	Clock hour, length, bevel, course, gaps, tip	10 o'clock, 2 mm, bevel up, parallel to iris, tip patent
Iris	Tube–iris touch; peripheral iridectomy (PI) and its patency; new vessels	CPN (colour, pattern normal); surgical PI at 12 o'clock, patent
Pupil, lens	Relative afferent pupillary defect (RAPD), before any drop; cataract progression	RTL (round, reacting to light) 3 mm, no RAPD; nuclear sclerosis grade 2
Tn by GAT (tension by Goldmann applanation tonometry)	Before gonioscopy; time; drops	14 mm Hg at 10 am
Gonioscopy	Tube tip; PAS; NVA	Cross diagram, deepest structure seen
Fundus, fields	Disc (card G1); folds; effusion; confrontation	Cup–disc ratio (CDR) 0.8; no folds

Step 1 — Visual acuity

Do: Test each eye with the Snellen chart, then pinhole, then near vision. Ask him to cover the other eye without pressing.

Record: "RE (right eye) 6/18, PH 6/12; LE (left eye) 6/9; N8 at 33 cm."

Q Why does useful vision matter here?

Potential for useful vision is a must before device surgery. An eye without it is treated by cyclodestruction instead.

Q Why may vision fall after a device?

Cataract progression, hypotony maculopathy and chorioretinal folds from prolonged hypotony, and corneal oedema or graft failure. Exclude glaucoma progression too.

Step 2 — Ocular motility and diplopia

Do: Cover test in the primary position, then movements in nine gazes, asking about double vision in each.

Record: "Orthophoric; EOM full in all gazes; no diplopia."

Q Why test motility in a device eye?

The plate lies between two recti under a bulky bleb. **Diplopia** comes from the bleb displacing the globe or from muscle fibrosis.

Q Why is the plate put superotemporally, not superonasally?

The **superotemporal quadrant** is preferred; inferonasal is next. A superonasal plate causes vertical strabismus (**pseudo-Brown's syndrome**) and can impact on the optic nerve.

Q What is Brown's syndrome?

Brown's syndrome is absent elevation in adduction with normal elevation in abduction; forced duction is positive. A superonasal plate can mimic it.

Step 3 — The plate and tube: conjunctival cover, patch graft, exposure

Do: Ask him to look down and away from the plate; lift the upper lid gently. Inspect the bleb, the patch graft and the conjunctiva over the tube. Then look at 12 o'clock for the old trabeculectomy bleb.

Record: "Superotemporal bleb over the plate; patch graft covered; no tube or plate exposure; no dellen. Old bleb at 12 o'clock: flat, opaque, highly vascular with corkscrew vessels. No microcysts; subconjunctival fibrosis present. Suggestive of a failed bleb."

Q How does a device bleb differ from a trabeculectomy bleb?

A **device bleb** is more posterior, with an organised capsule separable from Tenon's capsule. It has no conjunctival microcysts.

Q Why a scleral patch graft, and its complications?

A **scleral patch graft** covers the 3–5 mm of tube exposed to lid contact. It prevents tube erosion, hypotony, tube–corneal touch and scleral perforation. Complications: dellen and scleral melt. A dellen is treated with artificial tears and patching.

Q Why does tube exposure matter, and how is erosion managed?

Tube exposure is associated with endophthalmitis. Erosion follows graft melting with poor lid closure or dry eye. A **stent** (Storz silastic tube) lengthens and protects the tube.

Q How do you describe the old trabeculectomy bleb?

Height, clarity, horizontal extent, location over the limbus, vascularity, microcysts, subconjunctival fibrosis and sutures. A flat, vascular, fibrosed bleb without microcysts suggests a **failed bleb**. The signs of a **failing bleb** are reduced height, a thicker wall, vascularisation, loss of microcysts and raised pressure.

Step 4 — Anterior chamber and the tube: position, touch, occlusion

Do: At the slit lamp, note the tube's clock hour, length, bevel, course, gaps from endothelium and iris, and tip. Grade the chamber by Van Herick.

Record: "Tube at 10 o'clock, 2 mm, bevel up, parallel to the iris, clear of cornea and iris, tip patent; AC VH grade IV (ND), quiet."

Q What is the correct position of the tube?

It enters through a **23-gauge needle tract** at the posterior edge of the blue limbus, **bevel up**. About **1.5–2 mm** lies inside, parallel to the iris; this prevents migration.

Q When does the tube migrate?

In **postoperative hypotony** it moves forward onto the cornea, lens or iris. With poor plate fixation, the plate slides back and the tube extrudes.

Q What blocks the tube?

Blood, fibrin, iris or vitreous. Vitreous in the chamber is noted before surgery. In uveitis, steroid cover stops exudates blocking the lumen.

Q How do you examine the old iridectomy?

A **surgical iridectomy** has sharp, well-defined edges and normal surrounding iris. A laser iridotomy has ragged edges with pigment dispersion. On **retroillumination**, a light reflex seen through the defect means it is patent.

Step 5 — Cornea and endothelium

Do: Examine the whole cornea, above all near the tube tip, for oedema and Descemet's folds. Stain with fluorescein 1% for surface toxicity.

Record: "Cornea clear; no oedema near the tube; no tube–corneal touch."

Q Why does the cornea decompensate after a device?

Endothelial cell loss, from needle-tip trauma at surgery or tube–corneal touch. Oedema and graft failure occur with or without touch.

Q How do you assess the endothelium?

Specular microscopy gives the endothelial cell density per square millimetre. A normal cornea has 60–80% hexagonal cells; pleomorphism means stress.

Step 6 — Tonometry

Do: Goldmann applanation before gonioscopy and dilatation, with anaesthetic, fluorescein and cobalt-blue light; align the green mires.

Record: “Tn by GAT 14 mm Hg (RE), 16 mm Hg (LE) at 10 am; timolol 0.5% BD (twice daily).”

Q Why tonometry before gonioscopy and dilatation?

Gonioscopy presses on the cornea and lowers the next reading. **Dilatation** raises the pressure by 4–5 mm Hg for a while.

Q What causes a high pressure with a device?

A **tied tube**, a **blocked tube**, a **thick fibrous capsule**, or the hypertensive phase. Exclude poor compliance.

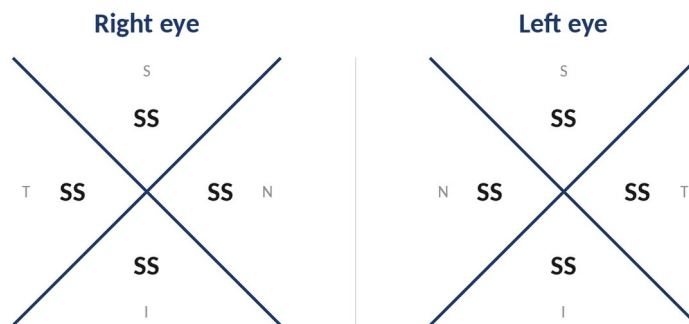
Q What causes a low pressure?

Hypotony from a leak around the tube or excessive run-off. Look for a flat chamber, choroidal effusion and hypotony maculopathy.

Step 7 — Gonioscopy

Do: After tonometry, use a Goldmann 2-mirror lens; a 4-mirror lens for indentation. Note the deepest structure in each quadrant.

Record: A cross diagram for each eye with the deepest structure seen.



Deepest structure seen in each quadrant (SS = scleral spur) — open angles in all quadrants. RE: tube tip free; no PAS or NVA

Q What do you look for on gonioscopy in a device eye?

The **tube tip** (free or blocked), **PAS** (areas to avoid for a tube) and **new vessels in the angle**.

Q What is done about angle new vessels before surgery?

They are treated first, to halt new growth, by **panretinal photocoagulation** or goniotomy.

Step 8 — Optic disc and fundus

Do: Dilate after tonometry and gonioscopy. Examine the disc with a +90 dioptre (D) lens and the periphery with the indirect ophthalmoscope.

Record: “Disc vertically oval, CDR 0.8, rim thin superiorly and inferiorly; no folds; periphery normal.”

Q Which posterior-segment signs follow a device?

Chorioretinal folds and hypotony maculopathy from prolonged hypotony; choroidal effusion; retinal detachment after scleral perforation or with diabetic retinopathy.

Q How is a scleral perforation at surgery recognised and treated?

A **gush of vitreous gel** is seen. Cryotherapy is applied to the area under the indirect ophthalmoscope.

Step 9 — Fields

Do: Confrontation in each eye; then request automated perimetry.

Record: “Fields by confrontation: nasal constriction in RE; full in LE.”

Q Why do fields matter after a device?

Fields show whether glaucoma still progresses at this pressure. Progression despite treatment is the trigger to step up.

Q How often do you repeat them?

Every 6 months in mild to moderate disease; every 3 months in advanced disease.

Summary and complete diagnosis

Mr X, a ___-year-old man with primary open-angle glaucoma for ___ years, had a failed trabeculectomy. A ___ device went into the ___ eye ___ years ago. He uses ___. The tube enters at ___ o'clock, ___ mm, bevel ___, tip ___. The plate lies ___ under a ___

bleb; the graft is _____. The old trabeculectomy bleb is _____. Motility _____. Tension by GAT ____ / ____ mm Hg at _____. Gonioscopy _____. CDR ____ / _____. Fields _____.

eye → POAG → stage → operations → tube → IOP control → lens → other eye

- “Right eye primary open-angle glaucoma, severe stage, status post trabeculectomy with failed bleb and glaucoma drainage device implantation, with a well-positioned patent tube, intraocular pressure under control on medical management, and grade 2 nuclear sclerosis; left eye primary open-angle glaucoma, moderate stage, on medical management.”
- “Right eye primary open-angle glaucoma status post Ahmed glaucoma valve implantation with tube–corneal touch and localised corneal oedema.”
- “Right eye primary open-angle glaucoma status post drainage device implantation, intraocular pressure above target, suggestive of a thick fibrous capsule.”

Differential diagnosis

With a device in place, ask which glaucoma it was put in for. These refractory glaucomas are also treated with devices:

Condition	Points for	Points against	How to tell apart
Neovascular glaucoma	A device is often the first operation; diabetes	No NVI, no NVA	Undilated slit lamp for NVI; gonioscopy for NVA
Uveitic glaucoma	An indication; chronic iritis after a device	No recurrent red eye; quiet AC	Keratic precipitates, posterior synechiae, PAS
Angle-recession glaucoma	Unilateral; trauma is an indication	No injury; devices help little here	Gonioscopy: widened ciliary body band
Glaucoma after keratoplasty	An indication; pars plana tube	No graft	Graft seen at the slit lamp
Sturge–Weber syndrome	Devices used; Ahmed better than Molteno	No facial stain	Port-wine stain; dilated episcleral vessels

Exclude first: neovascular glaucoma — look for NVI before dilating, and NVA on gonioscopy.

How I will proceed — investigations

Test — what I do	Why	Expected finding	Significance — what it changes
Diurnal tonometry: GAT through the day	Confirm control; find the peak	Below target; swing under 5 mm Hg	Above target or wide swings → step up
Central corneal thickness (pachymetry)	GAT is calibrated for 520 µm: thin (< 500 µm) reads falsely low, thick (> 570 µm) falsely high	Normal	Correct by about 0.7 mm Hg per 10 µm. In corneal oedema, take the true pressure as higher
Specular microscopy	Endothelium near the tube	About 2500 cells/mm ² in late adult life	Falling count → watch for decompensation
Automated perimetry, 24-2	Stage; progression	Arcuate defects in the right eye	Progression → step up
Optical coherence tomography (OCT), nerve fibre layer	Structural baseline; compare old records	Thinning matching the field	Structural progression → step up
B-scan ultrasound, if the view is poor	Choroidal effusion or haemorrhage	No choroidal detachment	Dome-shaped elevation: anechoic = serous, echoic = blood

Say: “I will proceed with diurnal tonometry to confirm control. I will stage the damage with fields and OCT, and check the endothelium by specular microscopy.”

How I will manage

Aim

Keep the pressure at target, keep the device working and covered, and catch tube, corneal and motility problems early.

The ladder

1. **Device working, pressure at target:** I will continue his drops. Every visit, I will check the tube, cornea, plate cover, motility and pressure.
2. **Pressure above target:** I will look for poor compliance, a tied tube, a blocked tip or a thick capsule. I will add drops, never a miotic or an alpha agonist.
3. **Early high pressure with a tied tube:** this rise is a risk to a nerve with advanced damage. The tie holds for 2–3 weeks while the capsule forms. Then I will release it by argon laser suturolysis, or pull the rip cord.
4. **Tube exposure or erosion:** I will repair it urgently with a stent, because exposure is associated with endophthalmitis.

5. **Uncontrolled after repeated surgical failure, or poor visual potential:** I will advise transscleral diode cyclophotocoagulation.

Drugs in this case

Drug (strength · frequency)	Mechanism of action	Adverse effects	Contraindications
Timolol 0.5% · once or twice daily	Beta-blocker: less aqueous production	Ocular: irritation, dry eye, blepharitis. Systemic: bronchospasm, bradycardia, hypotension	Asthma, chronic obstructive pulmonary disease, cardiac failure, bradycardia
Dorzolamide 2% · three times daily	Carbonic anhydrase inhibitor: less aqueous production	Irritation, dry eye, burning	Sulpha allergy, renal failure, chronic liver disease
Bimatoprost 0.01% or 0.03% · at bedtime	Prostaglandin analogue: more uveoscleral outflow	Hyperaemia, longer darker lashes, brown iris, uveitis, macular oedema	Avoid in neovascular glaucoma and Sturge-Weber syndrome
Not after a device: pilocarpine, brimonidine	Miotic; alpha agonist	Increase vascularisation of the bleb capsule	Contraindicated after device surgery

Full profiles: card GX.

Laser and surgery in this case

Procedure (indication)	Key steps	Advantages	Disadvantages and complications
Release of the tube tie (tied non-valved tube)	Argon laser suturolysis of an 8-0 polyglactin or 10-0 prolene tie: 50 µm, 0.02–0.1 s, 250–1000 mW. Or pull the 4-0 chromic catgut rip cord	Opens the tube without re-operation	The tie must stay 2–3 weeks while the capsule forms
Stent (tube erosion)	A stent of Storz silastic tube is placed	Lengthens the tube and protects it from laceration	None given in your books; the exposure itself risks endophthalmitis
Transscleral diode cyclophotocoagulation (repeated failure; poor vision)	Local anaesthesia; 1500–2000 mW for 1.5–2 s; 12–24 burns behind the limbus over 360°, sparing 3 and 9 o'clock; fewer in a seeing eye. Retreat after at least 1 month	Less aqueous production; quick, easy; less energy than the neodymium:YAG (Nd:YAG) laser	Hypotony, phthisis bulbi, retinal detachment, uveitis, malignant glaucoma, sympathetic ophthalmia

Follow-up and counselling

- I will see him every 3 months, or monthly if the damage is advanced, with GAT at each visit.
- I will treat any blepharitis, because it raises the risk of late infection.
- He must report at once any pain, redness, discharge, falling vision or double vision.

What you must know

Definition

A **glaucoma drainage device** (aqueous shunt device) “consists of an alloplastic tube leading to a large equatorial reservoir that drains aqueous humour from the anterior chamber”. In plain words, a tube takes aqueous to a plate near the equator, where a bleb absorbs it.

Parts: a tube (anterior chamber or pars plana), a plate (explant) at the equator, and a valve if valved. Over them lie the fibrous capsule (bleb) and a scleral patch graft. All devices use materials fibroblasts cannot firmly adhere to.

Related terms:

- **Seton:** non-lumened; aqueous flows along its surface.
- **Shunt:** a hollow tube allowing two-way flow.
- **Valve:** one-way flow that needs an activating pressure.
- **Hydraulic conductivity:** drainage rises with capsule area; resistance rises with capsule thickness.
- **Hypertensive phase:** a transient pressure rise soon after surgery, with poor control although the device works.

Classification

Route of drainage	Examples	Merits and demerits
Through the limbus to the anterior subconjunctival space	Krupin–Denver valve	Fails more from subconjunctival fibrosis
Through the limbus to a posterior sub-Tenon's reservoir	Valved or flow-restricted: Ahmed, long Krupin. Non-valved: Molteno, Baerveldt	Protected tube exit; larger area, more drainage; less erosion under Tenon's capsule

Device	Valve	Plate	Tube
Molteno (the first)	Non-valved	Circular polypropylene; single, double or triple	0.63 mm outer, 0.3 mm inner

Device	Valve	Plate	Tube
Baerveldt	Non-valved	Silicone, fenestrated; single or double	0.64 mm outer, 0.3 mm inner
Ahmed glaucoma valve	Valved	Oval polypropylene, 184 mm ² (double 364 mm ²)	Silicone; outflow above 8–10 mm Hg

Pathogenesis

1. The plate stimulates a **fibrous capsule**, which becomes the wall of the bleb.
2. Aqueous drains down the tube into the space between plate and capsule.
3. It passes by passive diffusion into the periocular tissues.
4. Lymphatics and venous capillaries take it up.

A valved device works at once. A non-valved tube is tied for 2–3 weeks while the capsule forms and resists outflow.

Indications and preoperative checks

- **Refractory glaucomas:** uveitic, neovascular, congenital, refractory juvenile, after keratoplasty, aniridia, post-traumatic, iridocorneal endothelial (ICE) syndrome, pseudophakic, after vitrectomy.
- **Failed filtering surgery:** failed trabeculectomies, failure with antimetabolites, or conjunctiva too scarred to dissect. Also failed primary glaucomas.
- **Checks:** useful vision (a must); lid position; tear film; blepharitis; peripheral corneal clarity; vitreous in the AC; cataract; gonioscopy.
- **Drugs:** stop pilocarpine or echothiophate and warfarin; continue other glaucoma drops.

The operation — key steps

1. Limbus-based or fornix-based conjunctival flap.
2. **Superotemporal quadrant.**
3. Plate centred at the equator between two recti, front edge **8–10 mm** behind the limbus, sutured to episclera.
4. Non-valved tube tied for **2–3 weeks:** rip cord, laser-released tie, or nylon to a spacer.
5. AC tube: **bevel up**, through a **23-gauge needle tract** at the posterior edge of the blue limbus. About **1.5–2 mm** lies inside, parallel to the iris.
6. Or **pars plana tube:** complete vitrectomy; **3–4 mm** behind the limbus; **bevel down**.
7. Full-thickness **scleral patch graft** over the tube.
8. Reform the AC with sodium hyaluronate (Healon).

Trabeculectomy versus drainage device

Feature	Drainage device	Trabeculectomy
Place in open-angle glaucoma	After failed filtering surgery, or scarred conjunctiva	First operation after failed medical treatment
Bleb	Posterior; organised capsule; no microcysts	At the limbus; microcysts
Pressure fall	Smaller percentage	Larger
Neovascular glaucoma	Often primary; comparable results	Poor alone; up to 95% with mitomycin C and bevacizumab or photocoagulation
Silicone-oil-filled eye	Better outcomes; oil can migrate up the tube	Limited role
Angle recession	Limited benefit	Works with antimetabolites, less well than in POAG

Complications

Complication	Cause	Prevention or treatment
During surgery: hyphaema; lens or endothelial damage	Iris-root injury; wrong tube length; needle-tip trauma	Correct length, parallel to the iris
Scleral perforation	Needle injury while suturing the plate	Cryotherapy under indirect ophthalmoscopy
Early: pressure rise	Tie not yet absorbed	Laser release of the tie
Hypotony, flat AC, choroidal effusion	Leak around the tube; excess run-off	Prevent: needle tract, tie, Healon. Your books give no device-specific treatment
Suprachoroidal haemorrhage; aqueous misdirection	Hypotony with high preoperative pressure; choroidal swelling	Prevent hypotony
Inflammation; endophthalmitis	Chronic uveitis; contamination at surgery	Operate when quiet; steroid cover
Late: cataract; corneal oedema, graft failure	Hypotony; endothelial loss with or without touch	Tube parallel to the iris; patch graft
High pressure, open tube	Thick fibrous capsule; hypertensive phase	Drops (no miotic or alpha agonist); cyclophotocoagulation

Complication	Cause	Prevention or treatment
Hypotony maculopathy, chorioretinal folds	Excess run-off	—
Diplopia, strabismus	Bleb displacing the globe; muscle fibrosis	Superotemporal plate
Graft melt, erosion, endophthalmitis	Poor lid closure; dry eye; tube exposure	Stent (Storz silastic tube)
Tube occlusion; migration	Blood, fibrin, iris, vitreous; poor plate fixation	Vitrectomy for a pars plana tube; secure plate
Retinal detachment	Scleral perforation; diabetic retinopathy	—

Prognosis

- **Good:** useful vision; a quiet eye; good lid closure; in neovascular glaucoma, anti-vascular endothelial growth factor (anti-VEGF) and photocoagulation first.
- **Poor:** active inflammation; angle recession; silicone oil, which can migrate up the tube; dry eye; a thick fibrous capsule.

Recent advances

- (extra) Neovascular glaucoma with vitreous haemorrhage: inject anti-VEGF into the vitreous (bevacizumab 1.25 mg/0.05 ml). Do it 2–3 days before vitrectomy, photocoagulation and a valve; success is high.

Viva questions

Why did you ask...?

Q Why did you ask about previous trabeculectomy?

Multiple failed trabeculectomies and extensive conjunctival scarring are the main indications for a device.

Q Why did you ask about diabetes and retinal laser?

They point to **neovascular glaucoma**, where a device is often the primary operation. Diabetic retinopathy also raises the risk of retinal detachment.

Q Why did you ask about warfarin?

Warfarin is stopped before device surgery, with pilocarpine or echothiophate (to reduce postoperative inflammation). Other glaucoma drops continue until surgery.

Q Why did you ask about a sticky discharge?

Blepharitis raises the risk of late infection after device surgery. Endophthalmitis is associated with tube exposure, so a discharge sends me looking for a bare tube.

Diagnosis and differentials

Q Define a glaucoma drainage device.

A **glaucoma drainage device** “consists of an **alloplastic tube** leading to a large **equatorial reservoir** that drains aqueous humour from the anterior chamber”.

Q What are the indications for a device?

All refractory glaucomas, such as uveitic, neovascular, congenital and post-traumatic. Also multiple failed trabeculectomies, extensive conjunctival scarring and failed primary glaucomas.

Q What is the principle of a drainage device?

The plate stimulates a **fibrous capsule**, the bleb wall. Aqueous collects between plate and capsule, diffuses into periorbital tissues, and lymphatics and veins take it up.

Q Why did his trabeculectomy fail?

Subconjunctival fibrosis is the most common cause of bleb failure. Other causes are a fistula blocked by clot, iris, ciliary body, lens or vitreous, and poor surgical technique.

Q What are setons, shunts and valves?

A **seton** is non-lumened. A **shunt** is hollow, with two-way flow. A **valve** allows one-way flow and needs an activating pressure.

Q What decides how well a device drains?

Hydraulic conductivity: drainage is directly proportional to capsule area; resistance is directly proportional to capsule thickness.

Investigations

Q Why do diurnal pressures matter?

Diurnal variation shows the baseline, the swings and the peak. Keep the pressure below target with a swing under 5 mm Hg.

Q What do you check before device surgery?

Useful vision, lid position, scleral exposure, tear film, blepharitis and peripheral corneal clarity. Then the lens, vitreous in the chamber, and gonioscopy.

Management — drugs

Q Which drops are contraindicated after a device?

Miotics and alpha agonists, because they increase vascularisation of the bleb capsule.

Q Why avoid prostaglandin analogues in some device eyes?

In **neovascular glaucoma** they may increase inflammation. In **Sturge-Weber syndrome** they are of no benefit, as episcleral venous pressure is high; they may cause anterior uveal effusion.

Q Why steroid cover after a valve in uveitic glaucoma?

The **Ahmed valve** goes in only when inflammation is controlled. **Steroid cover** stops exudates blocking the tube early after surgery.

Management — laser and surgery

Q How does the Ahmed glaucoma valve work?

The **Ahmed glaucoma valve** has two silicone membranes acting as a one-way valve by the **Venturi effect** (Bernoulli's principle). It allows outflow only above 8–10 mm Hg. At a high pressure it opens fully; as the pressure falls, the opening narrows.

Q Compare valved and non-valved devices.

A **valved device** is a simple one-step procedure. Safety and efficacy otherwise appear equal; in Sturge-Weber syndrome the Ahmed does better than the Molteno.

Q Why is a non-valved tube tied?

The **tie** gives 2–3 weeks for the capsule to form and resist outflow. Without it, early drainage is unrestricted.

Q How does a device compare with trabeculectomy?

The **percentage pressure fall is smaller** with a device. A device follows failed filtering surgery; trabeculectomy follows failed medical treatment.

Q Why a device rather than a repeat trabeculectomy with mitomycin C?

After failed filtration, the options are a **repeat filtering procedure with antimetabolites** or a **drainage implant**. Mitomycin C is used at 0.2–0.5 mg/ml for 1–5 minutes. A device is chosen when the conjunctiva is too scarred to dissect, or when surgery with antimetabolites has already failed.

Q What if the pressure stays high despite the device?

Cyclodestruction: transscleral diode or endolaser cyclophotocoagulation reduces aqueous production. Risks: hypotony, phthisis bulbi, retinal detachment.

Q What are the indications for cyclophotocoagulation?

Cyclophotocoagulation relieves refractory pain from high pressure in a blind eye. It also treats repeated failure of other glaucoma surgeries and glaucoma after keratoplasty. It suits high-risk eyes where other surgery is not feasible.

Q When do you choose a pars plana tube?

A **pars plana tube** suits aphakic or pseudophakic eyes, eyes after keratoplasty, a shallow chamber or poor endothelium.

Q What are the advantages of the Baerveldt implant?

The **Baerveldt implant** is easy to implant, drains more than the **Molteno implant** and fits the globe. Its fenestrations lower the bleb and reduce diplopia.

Q Why a needle tract rather than a sclerectomy?

A **21- or 23-gauge needle tract** gives a tight fit and prevents aqueous leak around the tube.

Q Why reform the chamber with Healon?

Sodium hyaluronate reduces postoperative hypotony and a flat chamber. It shows the tube position and slows early drainage.

Complications

Q What are the complications of valve implantation?

Early or late failure, corneal decompensation, diplopia, hypotony, hyphaema, scleral perforation, tube erosion with endophthalmitis, and plate migration or expulsion.

Q What causes hypotony after device surgery?

Leak around the tube where it passes the eye wall, in both one- and two-stage surgery. A wide sclerotomy or excess run-off adds to it.

Q The chamber is flat after device surgery. How do you find the cause?

Tonometry first. A low pressure with choroidal effusion means hypotony from excess run-off. A diffusely shallow chamber with normal or high pressure suggests **aqueous misdirection** (malignant glaucoma). Ultrasound biomicroscopy shows the ciliary processes rotated forward. B-scan tells a serous effusion (anechoic) from a haemorrhage (echoic).

Q What are the phases after device surgery?

A **hypertensive phase** and a **hypotensive phase**. The hypertensive phase is a transient pressure rise soon after surgery, with poor control although the device works.

Examiner traps

- Calling the bleb over the plate a trabeculectomy bleb, or expecting microcysts.
- "Tube in the AC" without clock hour, length, bevel, course, gaps and tip.
- Not lifting the upper lid to see the plate, graft and any exposure.
- Skipping extraocular movements and diplopia.
- Missing the old trabeculectomy bleb at 12 o'clock and its iridectomy.
- Doing gonioscopy before tonometry.
- Advising pilocarpine or brimonidine after a device.
- Offering a device to an eye without useful vision.

Say-it script

Opening (about 1 minute)

SAY IT

Mr X, a 65-year-old man, has come for regular follow-up. He was apparently normal 10 years ago. Then he had defective vision in both eyes, gradual in onset and progressive, without pain or headache. He was found to have raised pressure in both eyes and was started on eye drops.

Six years ago he had a trabeculectomy in the right eye. The pressure rose again, and three years ago an Ahmed glaucoma valve was put in. He had double vision for two weeks after surgery, which settled.

He has no double vision now, and no redness, pain, watering or discharge. He has no headache with vomiting, coloured haloes, trauma or frequent change of glasses. He has had no retinal laser, injections into the eye, or red eye treated with steroids.

He has been diabetic for eight years, on tablets. He has no asthma or sulpha allergy and takes no warfarin. He uses timolol 0.5 percent twice daily in both eyes, regularly and without side effects.

Closing (2 minutes)

SAY IT

To summarise, Mr X is a 65-year-old man with primary open-angle glaucoma in both eyes. He had a failed trabeculectomy and then an Ahmed glaucoma valve in the right eye.

Visual acuity is 6/18, improving to 6/12 with pinhole, on the right and 6/9 on the left. Eye movements are full, without diplopia. In the right eye, a bleb lies over the plate in the superotemporal quadrant. The scleral patch graft is covered, with no tube exposure. The old trabeculectomy bleb at twelve o'clock is flat and vascular, suggestive of a failed bleb. The tube enters the chamber at ten o'clock, two millimetres long, bevel up and parallel to the iris. There is no tube-corneal touch, and the tip is patent. At ten in the morning, Goldmann applanation pressure is 14 millimetres of mercury on the right. It is 16 on the left. Gonioscopy shows open angles. The cup-disc ratio is 0.8 on the right and 0.6 on the left.

My diagnosis is: right eye primary open-angle glaucoma, severe stage, status post trabeculectomy with failed bleb and glaucoma drainage device implantation. The tube is well positioned and patent. Intraocular pressure is under control on medical management. There is grade 2 nuclear sclerosis. Left eye primary open-angle glaucoma, moderate stage, on medical management.

I will confirm control with diurnal tonometry, and stage the damage with fields and optical coherence tomography. I will continue his drops, avoiding miotics and alpha agonists. If the pressure rises despite this, I will consider cyclophotocoagulation.

Quick recall

- Device: alloplastic tube + equatorial plate; aqueous → capsule space → tissues → lymphatics and veins.
- Drainage $\propto$ capsule area; resistance $\propto$ capsule thickness.
- **Ahmed:** valved, **184 mm²**, outflow above **8–10 mm Hg**. **Molteno, Baerveldt:** non-valved, **tied 2–3 weeks**.
- Plate **superotemporal**, **8–10 mm** behind the limbus; superonasal → pseudo-Brown's.
- AC tube: **bevel up**, **1.5–2 mm**, parallel to iris. Pars plana: **3–4 mm**, **bevel down**, after vitrectomy.
- Device bleb: posterior, **no microcysts**; IOP fall smaller than trabeculectomy.
- After a device: **no miotics**, **no alpha agonists**.
- Erosion → **stent**; exposure → endophthalmitis.
- Uncontrolled or no useful vision → **cyclophotocoagulation**.

Mnemonic: none in your mnemonics file for this case.

Read more: Baidya p.109–110, 151, 157, 165, 173–180, 251, 421, 441, 596, 740 · Namrata p.175–178, 186, 194–195, 198–199, 339 · Aravind 4.1, 4.6, 4.12, 4.18, 4.19, 4.20, 4.21, glaucoma model case sheet p.661–664

G11 Other glaucomas the examiner may ask — childhood and secondary glaucomas

Extra — not on the department checklist · asked in viva and ward rounds · Viva sheet

Why this card

- These conditions are not on your department checklist. Examiners still ask them in the viva, on ward rounds and as spot cases.
- Each is also a differential on the checklist glaucoma cards, so it sharpens those answers.

Keywords the examiner listens for: Haab's striae · corneal diameter · examination under anaesthesia · goniotomy · trabeculotomy · port-wine stain · episcleral venous pressure · phacolytic · phacomorphic · Posner-Schlossman syndrome · steroid responder · Krukenberg spindle · reverse pupillary block · angle recession · ciliary block

Buphthalmos (primary congenital glaucoma)

Definition: Buphthalmos is "a term that is applied to congenital glaucomas with enlargement of globe that appear during the first 3 years of life". In plain words, raised pressure stretches the soft infant eye, so the cornea and globe enlarge ("ox-eye"). The main cause is **primary congenital glaucoma**, from a malformed angle (**isolated trabeculodysgenesis**).

Signs to show

1. **Observe first.** The classic triad is **epiphora** (watering), **photophobia** and **blepharospasm**. Shine a torch: the child screws the eyes shut. The photophobia comes from corneal oedema and the breaks in Descemet's membrane.
2. **Globe and sclera.** The eyes are large, bilateral in 65–80%. The sclera looks **bluish** because it is stretched.
3. **Cornea.** The horizontal **corneal diameter** is over 12 mm, and the cornea is hazy from oedema. **Haab's striae** are horizontal breaks in Descemet's membrane, or breaks concentric to the limbus. The oedema clears with treatment; the striae stay for life. Severe cases may show acute hydrops.
4. **Anterior chamber deep; iris normal or with loss of crypts; lens normal.**
5. **Examination under anaesthesia (EUA)** gives the rest, in both eyes:

What I measure	How	Normal infant	Suggests congenital glaucoma
Intraocular pressure (IOP)	Perkins hand-held applanation tonometer or Tonopen	11–14 mm Hg	Over 14 mm Hg, or over 5 mm Hg difference between the eyes
Horizontal corneal diameter	Both eyes, compared	10–10.5 mm; grows 0.5–1.0 mm in the first year	Over 12 mm
Angle	Direct gonioscopy, infant Koeppel lens, child supine	No vessels in the angle	High iris insertion; compacted meshwork; Loch Ness monster vessels (loops from the major arterial circle over the iris root); Lister's morning mist (fine fluffy tissue on the peripheral iris)
Optic disc	Ophthalmoscopy	Small physiological cup; pink disc	Cup-disc ratio over 0.3, or asymmetry between the eyes
Refraction	Cycloplegic refraction	—	A myopic shift as the globe enlarges

Most general anaesthetics lower IOP, but ketamine and suxamethonium raise it.

Diagnosis line

- "A 6-month-old boy with **primary congenital glaucoma (infantile type) in both eyes**, with buphthalmos, corneal oedema and Haab's striae."

How to tell it apart

Condition	How to tell apart
Megalocornea	Non-progressive, symmetric enlargement over 12 mm, with normal corneal thickness and contour. Buphthalmos adds raised IOP, a cloudy cornea, Haab's striae and cupping.
Congenital hereditary endothelial dystrophy (CHED)	The cornea is two to three times thicker but not enlarged, with symmetric clouding and no Descemet breaks. The eye is white, without photophobia; IOP is normal.
Birth trauma (forceps injury)	Descemet tears are vertical or oblique . Oedema appears at once and clears by itself. Corneal size and IOP are normal, and there is no photophobia. Soft-tissue injury may show.

How I will manage

1. **Confirm under anaesthesia, and examine the siblings.** About 10% of cases are inherited, autosomal recessive, in the **CYP11B1** gene.

2. **Drops only bridge the gap; treatment is surgical.** Timolol, a beta-blocker, reduces aqueous production; avoid it in asthma and bradycardia. A carbonic anhydrase inhibitor also reduces production; avoid it in sulpha allergy, renal failure and chronic liver disease. A prostaglandin analogue increases uveoscleral outflow. I give **no alpha-2 agonists under 2 years**: they cause respiratory depression, coma, hypotension and bradycardia.
3. **Goniotomy if the cornea is clear.** Under a direct goniolens, a Swan knife cuts the anterior meshwork just behind Schwalbe's line, over about 120°. Advantage: aqueous flows straight into Schlemm's canal. Disadvantage: it needs a clear cornea; risks are hyphaema, iridodialysis and a shallow chamber.
4. **Trabeculotomy if the cornea is hazy.** Ab externo, I find Schlemm's canal where blue cornea meets white sclera. I derooft it and rotate the probe into the anterior chamber, on both sides. Advantage: it can be done when the cornea is hazy. If goniotomy or trabeculotomy fails, I move on:
 - **Trabeculectomy**, with or without mitomycin C, usually combined with trabeculotomy.
 - A **drainage device** (Molteno, Ahmed or Baerveldt). Risks: the tube touching the cornea, implant migration and cataract.
 - **Cyclodestruction** (diode cyclophotocoagulation) for refractory eyes. Risks: hypotony and retinal detachment.
5. **Follow up for life and counsel the parents.** I repeat the EUA, and treat refractive error, amblyopia, squint and corneal scarring (penetrating keratoplasty). A second child is affected in about 3%, and up to 25% if two already are. Full profiles: card GX.

Viva

Q What does "buphthalmos" mean, and how common is it?

Buphthalmos means "ox-eyed": enlarged eyes with raised IOP in the first 3 years. It affects 1 in 3,300 live births in India and 1 in 10,000 in the West; 65% are boys. About 90% are sporadic.

Q What will you ask the parents, and why?

The triad: watering, photophobia and screwing up the eyes in light; also large eyes or a hazy white cornea. **Birth history**: a forceps delivery gives a hazy cornea with vertical Descemet tears. **Family history**: about 10% are inherited, so I ask about siblings and examine them.

Q What else causes a cloudy or large cornea in an infant?

Clouding: CHED, mucopolysaccharidoses (no IOP rise, no enlargement), sclerocornea, forceps injury. **Enlargement**: axial myopia, megalocornea, keratoglobus (normal corneal diameter), Marfan syndrome, homocystinuria. **Watering**: nasolacrimal duct obstruction, ophthalmia neonatorum. The key is the **corneal diameter** and IOP at **examination under anaesthesia**.

Q How is childhood glaucoma classified?

Primary congenital glaucoma is typed by when the IOP rises. **Newborn**: intrauterine (25%); **infantile**: before 3 years (65%); **juvenile**: after 3 years (10%). **Secondary**: systemic disorders (chromosomal, metabolic such as Lowe and Zellweger syndromes, Sturge-Weber syndrome). Ocular developmental disorders (anterior segment dysgenesis, aniridia, congenital ectropion uveae, nanophthalmos). Ocular disease (retinoblastoma, retinopathy of prematurity, persistent hyperplastic primary vitreous, trauma, uveitis).

Q What is the pathogenesis, and what is Barkan's membrane?

In **isolated trabeculodysgenesis** the angle recess is absent. The iris inserts flat (commoner) or concave straight into a thick trabeculum, so outflow falls. **Barkan's membrane** is a proposed sheet of unresorbed mesodermal tissue over the angle. Its existence is debated, but **goniotomy** is based on cutting it.

Q What is the Hoskins classification?

The **Hoskins classification** groups developmental glaucomas by anatomy; it guides treatment and prognosis. (A) **Isolated trabeculodysgenesis**: flat or concave iris insertion. (B) **Iridodysgenesis**: aniridia, iris hypoplasia, congenital iris ectropion. (C) **Iridocorneal dysgenesis**: Axenfeld-Rieger or Peters anomaly, sclerocornea, micro- or megalocornea.

Q Is the damage reversible, and what is the prognosis?

Cupping progresses faster in infants than in adults, but can reverse if IOP is lowered early. The immature lamina cribrosa bows back under pressure and springs back when it falls. **Haab's striae** and corneal scars stay. The earlier the onset, the more severe the damage.

Sturge-Weber syndrome

Definition: Sturge-Weber syndrome (encephalotrigeminal angiomas) is "a rare congenital neuro-oculocutaneous disorder present at birth". Its classic triad is a facial **port-wine stain** that respects the midline, ipsilateral glaucoma and intracranial angiomas. It is not inherited.

Signs to show

1. **Face.** A one-sided red **port-wine stain** (naevus flammeus) over the ophthalmic (V1) and maxillary (V2) divisions. It rarely crosses the midline and thickens with age. Look for opposite-side hemiparesis; ask about seizures (75% in the first year).
2. **Lids.** The stain involves the lid, upper more than lower, with mechanical ptosis. Lid involvement raises the glaucoma risk.
3. **Episclera and conjunctiva.** A dense **episcleral vascular plexus** and dilated conjunctival vessels.
4. **Cornea, iris and lens.** Corneal oedema, or buphthalmos if the glaucoma is congenital. Heterochromia of the iris; cataract or a subluxated lens.

5. **IOP and gonioscopy.** IOP is raised on the stain's side in about half. In infants the angle may look like congenital glaucoma, or show blood in Schlemm's canal from raised venous pressure. In adults it usually looks normal.
6. **Fundus.** A diffuse **choroidal haemangioma** gives a dark red "**tomato-ketchup**" fundus. It can cause serous retinal detachment and cystoid macular oedema. Look for cupping.

Diagnosis line

- "A 12-year-old boy with **Sturge-Weber syndrome** (right V1 and V2 port-wine stain) and **associated right-eye glaucoma**."

How to tell it apart

Condition	How to tell apart
Primary congenital glaucoma	Usually bilateral, with no port-wine stain and rarely any systemic disease. Sturge-Weber glaucoma is usually one-sided, under the stain, with neurological involvement.
Primary open-angle glaucoma	In adults both show an open, normal-looking angle. Sturge-Weber adds the facial stain, dilated episcleral vessels and a one-sided rise.

How I will manage

1. **Look for brain involvement.** A skull X-ray or computed tomography shows **tramline (gyriform) calcification**; magnetic resonance imaging shows the pial angioma.
2. **Drugs for late-onset glaucoma.** A beta-blocker, an alpha-2 agonist or a carbonic anhydrase inhibitor, alone or combined. I avoid **prostaglandin analogues**: the venous pressure is already high, and they can cause anterior uveal effusion.
3. **Goniotomy first under 2 years.** Advantage: no risk of intraoperative choroidal effusion. Disadvantage: it needs a clear cornea.
4. **Filtering surgery with care.** Advantage: trabeculectomy makes an outflow path that does not depend on the episcleral veins. Disadvantage: it risks choroidal effusion, expulsive haemorrhage and bleb failure. So I prefer combined trabeculotomy-trabeculectomy, with one or more sclerotomies first. A drainage device causes fewer intraocular complications; an **Ahmed valve** gives better long-term control than a Molteno tube. Cyclophotocoagulation is for refractory eyes or poor visual potential; it risks hypotony and phthisis.
5. **Treat the stain; follow up for life.** Pulsed dye laser works best early, before the skin thickens, in sittings 2-4 weeks apart. Full profiles: card GX.

Viva

Q What is the mechanism of glaucoma in Sturge-Weber syndrome?

It depends on the onset. In infants (about 60%), an **angle anomaly** blocks outflow, as in congenital glaucoma. In late onset (about 40%), raised **episcleral venous pressure** is the cause. Hypersecretion by the ciliary body or the choroidal haemangioma also contributes.

Q Which patients with a port-wine stain get glaucoma?

Glaucoma occurs in 30-70% when the **port-wine stain** covers both V1 and V2; V2 alone carries the lowest risk. Lid involvement raises the risk further.

Q Why prefer combined surgery?

Combined **trabeculotomy**-trabeculectomy treats both causes: the angle anomaly and the raised venous pressure. Plain trabeculectomy carries more choroidal effusion and expulsive haemorrhage here.

Q What is the visual outcome?

The **visual outcome** is variable, so follow-up is long term. Vision and field suffer from the glaucoma and its surgery, and from the choroidal haemangioma. An occipital leptomeningeal angioma can also cause loss.

Lens-induced glaucomas

Definition: **Lens-induced glaucomas** are "a group of secondary glaucoma that share the lens as a common pathogenic cause". In plain words, the lens either clogs the meshwork or closes the angle. **Open-angle:** phacolytic, lens-particle and phacoanaphylactic. **Closed-angle:** phacomorphic and ectopia lentis.

Signs to show

1. **Phacolytic glaucoma.** An elderly patient with long poor vision has sudden pain and redness. Signs: hyperaemia and microcystic corneal oedema. The chamber is **deep**, with cells and flare but **no keratic precipitates**. **White flocculent material** floats in it, sometimes as a pseudohypopyon. The angle is open. The lens is a **mature, hypermature or morgagnian** cataract, with a wrinkled capsule. IOP is over 35 mm Hg.
2. **Phacomorphic glaucoma.** A **unilateral intumescent cataract** with a **shallow central chamber**, unlike the fellow eye. The angle is closed.
3. **Lens-particle glaucoma.** After cataract surgery or capsulotomy: **free cortical material**, dense flare and cells, synechiae and raised IOP.
4. **Phacoanaphylactic glaucoma.** After surgery or a penetrating injury: keratic precipitates on the cornea **and** lens surface, low-grade vitritis and residual lens matter.
5. **Ectopia lentis.** A forward-displaced lens causes pupillary block and a shallow chamber. Look for Marfan syndrome or homocystinuria.

Diagnosis line

- “Right eye **phacolytic glaucoma** with hypermature (morgagnian) cataract.”
- “Left eye **phacomorphic glaucoma** with intumescent mature cataract.”

How to tell it apart

Condition	How to tell apart
Primary angle-closure glaucoma (versus phacomorphic)	It occurs in hypermetropic eyes with normal lenses, and both chambers are shallow. Phacomorphic glaucoma affects one eye with a swollen cataract.
Acute angle closure (versus phacolytic)	Phacolytic glaucoma mimics an acute attack. But its chamber is deep, its angle open, and white flocculent material floats in the aqueous.
Uveitic glaucoma (versus phacolytic)	Uveitis shows keratic precipitates and posterior synechiae. Phacolytic glaucoma has no keratic precipitates and a hypermature cataract.

How I will manage

1. **Lower the IOP first.** I use a beta-blocker, **apraclonidine** (an alpha-2 agonist that reduces aqueous production) and a carbonic anhydrase inhibitor. If needed, I give intravenous **mannitol** 20%, 1–2 g/kg over 20–30 minutes. Mannitol draws water out of the vitreous; avoid it in anuria, severe dehydration or cardiac decompensation. No pilocarpine in phacolytic glaucoma.
2. **Phacolytic and phacomorphic: remove the lens once IOP is controlled.** Cataract extraction is definitive. In other uveitis the eye must be quiet for 3 months before cataract surgery. In lens-induced inflammation, surgery can be done at once. In phacomorphic glaucoma I add topical steroids and cycloplegics; a laser peripheral iridotomy can help.
 - **Morgagnian cataract:** high pressure inside the lens makes the capsulorhexis run out. I nick the capsule, aspirate the milky fluid, use a dispersive viscoelastic and avoid hydrodissection.
3. **Lens-particle glaucoma.** Aqueous suppressants, topical steroids and cycloplegics (against posterior synechiae); surgery removes the lens debris.
4. **Ectopia lentis with pupillary block.** Laser iridotomy is the treatment of choice. Cycloplegics flatten the lens and pull it back; miotics worsen the block. A lens in the anterior chamber is removed. Full profiles: card GX.

Viva

Q What is the mechanism of phacolytic glaucoma?

In an ageing, hypermature lens, high-molecular-weight lens protein leaks through microscopic capsular openings. Large phagocytes engulf it. Protein and phagocytes clog the meshwork, so **phacolytic** glaucoma is open-angle.

Q How does phacomorphic glaucoma develop?

Phacomorphic glaucoma is “secondary angle closure glaucoma due to lens intumescence”. A rapidly maturing or traumatic cataract swells and pushes the iris root against the cornea.

Q What causes ectopia lentis?

Without systemic disease: familial ectopia lentis, ectopia lentis et pupillae and aniridia. **With systemic disease:** Marfan syndrome, homocystinuria, Weill–Marchesani syndrome, Stickler syndrome and Ehlers–Danlos syndrome.

Q When do you suspect phacoanaphylactic glaucoma?

When granulomatous inflammation follows surgery or injury, because the patient is sensitised to his own lens protein. Criteria: polymorphonuclear leucocytes in the aqueous or vitreous, and too little lens protein to explain the glaucoma.

Uveitic glaucoma

Definition (working description): **Uveitic glaucoma** is a secondary glaucoma in which intraocular inflammation raises the IOP. The mechanism may be open-angle or closed-angle. In plain words, inflammation clogs the meshwork or closes the angle; steroid treatment can add a rise.

Signs to show

1. **Conjunctiva.** Circumciliary congestion, minimal in Posner–Schlossman syndrome.
2. **Cornea.** **Keratic precipitates**; band keratopathy in chronic cases.
3. **Anterior chamber and iris.** Flare and cells; posterior synechiae; **iris bombé** if they are complete.
4. **Lens.** Complicated cataract.
5. **IOP.** Raised; it can be low if the inflamed ciliary body makes less aqueous, or with a detachment.
6. **Gonioscopy.** An open angle with debris, or **peripheral anterior synechiae** closing it.

Diagnosis line

- “Right eye **uveitic glaucoma** (open angle) with acute anterior uveitis and complicated cataract.”

How to tell it apart

Condition	How to tell apart
Posner–Schlossman syndrome (glaucomatocyclitic crisis)	One eye, recurrent crises of high IOP in a white eye. Mild flare, fine white keratic precipitates, an open angle without synechiae. Normal between attacks.

Condition	How to tell apart
Fuchs heterochromic iridocyclitis	Chronic, non-granulomatous, one-sided and white; the affected iris is usually lighter. Stellate keratic precipitates , cataract, and fine vessels bridging the angle. Posterior synechiae are rare. The glaucoma comes from trabecular sclerosis.
Steroid-induced glaucoma in a treated uveitis	A quiet chamber; the pressure falls when the steroid is stopped or weakened. Active uveitis shows cells, flare and fresh keratic precipitates.

How I will manage

1. **Control the inflammation.** Topical **prednisolone acetate 1%**, titrated; periocular or systemic steroid if needed. **Cycloplegics** (atropine 1%) prevent posterior synechiae and relieve ciliary spasm.
2. **Lower the pressure with aqueous suppressants.** A beta-blocker, alpha-2 agonist or carbonic anhydrase inhibitor is first line. **Miotics and prostaglandin analogues are contraindicated:** miotics increase inflammation, and prostaglandins can cause mild anterior uveitis.
3. **Treat pupillary block.** Laser iridotomy is safer than surgical iridectomy, but often fails from inflammation, fibrin or new vessels.
4. **Surgery in a quiet eye, when drugs fail.** Trabeculectomy needs heavy steroid cover; subconjunctival 5-fluorouracil clearly improves its success. An **Ahmed valve** is placed once inflammation is controlled. It needs steroid cover, or early inflammatory exudate blocks the tube. Laser trabeculoplasty is contraindicated. Full profiles: card GX.

Viva

Q What are the mechanisms of uveitic glaucoma?

Open-angle: trabeculitis, fibrinous aqueous, raised episcleral venous pressure, more aqueous secretion, steroid response.

Closed-angle: iris bombé from posterior synechiae, or peripheral anterior synechiae. A swollen ciliary body can also rotate forward and close the angle.

Q Which uveitic conditions raise the IOP?

Viral uveitis, toxoplasmosis, sarcoidosis, **Fuchs heterochromic iridocyclitis** and **Posner-Schlossman syndrome**. Steroid treatment and lens-induced uveitis also raise it.

Q What is Posner-Schlossman syndrome?

Posner-Schlossman syndrome (glaucomatocyclitic crisis) is recurrent, one-sided mild anterior uveitis with a marked IOP rise. Crises last hours to weeks. Treat with anti-glaucoma drugs (apraclonidine especially) and cycloplegics.

Q Why is laser trabeculoplasty contraindicated in uveitic glaucoma?

Laser trabeculoplasty does not work in an inflamed eye with open-angle glaucoma. It may cause a further significant rise in IOP.

Steroid-induced glaucoma

Definition: **Steroid-induced glaucoma** is “a form of open angle glaucoma occurring as an adverse effect [of] exogenous corticosteroid therapy or excess endogenous production of glucocorticoids”. In plain words, a steroid by any route can raise IOP in a susceptible person.

Signs to show

1. **Lids and cornea.** Lid skin atrophy or ptosis; increased corneal thickness.
2. **Anterior chamber** quiet; **pupil** dilated by topical steroid.
3. **Lens.** **Posterior subcapsular cataract**, another steroid effect.
4. **IOP.** High while the steroid continues; it may be normal after it stops.
5. **Gonioscopy** open in all quadrants; **disc** cupped.

Diagnosis line

- “Both eyes **steroid-induced glaucoma** in a known case of vernal keratoconjunctivitis on topical dexamethasone, with posterior subcapsular cataract.”

How to tell it apart

Condition	How to tell apart
Primary open-angle glaucoma	Bilateral, high IOP, no steroid history. Steroid glaucoma may be one-sided, with normal or high IOP.
Normal-tension glaucoma	Thin corneas and normal IOP. Steroid glaucoma after the steroid stops looks similar, so ask about steroids in every form.
Glaucomatocyclitic crisis	One eye, with keratic precipitates and cells during the attack. Steroid glaucoma has a quiet chamber.

How I will manage

1. **Stop the steroid.** IOP usually returns to normal within weeks to months. If it cannot stop, I switch to fluorometholone or loteprednol, or a non-steroidal drug. The rise follows potency: dexamethasone 0.1% raises IOP by a mean 22 mm Hg; fluorometholone 0.1%, by 6.
2. **Anti-glaucoma drugs** while the pressure settles. After intravitreal triamcinolone I continue them for about 6 months, until the crystals clear.
3. **Remove a depot** if the rise is intractable: pars plana vitrectomy with trabeculectomy.
4. **Laser trabeculoplasty** when drugs fail or are not tolerated; **trabeculectomy**, with or without an antimetabolite, when both fail. Full profiles: card GX.
5. **Prevent it.** I record a baseline IOP and check it every 2 weeks on steroids.

Viva

Q What is a steroid responder, and how is the response graded?

A **steroid responder** shows a rise in IOP on repeated steroid drops. **Becker** (four times daily, 6 weeks, final IOP): low under 20, intermediate 20–30, high over 31 mm Hg. **Armaly** (three times daily, 4 weeks, rise): low under 6, intermediate 6–15, high over 15 mm Hg.

Q How will you take the steroid history?

I ask specifically about **steroid eye drops, ointments and tablets**. I also ask about inhalers, nasal sprays, skin creams, and periocular or intravitreal injections and implants.

Q Which route is most dangerous?

A **periocular depot** of long-acting steroid is the most dangerous. Topical drops often raise IOP; systemic steroids are the least likely to. Non-steroidal anti-inflammatory drugs do not raise IOP.

Q How soon does the IOP rise?

With a potent topical steroid four times daily, in 2–6 weeks. Intravitreal triamcinolone 4 mg takes 4–8 weeks. Inhaled and skin steroids take about 3 months; oral prednisolone, 2–15 months.

Q Who is at risk?

Patients with **primary open-angle glaucoma** or a family history of it, and children under 10. Also high myopes, diabetics and patients with rheumatoid arthritis. Infants on steroids can develop a picture like congenital glaucoma.

Q What is the mechanism?

Steroids raise **outflow resistance** in the trabecular meshwork. Glycosaminoglycans accumulate, extracellular matrix increases, and meshwork cells phagocytose less debris.

Pigmentary glaucoma

Definition (working description): **Pigmentary glaucoma** is the open-angle glaucoma that develops in **pigment dispersion syndrome**. Pigment rubbed off the back of the iris clogs the trabecular meshwork. In plain words, the mid-peripheral iris bows back and rubs on the zonules. About 35% of patients with pigment dispersion syndrome develop glaucoma.

Signs to show

1. **Patient.** A young myopic man, 20–30 years; usually both eyes.
2. **Cornea.** A **Krukenberg spindle**: pigment on the corneal endothelium.
3. **Anterior chamber.** Deep, with the mid-peripheral iris bowed back (**reverse pupillary block**).
4. **Iris.** **Mid-peripheral transillumination defects**; pigment granules.
5. **Lens.** Pigment on the posterior lens surface (Zentmayer's ring).
6. **Gonioscopy.** A wide open angle with a **dense, homogeneous dark-brown band** around the meshwork.
7. **Fundus.** Look for lattice degeneration and retinal detachment.

Diagnosis line

- "Both eyes **pigmentary glaucoma** with Krukenberg spindles and mid-peripheral iris transillumination defects."

How to tell it apart

Condition	How to tell apart
Pseudoexfoliation glaucoma	Older (over 60 years), both sexes. White powdery material on the pupil margin and lens; transillumination defects at the pupillary margin; blotchy angle pigment with Sampaolesi's line.
Primary open-angle glaucoma	No Krukenberg spindle, no mid-peripheral defects, no dense pigment band.

How I will manage

1. **Drops first.** Timolol 0.5% twice daily and dorzolamide 2% three times daily reduce aqueous production. Avoid timolol in asthma; avoid dorzolamide in sulpha allergy, renal failure and chronic liver disease. **Pilocarpine 1%** limits iris movement and pigment release. But young patients tolerate it poorly, and it raises the risk of retinal detachment.
2. **Laser iridotomy** relieves the **reverse pupillary block**, so the iris flattens. Advantage: it treats the cause, the backward bowing of the iris. Disadvantage: iridotomy risks such as an IOP spike and uveitis.

3. **Laser trabeculoplasty.** Advantage: a good initial response. Disadvantage: it may not last.
4. **Trabeculectomy** with an antimetabolite, or a drainage device. Full profiles: card GX.

Viva

Q What is the mechanism?

By **Campbell's mechanical theory**, the concave mid-peripheral iris rubs against the zonules and sheds pigment. This backward bowing is **reverse pupillary block**, which laser iridotomy relieves. Pigment and reduced outflow raise IOP. Later the meshwork cells degenerate and the meshwork scars.

Q What is a Krukenberg spindle, and where else is it seen?

A **Krukenberg spindle** is pigment on the corneal endothelium. It also occurs in chronic angle closure, in Vogt's triad with glaukomflecken and sectoral iris atrophy.

Q What else causes a heavily pigmented angle?

Ageing, **pseudoexfoliation**, trauma, and previous surgery or laser iridotomy. Also uveitis, a past angle-closure attack, ocular melanosis and a sulcus intraocular lens.

Q How does it differ from pseudoexfoliation glaucoma?

Pigmentary glaucoma affects younger men (2:1), lessens with age, has no systemic link and is mostly treated medically. Pseudoexfoliation affects older people, worsens with age and mostly needs surgery.

Angle-recession glaucoma

Definition: "Angle recession glaucoma is a secondary open angle glaucoma that is associated with blunt trauma to eye. Angle recession is a tear between the longitudinal and circular muscles of the ciliary body." It is a gonioscopy diagnosis. In plain words, the pressure can rise months or years after a blunt blow.

Signs to show

1. **Old injury.** Lid or periocular scars; the injury itself may be forgotten.
2. **Cornea and anterior chamber.** Early cases may show blood staining or pigment on the endothelium, and a hyphaema. The chamber is deep and irregular.
3. **Iris and pupil.** Iridodialysis (a D-shaped pupil), sphincter tears, traumatic mydriasis.
4. **Lens.** Subluxation, traumatic cataract, Vossius ring.
5. **IOP.** Raised in one eye; it may be low early, from ciliary shutdown.
6. **Gonioscopy**, deferred 4–6 weeks after injury, because a hyphaema masks the recession early. A **widened ciliary body band** is the key sign, with a prominent scleral spur and irregular pigment. Compare the other eye: simultaneous bilateral Koeppe gonioscopy shows subtle recession best.
7. **Fundus.** Retinal dialysis or detachment; glaucomatous cupping.

Diagnosis line

- "Right eye **angle-recession glaucoma** following blunt trauma ___ years ago."

How to tell it apart

Condition	How to tell apart
Cyclodialysis	In recession the longitudinal muscle stays attached; in cyclodialysis the whole ciliary body is detached.
Other one-sided glaucomas (pseudoexfoliation, neovascular, phacolytic)	Look for pseudoexfoliative material, iris new vessels or lens matter. A trauma history and a widened band confirm recession.

How I will manage

1. **Acute injury.** I use timolol 0.5% twice daily or brimonidine 0.2%, with topical atropine and a steroid for pain and inflammation.
2. **Chronic glaucoma: prostaglandin analogues are the drug of choice.** Miotics are ineffective because the meshwork is scarred.
3. **Laser trabeculoplasty fails in most eyes**, because the angle is distorted and scarred.
4. **Trabeculectomy with an antimetabolite** when drugs fail. Advantage: it controls IOP. Disadvantage: success is lower than in primary open-angle glaucoma. Drainage devices give only limited benefit. Full profiles: card GX.

Viva

Q How common is angle recession, and how many get glaucoma?

Angle recession follows 20–94% of blunt injuries and 85% of traumatic hyphaemas. Only about 5.5% develop glaucoma; the risk is higher beyond 180°. Glaucoma peaks within 1 year and again after 10 years.

Q How does angle recession happen?

Blunt trauma compresses the globe front to back. The equator expands, the limbus stretches and the lens–iris diaphragm is pushed back, tearing the ciliary muscle.

Q What causes the glaucoma?

Trabecular damage, not the recession itself. That is why miotics fail.

Q What are the seven rings of trauma?

Iris sphincter tear, iridodialysis, angle recession, cyclodialysis, trabecular tear, zonular dialysis with lens subluxation, and retinal dialysis or tear.

Malignant glaucoma (aqueous misdirection)

Definition (working description): Malignant glaucoma (ciliary block glaucoma) is a secondary angle closure without pupillary block. It usually follows eye surgery or laser. Aqueous flow is blocked at the secreting part of the ciliary body, so aqueous is forced backwards into the vitreous. In plain words, fluid trapped behind pushes the iris and lens forward against the cornea.

Signs to show

1. **History.** Recent trabeculectomy, cataract surgery, laser iridotomy or cyclophotocoagulation.
2. **Anterior chamber.** Diffusely shallow, centre and periphery.
3. **Bleb** absent; **Seidel test negative**.
4. **IOP.** High or normal.
5. **Ultrasound biomicroscopy.** The ciliary processes are rotated forward, and all anterior structures press against the cornea.

Diagnosis line

- "Right eye **malignant glaucoma**, third day after trabeculectomy."

How to tell it apart

Condition	How to tell apart
Pupillary block	The iris bows forward and the iridotomy is not patent; laser iridotomy relieves it. In malignant glaucoma the iridectomy is patent and the centre of the chamber is shallow too.
Wound leak	A soft eye, a poor bleb and a positive Seidel test.
Excessive filtration	Very low IOP, a good bleb and a negative Seidel test.

How I will manage

1. **Strong topical mydriatics** first.
2. **If they fail, intravenous mannitol** 20%, 1–2 g/kg over 20–30 minutes. It shrinks the vitreous by drawing water into the plasma.
3. **If that fails, neodymium:yttrium-aluminium-garnet (Nd:YAG) laser hyaloidotomy** through a patent iridectomy.
4. **If the laser fails, pars plana anterior vitrectomy** with a vitreous cutter removes the trapped fluid.
5. **Prevent it.** After past malignant glaucoma, or in angle closure, I suture the trabeculectomy flap more tightly. Full profiles: card GX.

Viva

Q What is the mechanism?

Ciliary block: aqueous is forced into the vitreous, and the anterior hyaloid face stops it coming forward.

Q How does the Nd:YAG laser help?

The **anterior hyaloid face** is the barrier that stops the trapped aqueous coming forward. The laser breaks it through a patent iridectomy, which relieves the ciliovitreal compression.

Q How does ultrasound biomicroscopy help?

Ultrasound biomicroscopy shows where the angle closes. The four sites are the iris (pupillary block), ciliary body (plateau iris), lens (phacomorphic) and behind the iris (malignant).

Q What causes a shallow chamber with raised IOP after trabeculectomy?

Malignant glaucoma, pupillary block from an incomplete iridectomy, or delayed suprachoroidal haemorrhage. A soft eye points instead to a wound leak, excessive filtration or a serous choroidal detachment. After cataract surgery, the same three causes give a shallow chamber with raised IOP.

Quick recall

- **Buphthalmos:** epiphora, photophobia, blepharospasm; cornea over 12 mm (normal 10–10.5); IOP over 14 mm Hg (normal 11–14); cup over 0.3.
- **Haab's striae** are horizontal or concentric to the limbus; forceps tears are vertical or oblique.
- **Goniotomy** if the cornea is clear; **trabeculotomy** if hazy. No alpha-2 agonists under 2 years.
- **Sturge-Weber:** V1 and V2 stain; raised episcleral venous pressure; no prostaglandins.
- **Lens-induced:** lower IOP, then remove the lens. **Phacolytic:** deep chamber, white flocculent material, open angle.
- **Uveitic:** steroid, cycloplegic, aqueous suppressant; no miotics or prostaglandins.
- **Steroid responders:** Becker final IOP under 20 / 20–30 / over 31; Armaly rise under 6 / 6–15 / over 15 mm Hg.
- **Pigmentary:** Krukenberg spindle, mid-peripheral defects; laser iridotomy.
- **Angle recession:** widened ciliary body band; miotics fail.
- **Malignant:** diffuse shallow chamber, IOP high or normal, no bleb, Seidel negative → mydriatics → mannitol → Nd:YAG → vitrectomy.

Mnemonic: Congenital glaucoma — **ABCDDE:** Axial myopia · Buphthalmos · Cloudy cornea · Descemet's breaks · Diameter enlarged · Disc cupping · Examination under anaesthesia.

Angle pigment — **PIGMENT:** Pseudoexfoliation and pigment dispersion · Iritis · Glaucoma after angle closure · Melanosis · Endocrine · Naevus · Trauma.

Read more: Baidya p.150–151, 155, 166, 169–176, 367, 383–384, 442, 737, 739 · Namrata p.142–143, 154, 174, 181, 184–191, 197–202, 205, 207 · Aravind 2.16, 3.1, 3.3, 4.1, 4.2, 4.5, 4.6, 4.8, 4.9, 4.10, 4.12, 4.13, 4.14, 4.15, 4.17, 4.18, 4.21, 5